Algebraic Topology

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Contents

0 Introduction

The fundamental question we want to answer in topology is when two topological spaces are *equivalent*. Showing equivalence can be done by construction, but showing two spaces aren't equivalent is somewhat harder. This is what algebraic topology answers. We note that it is also useful to keep examples from algebraic topology when learning more abstract areas such as category theory.

§0.1 Conventions

- 'map' is a continuous map.
- $I := [0, 1]$, the unit interval.

§0.2 Recap

Definition 0.2.1. (X, Ω) is a **topological space** where X is a set, $\Omega \subset \mathcal{P}(X)$ such that

- $\emptyset, X \in \Omega$
- $\{U_i\}_{i \in I} \subset \Omega \implies \bigcup_{i \in I} U_i \in \Omega$
- $U, V \in \Omega \implies U \cap V \in \Omega$

$U \in \Omega$ are called **open sets**. $X \setminus U$ are called **closed sets** (where U is open).

Example 0.2.2
The trivial topology on any X, is $\Omega = \{\emptyset, X\}$.

Example 0.2.3
The discrete topology on any X, is $\Omega = \mathcal{P}(X)$.

Definition 0.2.4. Given $(X, \Omega_X), (Y, \Omega_Y)$, $X \times Y$ is a topological space with **product topology**, which is the topology generated by $U \times V$, $U \in \Omega_X$ and $V \in \Omega_Y$.

Remark 0.2.5 — Every open subset in the product topology is a (possibly infinite union) of rectangles $U \times V$ (but not all opens are rectangles).

Definition 0.2.6. (X, Ω), $\mathcal{B} \subset \Omega$ **generates** the topology if $\mathcal{B}$ is a basis of the topology. That is, if $\forall U \in \Omega$, $\exists \{U_i\}_{i \in I} \subset \mathcal{B}$ such that $U = \bigcup_{i \in I} U_i$.

Definition 0.2.7. Given $(X, \Omega_X), (Y, \Omega_Y)$, their **disjoint union** is $X \sqcup Y = (X \times \{0\}) \cup (Y \times \{1\})$. This is generated by

$$\{U \times \{0\} \mid U \in \Omega_X\} \cup \{V \times \{1\} \mid V \in \Omega_Y\}.$$

Remark 0.2.8 — Topologically speaking X is the same as $X \times \{0\}$ and Y is the same as $Y \times \{1\}$, but we make an arbitrary choice of $0, 1$ to make sure they do not overlap in the union.

Example 0.2.9

$X \cup X = X$, whereas $X \sqcup X = X \times X$.

Definition 0.2.10. $(X, \Omega_X), A \subset X$, A is a topological space with the **subspace topology** $\{U \cap A \mid U \in \Omega_X\}$.

Example 0.2.11

Consider $[0, 1] \subset \mathbb{R}$ with the subspace topology (where $\mathbb{R}$ is taken with the topology induced by the standard metric). Then $(0, 1] \subset \mathbb{R}$ is not open, but $(0, 1] = (0, 2) \cap [0, 1] \subset [0, 1]$ is open in the subspace topology.

Definition 0.2.12. Given a topological space (X, Ω_X) and an equivalence relation $\sim$, $Y := X/\sim = \{[x] \mid x \in X\}$, where $[x]$ is an equivalence class, is a topological space with the **quotient topology**

$$\Omega_Y = \{U \subset Y \mid \pi^{-1}(U) \in \Omega_X\},$$

where $\pi \colon X \to X/\sim = Y$.

We often consider the quotient space defined by a subset $A \subset X$. We denote this by X/A, where the equivalence relation is $x \sim y$ if and only if $x = y$ or $x, y \in A$.

Remark 0.2.13 — π is an example of an **identification map**; that is, a surjective map $f \colon X \to Y$ such that $U \subset Y$ is open if and only if $f^{-1}(U) \subset X$ is open.

Proposition 0.2.14

A surjective map $f \colon X \to Y$ is an identification map if and only if $\forall g \colon Y \to Z$, g is continuous if and only if $g \circ f$ is continuous.

$$\begin{tikzcd} X \arrow[d, "f"] \arrow[dr, "g \circ f"] & \\ Y \arrow[r, "g"] & Z \end{tikzcd}$$

Proof. Exercise. □

Here are now some examples of topological spaces.

Example 0.2.15 (Unit Sphere)

$\mathbb{S}^n \subset \mathbb{R}^{n+1}$, where

$$\mathbb{S}^n = \left\{ x \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} x_i^2 = 1 \right\}.$$

Note that $\mathbb{S}^0 = \{\text{point}\} \sqcup \{\text{point}\}$.

Example 0.2.16 (Unit Ball)

$\mathbb{D}^n \subset \mathbb{R}^n$, where

$$\mathbb{D}^n = \left\{ x \in \mathbb{R}^n \mid \sum_{i=1}^{n} x_i^2 \leq 1 \right\}.$$

Note that $\mathbb{D}^n/\mathbb{S}^{n-1} \cong \mathbb{S}^n$.

Example 0.2.17 (n-Torus)

$\mathbb{T}^n = \mathbb{S}^1 \times \cdots \times \mathbb{S}^1$ (n-times).

Example 0.2.18

We also have $\mathbb{T}^1 = \mathbb{R}^2/\mathbb{Z}^2$.

Definition 0.2.19. $f\colon X \to Y$ is a **homeomorphism** is it is continuous and there exists a continuous map $g\colon Y \to X$ such that

$$f \circ g = id_Y, \quad g \circ f = id_X.$$

Example 0.2.20

Let us show that the topological torus $\mathbb{S}^1 \times \mathbb{S}^1 \subset \mathbb{R}^4$ is homeomorphic to the torus embedded in $\mathbb{T}^2 \subset \mathbb{R}^3$, parametrized by

$$\gamma(\theta, \phi) = ((b + a\cos\theta)\sin\phi, (b + a\sin\theta)\cos\phi, a\sin\theta),$$

for $\theta, \phi \in [0, 2\pi)$ and fixed $0 < a < b$. It is clear we can construct the homeomorphism $f\colon \mathbb{S}^1 \times \mathbb{S}^1 \to \mathbb{T}^2$

$$(x_1, y_1, x_2, y_2) \mapsto ((b + ax_1)y_2, (b + ax_2)y_1, ay_1).$$

We note that these are different spaces (they are embedded in Euclidean spaces of different dimension, and taken with the respective subspace topology), but they are homeomorphic, so they are considered to be the same.

Example 0.2.21

$\mathbb{D}^n/\partial\mathbb{D}^n \cong \mathbb{S}^n$.

Example 0.2.22

$X = [0,1] = \mathbb{D}^1$, $A = \{0,1\} = \partial\mathbb{D}^1 \cong \mathbb{S}^0$. The map $f\colon [0,1] \to S^1$, $t \mapsto (\cos 2\pi t, \sin 2\pi t)$, it is surjective and injective in the quotient X/A. This then gives that

$$X/A = [0,1]/\{0,1\} \cong \mathbb{S}^1.$$

Definition 0.2.23. We say that f is **factoring** over the quotient, i.e. we are eliminating all points that cause a problem in the map $f\colon X \to Y$ in order to make $\tilde{f}\colon X/\sim \to Y$ a homeomorphism.

Note. These notes are a combination of the lectures, Lee's *Introduction to Topological Manifolds*, and Hatcher's *Algebraic Topology*.

§0.3 Preliminary Results

Lemma 0.3.1 (Pasting/ Gluing Lemma)

Let X, Y be topological spaces, and let $\{A_i\}$ be either an arbitrary open cover of X or a finite closed cover of X. Suppose that we're given continuous maps $f_i\colon A_i \to Y$ that agree on overlaps: $f_i|_{A_i \cap A_j} = f_j|_{A_i \cap A_j}$. Then, there exists a unique continuous map $f\colon X \to Y$ whose restriction to each A_i is equal to f_i.

Proof. In either case, it follows from elementary set theory that there exists a unique map f such that $f|_{A_i} = f_i$, for each i. If the sets A_i are open, the continuity of f follows immediately from the local criterion for continuity. On the other hand, suppose $\{A_1, \ldots, A_k\}$ is a finite closed cover of X. To prove that f is continuous, it suffices to show that the preimage of each closed subset $K \subseteq Y$ is closed. It is easy to check that for each i, $f^{-1}(K) \cap A_i = f_i^{-1}(K)$, and $f^{-1}(K)$ the union of these sets for $i = 1, \ldots, k$. Since $f_i^{-1}(K)$ is closed in A_i by continuity of f_i, and A_i is closed in X by hypothesis, it follows that $f_i^{-1}(K)$ is also closed in X. Thus $f^{-1}(K)$ is the union of finitely many closed subsets, and hence closed. $\square$

Definition 0.3.2. The **diameter** of a bounded set S in a metric space is defined to be $\operatorname{diam}(S) = \sup\{d(x, y) \mid x, y \in S\}$.

Definition 0.3.3. If $\mathcal{U}$ is an open cover of a metric space, the number $\delta > 0$ is called a **Lebesgue number** for the cover if every set whose diameter is less than δ is contained in one of the sets $U \in \mathcal{U}$.

Lemma 0.3.4 (Lebesgue Covering/ Number Lemma)

Every open cover of a compact metric space has a Lebesgue number.

Proof. Let $\mathcal{U}$ be an open cover of the compact metric space M. Each point $x \in M$ is in some set $U \in \mathcal{U}$. Since U is open, there is some $r(x) > 0$ such that $B_{2r(x)}(x) \subseteq U$. The balls $\{B_{r(x)}(x) \mid x \in\}$ form an open cover of M, so finitely many of them, say $B_{r(x_1)}(x_1), \ldots, B_{r(x_n)}(x_n)$ cover M.

We will now show that $\delta = \min\{r(x_1), \ldots, r(x_n)\}$ is a Lebesgue number for $\mathcal{U}$. Suppose that $S \subseteq M$ is a nonempty set whose diameter is less than δ. Let y_0 be any point of S. Then, there is some x_i such that $y_0 \in B_{r(x_i)}(x_i)$. It suffices to show that $S \subseteq B_{2r(x_i)}(x_i)$, since the latter set is, by construction, contained in some $U \in \mathcal{U}$. If $z \in S$, the triangle inequality provides

$$d(z, x_i) \leq d(z, y_0) + d(y_0, x_i) < \delta + r(x_i) < 2r(x_i),$$

which proves the claim. $\square$

1 Homotopy and The Fundamental Group

§1.1 Homotopy

Definition 1.1.1. (X, A) **pair of topological spaces** if $A \subset X$. If $A = \{\text{point}\}$, then the pair is called a **pointed space**.

Definition 1.1.2. Let X, Y be topological spaces and $f, g \colon X \to Y$ be continuous. A **homotopy** between f and g is a continuous map

$$\begin{aligned} F \colon X \times I &\to Y \\ (x, t) &\mapsto f_t(x) \end{aligned}$$

such that for all $x \in X$, $F(x, 0) = f_0 = f$ and $F(x, 1) = f_1 = g$. We say that f and g are **homotopic**, writing $f \simeq g$.

Proposition 1.1.3

Homotopy equivalence is an equivalence relation.

Proof. Reflexive: $F(x, t) = f(x)$, so $f \simeq f$.

Symmetric: if $F(x, t)$ is a homotopy equivalence between f and g, then we can consider $G(x, t) = F(x, 1 - t)$, so that $g \simeq f$.

Transitive: Let $f, g, h \colon X \to Y$, and consider the homotopies F, G such that $f \simeq g$ and $g \simeq h$, respectively. Then, we can define

$$H(x, t) = \begin{cases} F(x, 2t) & t \leq \frac{1}{2} \\ G(x, 2t - 1) & t \geq \frac{1}{2} \end{cases},$$

which is continuous by the pasting/ gluing lemma. □

Proposition 1.1.4

The homotopy relation is preserved by composition: if

$$f_0, f_1 \colon X \to Y \quad \text{and} \quad g_0, g_1 \colon Y \to Z$$

are continuous maps with $f_0 \simeq f_1$ and $g_0 \simeq g_1$, then $g_0 \circ f_0 \simeq g_1 \circ f_1$.

Proof. Suppose $F \colon f_0 \simeq f_1$ and $G \colon g_0 \simeq g_1$ are homotopies, and define $H \colon X \times I \to Z$ by $H(x, t) = G(F(x, t), t)$. At $t = 0$, $H(x, 0) = G(f_0(x), 0) = g_0(f_0(x))$, and at $t = 1$, $H(x, 1) = G(f_1(x), 1) = g_1(f_1(x))$. Thus, H is a homotopy that proves our result. □

Example 1.1.5 (Straight-Line Homotopy)
Let $B \subseteq \mathbb{R}^n$ be a convex set, and X be any topological space. Suppose that $f, g \colon X \to B$ are any two continuous maps. Then, we can define a homotopy $H \colon f \simeq g$ by
$$H(x,t) = f(x) + t(g(x) - f(x)).$$

Definition 1.1.6. Let X, Y be topological spaces and $f \colon X \to Y$ is a **homotopy equivalence** if there exists a continuous map $g \colon Y \to X$ such that
$$f \circ g \simeq id_Y, \quad g \circ f \simeq id_X.$$
We then say that X and Y are **homotopic**, writing $X \simeq Y$.

Remark 1.1.7 — This is also an equivalence relation, but we won't show it here.

Example 1.1.8 (Homotopy equivalence $\not\Longrightarrow$ Homeomorphism.)
Brouwer's Invariance of Domain theorem first tells us that $\mathbb{R}^n$ is not homeomorphic to $\mathbb{R}^m$, unless $n = m$. However, $\mathbb{R}^n \simeq \mathbb{R}^m$, since $\mathbb{R}^n \simeq \{0\}$, for all n. Let $f \colon \mathbb{R}^n \to \{0\}, x \mapsto 0$ and $g \colon \{0\} \hookrightarrow \mathbb{R}^n, 0 \mapsto 0$. The homotopy we are searching for is
$$F(x,t) = tx.$$
As a consequence of this homotopy, we have, in general, for all topological spaces $X \times \mathbb{R} \simeq X$.

Example 1.1.9
Another example of homotopic spaces that are not homeomorphic is the real line $\mathbb{R}$ and the singleton $\{0\}$. These are homotopically equivalent since the singleton $\{0\}$ is a contraction of the real line, but the two spaces are not homeomorphic since the latter is a finite set, and $\mathbb{R}$ is an infinite set.

Example 1.1.10

Let $X = \mathbb{S}^1$, $Y = \mathbb{R}^2 \setminus \{0\}$, and $i\colon \mathbb{S}^1 \to \mathbb{R}^2 \setminus \{0\}$ be the standard inclusion. Define the following map:

$$\begin{aligned} r\colon \mathbb{R}^2 \setminus \{0\} &\to \mathbb{S}^1 \\ x &\mapsto \frac{x}{|x|} \end{aligned}$$

Then, $r \circ i = id_{\mathbb{S}^1}$, so we may take the constant homotopy. On the other hand, the composition $i \circ fr\colon \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}^2 \setminus \{0\}$ is $x \mapsto \frac{x}{|x|}$. Define a homotopy

$$\begin{aligned} H\colon (\mathbb{R}^2 \setminus \{0\}) \times I &\to \mathbb{R}^2 \setminus \{0\} \\ (x,t) &\mapsto \frac{x}{t + (1-t) \cdot |x|}, \end{aligned}$$

which is well define as if $t + (1-t) \cdot |x| = 0$, then $|x| = \frac{t}{t-1} \leq 0$, which is not possible. This is a homotopy from $i \circ r$ to $id_{\mathbb{R}^2 \setminus \{0\}}$, and so i is a homotopy equivalence with homotopy inverse r; $\mathbb{S}^1 \simeq \mathbb{R}^2 \setminus \{0\}$.

Example 1.1.11

Let $Y = \mathbb{R}^n$, $X = \{0\}$, and $i\colon \{0\} \to \mathbb{R}^n$ be the inclusion. Let $r\colon \mathbb{R}^n \to \{0\}$ be the only map there is. Then, $r \circ i = id_{\{0\}}$ and $i \circ r\colon \mathbb{R}^n \to \mathbb{R}^n$ is the constant map to 0. Define $H\colon \mathbb{R}^n \times I \to \mathbb{R}^n$ by $H(x,t) = t \cdot x$, which is a homotopy from $i \circ r$ to $Id_{\mathbb{R}^n}$. Therefore, i is a homotopy equivalence.

This is an important property in this course, requiring a definition:

Definition 1.1.12. A topological space X is **contractible** if it is homotopy equivalent to a point; $X \simeq \{\text{point}\}$.

> **Exercise 1.1.13.** Show that $\mathbb{S}^0$ is not contractible.

Solution. The solution follows from the more general claim:

Claim 1.1.14. Two discrete spaces are homotopy equivalent if and only if the have the same number of points.

Proof. This is because continuous maps from a discrete set to a discrete set must be constant, so we must attain a bijection. □

□

Example 1.1.15

$\mathbb{S}^n$ is not contractible.

§1.1.i Retractions

Definition 1.1.16. Let (X, A) be a pair. A **retraction** of X onto A is a continuous map $r\colon X \to A$ such that $r(X) = A$ and $r\,|_A = Id_A$.

Definition 1.1.17. Let (X, A) be a pair of spaces. A **deformation retract** of X onto A is a retraction that is homotopic to the identity. i.e., there exists a continuous map

$$\begin{aligned} F\colon X \times I &\to X \\ (x, t) &\mapsto f_t(x) \end{aligned}$$

such that $f_0 = id_X$ and f_1 is a retraction, with the property that

$$f_t\,|_A = id_A, \quad \forall t \in [0, 1].$$

Remark 1.1.18 — We may refer to the final condition as a 'strong deformation retract.'

Example 1.1.19

$\mathbb{R}^n$ deformation retracts to a point (the origin, by $f_t(x) = t \cdot x$).

Example 1.1.20

Note that as the definition suggests, a retraction need not to be a deformation retract. As a simple example, consider the retraction of a space to a point. This is always a retract, but only a deformation retract when the space is contractible.

Definition 1.1.21. A continuous map is **null-homotopic** if it is homotopy equivalent to the constant map.

§1.2 Paths

Definition 1.2.1. Let X be a topological space and $x_0, x_1 \in X$. A **path** from x_0 to x_1 is a continuous map $f\colon I \to X$ such that $f(0) = x_0$ and $f(1) = x_1$.

Remark 1.2.2 — A path doesn't equal to the image of the path, as we see in the example below.

Example 1.2.3

Consider $f, g\colon I \to \mathbb{S}^1$ given by

$$f(t) = e^{2\pi it}, \quad g(t) = e^{4\pi it}.$$

These have the same image, but different paths.

Definition 1.2.4. A **loop** with basepoint x_0 is a path $f\colon I \to X$ such that $f(0) = f(1) = x_0$.

Definition 1.2.5. Given $f, g\colon I \to X$, with $f(1) = g(0)$, then we define the **product path** as the concatenation:

$$(f \cdot g)(s) = \begin{cases} f(2s) & s \in [0, \frac{1}{2}] \\ g(2s-1) & s \in [\frac{1}{2}, 1] \end{cases}$$

Remark 1.2.6 — By the pasting lemma, this is continuous.

Definition 1.2.7. Given a path $f\colon I \to X$, we define its **inverse** $f^{-1} = f(1-t)$.

Remark 1.2.8 — We may also denote the inverse as $\overline{f}$.

Definition 1.2.9. The **constant path** at x_0 $c_{x_0}\colon I \to X$ is the map $c_{x_0}(t) \equiv x_0$.

Definition 1.2.10. X is **path connected** if there exists a path between any two points in X.

Definition 1.2.11. X is **connected** if it cannot be written as a union $U \cup V$, where U, V are disjoint closed subsets of X.

Example 1.2.12 (Topologist's Sine Curve)
The following space is connected, but not path connected.

$$T = \{(x, \sin\frac{1}{x}) \mid x \in (0,1]\} \cup \{(0,0)\}.$$

Exercise 1.2.13. If X is path connected, then every path is homotopic to the constant path.

Definition 1.2.14. Let $f, g\colon I \to X$ have the same endpoints. i.e., $f(0) = g(0) = x$ and $f(1) = g(1) = y$. Then they are **homotopic with respect to the endpoints**, or **path-homotopic**, if there exists a homotopy

$$\begin{aligned} F\colon I \times I &\to X \\ (s,t) &\mapsto f_t(s) \end{aligned}$$

such that $F(s,0) = f_0(s) = f(s)$, $F(s,1) = f_1(s) = g(s)$, $F(0,t) = f_t(0) = x$ and $F(1,t) = f_t(1) = y$.

Notation 1.2.15. If f and g are path-homotopic with respect to their endpoints we may denote it by $f \sim g$.

Remark 1.2.16 — Path homotopy is an equivalence relation, but we won't prove it here.

Lemma 1.2.17 (Path Multiplication)

Let $f, g, f', g' \colon I \to X$ be paths such that

$$\begin{aligned} f(0) = g(0) &= x \\ f(1) = g(1) = f'(0) = g'(0) &= y \\ f'(1) = g'(1) &= z. \end{aligned}$$

Then, if $f \simeq f'$ and $g \simeq g'$, then

$$f \cdot g \simeq f' \cdot g'.$$

Proof. Let $F, G \colon I \times I \to X$, with $f_0 = f, f_1 = f'$, and $g_0 = g, g_1 = g'$. Then, define $H \colon I \times I \to X$ by

$$H(s,t) := \begin{cases} F(2s,t) & s \leq \frac{1}{2} \\ G(2s-1,t) & s > \frac{1}{2} \end{cases}.$$

This is a path homotopy between $f \cdot g$ and $f' \cdot g'$. □

Proposition 1.2.18

Any reparametrization of a path f is path-homotopic to f.

Proof. Consider a path $\phi \colon I \to I$ such that $\phi(0) = 0, \phi(1) = 1, f \colon I \to X$. $f \circ \phi$ is the reparametrization and $f \circ \phi \simeq f$ by

$$f_t(s) = f((1-t)\phi(s) + ts).$$

□

§1.3 The Fundamental Group

Definition 1.3.1. The **fundamental group** of (X, x_0) is

$$\pi_1(X, x_0) = \{[f] \mid f \colon I \to X, f(0) = f(1) = x_0\},$$

where $[f]$ denotes the equivalence classes for homotopy equivalence with respect to the basepoint. That is,

$$[f] = \{g \colon I \to X \mid g(0) = g(1) = x_0, g \simeq f\}.$$

Proposition 1.3.2

$\pi_1(X, x_0)$ is, indeed, a group with product given by concatenation (or path-product)

$$[f] \cdot [g] = [f \cdot g].$$

Its unit element is the equivalence class of the constant loop $[c_{x_0}]$, and for every element $[f]$, its inverse is given by

$$[f]^{-1} = [\overline{f}],$$

where $\overline{f}(t) = f(1-t)$.

Proof. First, we need to check

$$[f] \cdot [c_{x_0}] = [c_{x_0}] \cdot [f] = [f].$$

In other words, we need to check that $f \cdot c_{x_0} \simeq c_{x_0} \cdot f \simeq f$. Such an c_{x_0} can be constructed easily by staying stationary after or before making one whole loop. The first homotopy is defined as

$$F_1(s,t) = \begin{cases} x_0 & t \geq 2s \text{ or } t \leq 2s-1 \\ f(2s-t) & \text{otherwise} \end{cases},$$

and the second is defined as

$$F_2(s,t) = \begin{cases} x_0 & t \geq 2s \\ f\left(\frac{2s-t}{2-t}\right) & \text{otherwise} \end{cases}.$$

Next, we show the inverse condition. We can define the homotopy $f \cdot \overline{f} \simeq e$ by

$$F_3(s,t) = \begin{cases} x_0 & t \geq 2s \text{ or } t \geq 2-2s \\ f(2s-t) & 1 \geq 2s \text{ or } t \leq 2s \\ \overline{f}(2s+t-1) & \text{otherwise} \end{cases}$$

Inverting the roles of f and $\overline{f}$ we obtain the homotopy $\overline{f} \cdot f \simeq e$.

For associativity, we must show that $(f \cdot g) \cdot h \simeq f \cdot (g \cdot h)$. The first path follows f and then g at quadruple speed for $s \in [0, \frac{1}{2}]$, and then follows h at double speed for $s \in [\frac{1}{2}, 1]$, while the second follows f at double speed and then g and h at quadruple speed. The two paths are therefore reparametrizations of each other and thus homotopic. □

Example 1.3.3

$\pi_1(\mathbb{R}^2, 0) = 0$, since every loop is homotopic to the constant loop.

Example 1.3.4

Similarly, Let $X \subset \mathbb{R}^n$ be a convex set and $x_0 \in X$. Then $\pi_1(X, x_0) = 0$.

Example 1.3.5

Any space with the trivial topology has trivial fundamental group (since any map $f\colon I \to X$ is continuous, so X will be path-connected and any two paths are homotopic).

Example 1.3.6

Any space with the discrete topology has trivial fundamental group (since any continuous map $f\colon I \to X$ has to be constant).

Proposition 1.3.7 (Change of Basepoint)
Let X be a path connected, with $x_0, x_1 \in X$ be two distinct points. Then, $\pi_1(X, x_0) \cong \pi_1(X, x_1)$.

Proof. Let $h\colon I \to X$, with $h(0) = x_0$ and $h(1) = x_1$, be a path. Now, consider

$$\begin{aligned} \beta_h\colon \pi_1(X, x_1) &\to \pi_1(X, x_0) \\ [f] &\mapsto [h \cdot f \cdot \overline{h}]. \end{aligned}$$

This is a well-defined map, since we know that path homotopy behaves well with respect to concatenation. Next, this is a group homomorphism, since

$$\beta_h([f] \cdot [g]) = [h \cdot f \cdot g \cdot \overline{h}] = [h \cdot f \cdot \overline{h}] \cdot [h \cdot g \cdot \overline{h}] = \beta_h([f]) \cdot \beta_h([g]).$$

We also have that $\beta_h([c_p]) = [c_p]$. Next, it is bijective with $\beta_h^{-1} = \beta_{h^{-1}}$, so β_h is an isomorphism. □

Notation 1.3.8. If X is path-connected space, we write $\pi_1(X)$ instead of $\pi_1(X, x_0)$.

Definition 1.3.9. A topological space X is **simply connected** if $\pi_1(X, x_0) = 0$ (trivial) and X is path connected.

Exercise 1.3.10. Show that any contractible space is simply connected.

Example 1.3.11
$\mathbb{R}^n$ is simply connected for any n. Indeed, it is contractible, so has trivial fundamental group and it is a path connected space.

Example 1.3.12
$\mathbb{S}^1$ is not simply connected (we'll show later that $\pi_1(\mathbb{S}^1, 1) \cong \mathbb{Z}$).

Proposition 1.3.13
X is simply connected if and only if there exists a unique homotopy class of paths between any two points.

Proof. ($\Longrightarrow$): Let $f, g\colon I \to X$ be two distinct paths between any two points. Assume that $f \cdot \overline{g} \simeq [e] \simeq g \cdot \overline{g}$. Then, we get that $f \simeq g$, since

$$f \simeq f \cdot \overline{g} \cdot g \simeq g \cdot \overline{g} \cdot g \simeq g.$$

($\Longleftarrow$): X is path connected, so $x_0 = x_1$ implies that $\pi_1(X, x_0) = 0$. □

Lemma 1.3.14 (Square Lemma)

Let $F\colon I \times I \to X$ be a continuous map, and let f, g, h, k be the paths in X defined by:

$$\begin{aligned} f(s) &= F(s, 0) \\ g(s) &= F(1, s) \\ h(s) &= F(0, s) \\ k(s) &= F(s, 1) \end{aligned}$$

Then, $f \cdot g \sim h \cdot k$.

Proof. Exercise. □

§1.3.i Induced Maps

Definition 1.3.15. Let $(X, A), (Y, B)$ be pairs of topological spaces. Then, a **map of pairs** is a map $f\colon (X, A) \to (Y, B)$ such that $f\colon X \to Y$ is a continuous map, and $f(A) \subseteq B$.

Remark 1.3.16 — We will particularly care about a map of pointed spaces:

$$\begin{aligned} f\colon (X, x_0) &\to (Y, y_0) \\ x_0 &\mapsto y_0 \end{aligned}$$

Definition 1.3.17. The **induced homomorphism** of the map of pointed spaces $f\colon (X, x_0) \to (Y, y_0)$ is the map

$$\begin{aligned} f_*\colon \pi_1(X, x_0) &\to \pi_1(Y, y_0) \\ [\alpha] &\mapsto [f \circ \alpha] \end{aligned}$$

Remark 1.3.18 — Sometimes, this is called a **push forward**.

Lemma 1.3.19

f_* is a group homomorphism.

Proof. First, note that this map is well defined. Assume that we have loops α, β such that $\alpha \simeq \beta$ and are based at $x_0 \in X$. Then, there exists a homotopy F between α, β with $f_0 = \alpha, f_1 = \beta$. Let $G = f \circ F\colon I \times I \to Y$ with $g_0 = f \circ \alpha, g_1 = f \circ \beta$, so $f \circ \alpha \simeq f \circ \beta$.

We next show that this is a group homomorphism.

$$\begin{aligned} f_*([\alpha] \cdot [\beta]) = f_*([\alpha \cdot \beta]) = [f \circ (\alpha \cdot \beta)] &= [(f \circ \alpha) \cdot (f \circ \beta)] \\ &= [f \circ \alpha] \cdot [f \circ \beta] = f_*([\alpha]) \cdot f_*([\beta]). \end{aligned}$$

□

Lemma 1.3.20

The induced homomorphism satisfies

1. $(id_{(X,x_0)})_* = id_{\pi_1(X,x_0)}$
2. If $f\colon (X,x_0) \to (Y,y_0), g\colon (Y,y_0) \to (Z,z_0)$, then
$$(g \circ f)_* = g_* \circ f_*.$$

Proof. (1): $(id_{(X,x_0)})_*([\alpha]) = [\alpha]$.

(2): $(g \circ f)_*([\alpha]) = [(g \circ f) \circ \alpha] = [g \circ (f \circ \alpha)] = g_*([f \circ \alpha]) = g_*(f_*([\alpha]))$. □

Remark 1.3.21 — The induced homomorphism is an example of a *functor* (very *very* basic category theory at the end of this chapter).

Proposition 1.3.22 (Topological Invariance of π_1)

If $f\colon (X,x_0) \to (Y,y_0)$ is a homeomorphism, then f_* is an isomorphism.

Proof. Let $g = f^{-1}$, so we have $g_* \circ f_* = id_{\pi_1(X,x_0)}$ and $f_* \circ g_* = id_{\pi_1(Y,y_0)}$. □

Remark 1.3.23 — This proposition is telling us that two homeomorphic spaces have the isomorphic fundamental groups.

Lemma 1.3.24

Suppose $\phi, \psi\colon X \to Y$ are continuous, and $H\colon \phi \simeq \psi$ is a homotopy. For any $p \in X$, let h be the path in Y from $\phi(p)$ to $\psi(p)$ defined by $h(t) = H(p,t)$ and let $\Phi_h\colon \pi_1(Y,\phi(p)) \to \pi_1(Y,\psi(p))$ be the isomorphism defined in the proof of proposition 1.3.7. Then the following diagram commutes:

$$\begin{tikzcd}
& \pi_1(Y,\phi(p)) \arrow[dd, "\Phi_h"] \\
\pi_1(X,p) \arrow[ur, "\phi_*"] \arrow[dr, "\psi_*"] & \\
& \pi_1(Y,\psi(p))
\end{tikzcd}$$

Proof. Let f be any loop in X based at p. What we need to show is

$$\begin{aligned}
&\psi_*[f] = \Phi_h(\phi_*[f]) \\
\iff\ &\psi \circ f \sim \overline{h} \cdot (\phi \circ f) \cdot h \\
\iff\ &h \cdot (\psi \circ f) \sim (\phi \circ f) \cdot h.
\end{aligned}$$

This follows easily from the square lemma (1.3.14) applied to the map $F\colon I \times I \to Y$ defined by $F(s,t) = H(f(s),t)$. □

Theorem 1.3.25

If $\phi\colon X \to Y$ is a homotopy equivalence, then for any point $p \in X$, $\phi_*\colon \pi_1(X,p) \to \pi_1(Y,\phi(p))$ is an isomorphism.

Proof. Suppose $\phi\colon X \to Y$ is a homotopy equivalence, and let $\psi\colon X \to Y$ be a homotopy inverse for it. Consider the sequence of maps

$$\pi_1(X,p) \xrightarrow{\phi_*} \pi_1(Y,\phi(p)) \xrightarrow{\psi_*} \pi_1(X,\psi(\phi(p))) \xrightarrow{\phi_*} \pi_1(Y,\phi(\psi(\phi(p)))).$$

We need to prove that the first ϕ_* above is bijective. As we mentioned earlier, ψ_* is not an inverse for it, because it does not map into the right group. Since $\psi \circ \phi \simeq id_X$, lemma 1.3.24 shows that there is a path h in X such that the following diagram commutes:

$$\begin{tikzcd}
 & \pi_1(X,p)) \arrow[d, "\Phi_h"] \\
\pi_1(X,p) \arrow[ur, "(Id_X)_*"] \arrow[dr, "(\psi\circ\phi)_*"'] & \\
 & \pi_1(X,\psi(\phi(p)))
\end{tikzcd}$$

Therefore, $\psi_* \circ \phi_* = \Phi_h$, which is an isomorphism. In particular, this means that the first ϕ_* in the sequence of maps above is injective and ψ_* is surjective. Similarly, the homotopy $\phi \circ \psi \simeq Id_Y$ leads to the diagram

$$\begin{tikzcd}
 & \pi_1(Y,\phi(p))) \arrow[d, "\Phi_k"] \\
\pi_1(Y,\phi(p))) \arrow[ur, "(Id_Y)_*"] \arrow[dr, "(\phi\circ\psi)_*"'] & \\
 & \pi_1(Y,\phi(\psi(\phi(p))))
\end{tikzcd}$$

from which it follows that $\phi_* \circ \psi_*\colon \pi_1(Y,\phi(p) \to \pi_1(Y,\phi(\psi(\phi(p))))$ is an isomorphism. This means in particular that ψ_* is injective; since we already showed that it is surjective, it is an isomorphism. Therefore, going back to $\psi_* \circ \phi_* = \Phi_h$, we conclude that

$$\phi_* = (\psi_*)^{-1} \circ \Phi_h\colon \pi_1(X,p) \to \pi_1(Y,\phi(p))$$

is also an isomorphism. □

Remark 1.3.26 — This is telling us that homotopy equivalent spaces have isomorphic fundamental groups.

Remark 1.3.27 — An important case of this is when we consider a pair (X,A) where X deformation retracts onto A. Indeed, if $r\colon X \to A$ is the deformation retract and $\iota\colon A \to X$ is the inclusion, we have

$$r \circ \iota = id_A, \quad \iota \circ r \sim id_X.$$

§1.4 Categories and Functors

Definition 1.4.1. A **category** $\mathcal{C}$ consists of objects $\mathrm{obj}(\mathcal{C})$ and for any ordered pair of objects (a, b) a set $\mathrm{Hom}_{\mathcal{C}}(a, b)$ whose elements are called morphisms, and composition maps

$$\circ\colon \mathrm{Hom}_{\mathcal{C}}(a, b) \times \mathrm{Hom}_{\mathcal{C}}(b, c) \to \mathrm{Hom}_{\mathcal{C}}(a, c)$$

such that:

- If $f \in \mathrm{Hom}_{\mathcal{C}}(a, b), g \in \mathrm{Hom}_{\mathcal{C}}(b, c), h \in \mathrm{Hom}_{\mathcal{C}}(c, d)$, then

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

- For every $a \in \mathrm{obj}(\mathcal{C})$, there exists a morphism $\mathrm{id}_a \in \mathrm{Hom}_{\mathcal{C}}(a, a)$ such that

$$f \circ \mathrm{id}_a = \mathrm{id}_b \circ f = f,$$

  for all $f \in \mathrm{Hom}_{\mathcal{C}}(a, b)$.

Definition 1.4.2. Let $\mathcal{C}, \mathcal{D}$ be categories. A **functor** $F\colon \mathcal{C} \to \mathcal{D}$ assigns to every object $a \in \mathrm{obj}(\mathcal{C})$ and object $F(a) \in \mathrm{obj}(\mathcal{D})$ and to every morphism $f \in \mathrm{Hom}_{\mathcal{C}}(a, b)$ a morphism $F(f) \in \mathrm{Hom}_{\mathcal{D}}(f(a), f(b))$ so that:

- $F(\mathrm{id}_a) = \mathrm{id}_{F(a)}$
- If $f \in \mathrm{Hom}_{\mathcal{C}}(a, b), g \in \mathrm{Hom}_{\mathcal{C}}(b, c)$, then $F(g \circ f) = F(g) \circ F(f)$.

Remark 1.4.3 — This is an example of a **covariant** functor. A **contravariant** functor reverses the direction of the arrows.

2 Covering Spaces

§2.1 Covering Spaces

Definition 2.1.1. A **covering space** of a space X is a pair $(\tilde{X}, p)$ of a space $\tilde{X}$ and a map $p\colon \tilde{X} \to X$ such that for any $x \in X$, there is an open neighborhood U of x such that $p^{-1}(U)$ is a disjoint union of open sets of $\tilde{X}$, each of which is mapped homeomorphically onto U. In this situation, we call p a **covering map**.

Note that for such an open neighborhood U, we usually write $p^{-1}(U) = \bigsqcup_{\alpha \in I} V_\alpha$ where each $p\,|_{V_\alpha}\colon V_\alpha \to U$ is a homeomorphism.

> **Remark 2.1.2** — In the situation above, we have that $p\colon \tilde{X} \to X$ is a covering map if $p\colon \tilde{X} \to X$ is surjective and in addition every point of X is contained in some open set that is evenly covered by the map p.

Definition 2.1.3. In the above definition, we call:

- the open sets V_α^β of $\tilde{X}$ **sheets**
- the set $p^{-1}(x)$, for each $x \in X$, a **fiber**

> **Remark 2.1.4** — If X is connected, then the number of sheets (V_α^β) over each point is constant. We call this the **degree** or **cardinality** of the covering.

Proof. Let U be an even covering by p; that is $p^{-1} = \bigsqcup\limits_{\alpha \in I} V_\alpha$, where each $V_\alpha \subset \tilde{X}$ are pairwise disjoint and $p\,|_{V_\alpha}\colon \to U$ is a homeomorphism. Letting $x \in U$, we note that $|p^{-1}(x) \cap V_\alpha| = 1$, so $|p^{-1}(x)| = |I|$. In other words, for any two points in the even covering U, we have that their cardinalities are the same. Now let $x \in X$ and define

$$A = \{x' \in X \mid |p^{-1}(x')| = |p^{-1}(x)|\}.$$

Then, $A \cap (X \setminus A) = \emptyset$ and $A \cup (X \setminus A) = X$ and A is nonempty (because $x \in A$), so it suffices to show that A and $X \setminus A$ are both open. First, A is open for the reason above. Next, for $X \setminus A$. We just need to analyze points that have different cardinalities. These would certainly not be contained in the same even cover, and so it follows. $\square$

Definition 2.1.5. If $|p^{-1}(x)| = n$, for all $x \in X$ then we call p an **n-fold covering**.

Definition 2.1.6. Given coverings $p_1\colon Y_1 \to X$ and $p_2\colon Y_2 \to X$, they are **isomorphic** if there exists a homeomorphism $h\colon Y_1 \to Y_2$ such that $p_1 = p_2 \circ h$. In particular, we have the following commutative diagram:

$$\begin{tikzcd}
Y_1 \arrow[r, "h"] \arrow[d, "p_1"'] & Y_2 \arrow[dl, "p_2"] \\
X &
\end{tikzcd}$$

Example 2.1.7

A homeomorphism is a covering map.

Example 2.1.8

If $p\colon \tilde{X} \to X$ and $q\colon \tilde{Y} \to Y$ are covering maps, so is the product $p \times q\colon \tilde{X} \times \tilde{Y} \to X \times Y$.

Example 2.1.9

Let $\mathbb{S}^1 \subset \mathbb{C}$ be the set of complex numbers with modulus 1, and consider the map $p\colon \mathbb{R} \to \mathbb{S}^1$ given by $p(t) = e^{2\pi t}$. For the upper half of $\mathbb{S}^1$, $U_{y>0} = \{x+iy \in \mathbb{S}^1 \mid y > 0\}$, we have

$$p^{-1}(U_{y>0}) = \bigsqcup_{j\in\mathbb{Z}} (j, j + \frac{1}{2}),$$

the disjoint union of $\mathbb{Z}$-many open intervals. Further,

$$p\,|_{(j,j+\frac{1}{2})}\colon (j, j+\frac{1}{2}) \to U_{y>0}$$

is a continuous bijection with inverse

$$\begin{aligned} U_{y>0} &\to (j, j+\frac{1}{2}) \\ x + iy &\mapsto j + \frac{\arccos(x)}{2\pi}. \end{aligned}$$

Similarly with $U_{y<0}$, $U_{x>0}$, $U_{x<0}$, and these four open sets cover $\mathbb{S}^1$, so p is a covering map.

Example 2.1.10

Let $\mathbb{S}^1 \subset \mathbb{C}$ and $p\colon \mathbb{S}^1 \to \mathbb{S}^1$ be given by $p(z) = z^n$. Let $\eta := e^{\frac{2\pi i}{n}}$, and for $y \in \mathbb{S}^1$ let ξ be some n-th root of y. Then,

$$p^{-1}(y) = \{\xi, \xi\cdot\eta, \xi\cdot\eta^2, \dots, \xi\eta^{n-1}\}$$

and $p^{-1}(\mathbb{S}^1 \setminus \{y\})$ is the complement of this set. Let

$$V_0 := \{z \in \mathbb{S}^2 \mid 0 < \arg(\frac{z}{\xi}) < \frac{2\pi}{n}\}$$

and $V_i := V_0\eta^i$ for $i = 0, 1, \dots, n-1$, so $p^{-1}(\mathbb{S}^1 \setminus \{y\}) = \bigsqcup_{i=0}^{n-1} V_i$,. Then, $p\,|_{V_i}\colon V_i \to \mathbb{S}^1 \setminus \{y\}$ is a homeomorphism.

Exercise 2.1.11. Given covering spaces $p_1\colon \tilde{X}_1 \to X_1$ and $p_2\colon \tilde{X}_2 \to X_2$, show that $p_1 \times p_2\colon \tilde{X}_1 \times \tilde{X}_2 \to X_1 \times X_2$ is a covering space.

§2.2 Lifts

Definition 2.2.1. Let $p\colon \tilde{X} \to X$ be a covering. A **lift** of a continuous map $f\colon Y \to X$ is a map $\tilde{f}\colon Y \to \tilde{X}$ such that $p \circ \tilde{f} = f$. In other words, we have the following commutative diagram.

$$\begin{tikzcd} & \tilde{X} \arrow[d, "p"] \\ Y \arrow[ur, "\tilde{f}"] \arrow[r, "f"] & X \end{tikzcd}$$

Lemma 2.2.2

Given a covering $p\colon \tilde{X} \to X$ and $\tilde{f}, \tilde{g}\colon Y \to \tilde{X}$, then the following holds.

1. $\tilde{f}$ is a lift of $p \circ \tilde{f}$.
2. If $\tilde{f} \sim \tilde{g}$, then $p \circ \tilde{f} \sim p \circ \tilde{g}$ (homotopies descend).
3. If $\alpha, \beta\colon I \to \tilde{X}$ are two paths, with $\alpha(1) = \beta(0)$, then $p(\alpha \cdot \beta) = (p \circ \alpha) \cdot (p \circ \beta)$ (paths descend).

Proof. 1: Follows from definition.

2: Any homotopy $\tilde{f}_t$ from $\tilde{f}$ to $\tilde{g}$ gives rise to the homotopy $p \circ \tilde{f}_t$ going from $p \circ \tilde{f}$ to $p \circ \tilde{g}$.

3. We have

$$p \circ (\alpha \cdot \beta)(t) = \begin{cases} p(\alpha(2t)) & t \leq \frac{1}{2} \\ p(\beta(2t-1)) & t \geq \frac{1}{2} \end{cases} = (p \circ \alpha) \cdot (p \circ \beta)$$

□

Theorem 2.2.3 (Unique Lifting Property)

Given a covering space $p\colon \tilde{X} \to X$ and a map $f\colon Y \to X$, if two lifts $\tilde{f}_1, \tilde{f}_2\colon Y \to \tilde{X}$ agree at one point of Y and Y is connected, then $\tilde{f}_1$ and $\tilde{f}_2$ agree on all of Y.

Proof. For a point $y \in Y$, let U be an evenly covered open neighborhood of $f(y)$ in X, so $p^{-1}(U)$ is decomposed into disjoint sheets each mapped homeomorphically onto U by p. Let $\tilde{U}_1$ and $\tilde{U}_2$ be the sheets containing $\tilde{f}_1(y)$ and $\tilde{f}_2(y)$, respectively. By continuity of $\tilde{f}_1$ and $\tilde{f}_2$, there exists a neighborhood N of y mapped into $\tilde{U}_1$ by $\tilde{f}_1$ and into $\tilde{U}_2$ by $\tilde{f}_2$. If $\tilde{f}_1(y) \neq \tilde{f}_2(y)$, then $\tilde{U}_1 \neq \tilde{U}_2$; thus, $\tilde{U}_1$ and $\tilde{U}_2$ are disjoint and $\tilde{f}_1 \neq \tilde{f}_2$ throughout the neighborhood N. On the other hand, if $\tilde{f}_1(y) = \tilde{f}_2(y)$, then $\tilde{U}_1 = \tilde{U}_2$, so $\tilde{f}_1 = \tilde{f}_2$ on N, since $p\tilde{f}_1 = p\tilde{f}_2$ and p is injective on $\tilde{U}_1 = \tilde{U}_2$. Therefore, the set of points where $\tilde{f}_1$ and $\tilde{f}_2$ agree is both open and closed in Y. □

Definition 2.2.4. Let $p\colon Z \to X$ be a map. Then, p has the **homotopy lifting property** if, given a homotopy $F\colon Y \times I \to X$ and a lift $g\colon Y \times \{0\} \to Z$ of f_0 so that $f_0 = p \circ g$, there exists a unique homotopy $\tilde{F}\colon Y \times I \to Z$ such that $\tilde{f}_0 = g$ and $p \circ \tilde{F} = F$.

This can be summarized in the commutative diagram below.

$$\begin{tikzcd}
 & & Z \arrow[d, "p"] \\
Y \times \{0\} \arrow[r, hook] \arrow[urr, bend left, "g=\tilde{f}_0"] & Y \times I \arrow[r, "F"] \arrow[ur, dashed, "\tilde{F}"] & X
\end{tikzcd}$$

Definition 2.2.5. A map $p\colon Z \to X$ has the **path lifting property** if for all paths $f\colon I \to X$ with $x_0 = f(0)$, $\tilde{x}_0 \in p^{-1}(x_0)$, then there exists a unique path $\tilde{f}\colon I \to Z$ such that $\tilde{f}(0) = \tilde{x}_0$ and $p \circ \tilde{f} = f$.

Lemma 2.2.6 (Local Homotopy Lifting Property)
Let $p\colon \tilde{X} \to X$ be a covering and let $F\colon Y \times I \to X$ be a homotopy let $g\colon Y \times \{0\} \to \tilde{X}$ be such that $p \circ g = f_0$. Then, for every $y_0 \in Y$, there exists an open neighborhood N with $y_0 \in N \subset Y$ and a unique homotopy (depending on N)

$$\tilde{F}_N\colon N \times I \to \tilde{X}$$

such that $p \circ \tilde{F}_N = F \mid_{N \times I}$ and $(\tilde{f}_N)_0 = g \mid_{N \times \{0\}}$. Moreover, if M is another such neighborhood, with $y_0 \in M \subset Y$, then

$$\tilde{F}_M \mid_{(M \cap N) \times I} = \tilde{F}_N \mid_{(M \cap N) \times I} = \tilde{F}_{M \cap N}.$$

Proof. Let $U = \{U_\alpha\}_\alpha$ be the cover of X such that, for each α, $p^{-1}(U_\alpha) = \cup_\beta V_\alpha^\beta$ and $p \mid_{V_\alpha^\beta}\colon V_\alpha^\beta \to U_\alpha$ is a homeomorphism. Therefore, for every β we have the inverse homeomorphism $q_{\alpha,\beta}$. By continuity of the map $F\colon Y \times I \to X$, we have that for any $(y_0, t_0) \in Y \times I$, there exists an open neighborhood $N \times (a, b) \subset Y \times I$ of (y_0, t_0) such that $F(y, t) \in U_\alpha$ for every $(y, t) \in N \times (a, b)$.

For every fixed y and as t ranges over I, we get a collection of subsets with this property, and since I is compact there exists finitely many such $N_i \times (a_i, b_i)$ covering $\{y\} \times I$. Now set $N := \cap_i N_i$ and choose a partition $0 = t_0 < t_1 < \cdots < t_n = 1$ such that for all i, there exists α with $F(N \times [t_i, t_{i+1}]) \subset U_\alpha$ by the Lebesgue covering lemma. We now claim that there is a sequence of maps $\tilde{F}_N^k$ such that

1. $\tilde{F}_N^k\colon N[0, t_k] \to \tilde{X}$ is a lift of $F \mid_{N \times [0, t_k]}$
2. $(\tilde{F}_N^k)_0 = g \mid_N$
3. $\tilde{F}_N^{k+1} \mid_{N \times [0, t_k]} = \tilde{F}_N^k$

Moreover, these properties determine the sequence of $\{\tilde{F}_N^k\}$ uniquely. We then set $F_N = F_N^n$. We construct the sequence of maps $\tilde{F}_N^k$ via induction. First, note that there is only one way to define $\tilde{F}_N^0$ on $N \times \{0\}$ so that $\tilde{F}_N^0 = g \mid_N$. Now, assume that we've defined the sequence up to $\tilde{F}_N^k$. By assumption, there exists an index α such that $F(N \times [t_k, t_{k+1}] \subset U_\alpha$. By making N smaller, if necessary, we can assume that $\tilde{F}_N^k \mid_{N \times \{t_k\}} \subset V_\alpha^\beta$, for some β. Now define

$$\tilde{E} = q_{\alpha,\beta} \circ F \mid_{N \times [t_k, t_{k+1}]} .$$

Then,

$$\tilde{E} \mid_{N \times \{t_k\}} = \tilde{F}_N^k \mid_{N \times \{t_k\}}$$

by construction. Now defining the extension

$$\tilde{F}_N^{k+1}(z,t) = \begin{cases} \tilde{F}_N^k(z,t) & t \le t_k \\ \tilde{E}(z,t) & t \in [t_k, t_{k+1}] \end{cases},$$

we see that this is continuous by the pasting lemma (0.3.1).

To conclude the uniqueness of $\tilde{F}_N$, we apply the unique lifting property. □

Proposition 2.2.7

A covering map has the homotopy lifting property.

Proof. Take a cover $Y = \cup_\alpha N_\alpha$ such that they have the local homotopy lifting property. Then, $Y \times I = \cup_\alpha N_\alpha \times I$ is a cover. Therefore, we get a family of homotopy lifts $F_{N_\alpha} \colon N_\alpha \times I \to \tilde{X}$, which coincide with the intersection in the previous lemma. Therefore, we have a homotopy $F \colon Y \times I \to \tilde{X}$ given by $F(y,t) = F_{N_\alpha}(y,t)$ if $y \in N_\alpha$. Then, by the pasting lemma (0.3.1), the resulting function F is continuous, concluding the proof. □

Theorem 2.2.8 (Monodromy Theorem)

Let $q \colon E \to X$ be a covering mop. Suppose f, g are paths in X with the same initial and terminal points, and $\tilde{f}_e, \tilde{g}_e$ are their lifts with the same initial point $e \in E$. Then,

1. $\tilde{f}_e \sim \tilde{g}_e$ if and only if $f \sim g$.
2. If $f \sim g$, then $\tilde{f}_e(1) = \tilde{g}_e(1)$.

Proof. If $\tilde{f}_e \sim \tilde{g}_e$, then $f \sim g$ because we can compose with q, and this preserves path homotopy. Conversely, if $f \sim g$, let $H \colon I \times I \to X$ be a path homotopy between them. Then, the homotopy lifting property implies that H lifts to a homotopy $\tilde{H} \colon I \times I \to E$ between $\tilde{f}_e$ and some lift of g starting at e, which must equal to $\tilde{g}_e$ by the unique lifting property. This proves the first statement. For the second, we note that $f \sim g$ implies the path homotopy of $\tilde{f}_e$ and $\tilde{g}_e$, so they must have the same terminal point. □

Theorem 2.2.9 (Injectivity Theorem)

Let $q \colon E \to X$ be a covering map. For any point $e \in E$, the induced homomorphism $q_* \colon \pi_1(E,e) \to \pi_1(X, q(e))$ is injective.

Proof. Suppose that $[f] \in \pi_1(E,e)$ is in the kernel of q_*. This means that $q_*[f] = [c_x]$, where $x = q(e)$. In other words, $q \circ f \sim c_x$ in X. By the monodromy theorem, any lifts of $q \circ f$ and c_x that start at the same point must be path homotopic in E. Now, f is a lift of $q \circ f$ starting at e, and the constant loop c_e is a lift of c_x starting at the same point. Therefore, $f \sim c_e$ in E, which means that $[f] = 1$. □

§2.3 Fundamental Group of The Circle

The main goal here will be the following theorem.

Theorem 2.3.1

$\pi_1(\mathbb{S}^1, 1) = (\{\omega_n \mid n \in \mathbb{Z}\}, \cdot) \cong \mathbb{Z}$, where $\omega_n \colon I \to \mathbb{S}^1, s \mapsto e^{2\pi i n s}$.

The following four maps are the tools we employ to prove this.

$$\begin{aligned}
\omega_n \colon I \to \mathbb{S}^1, &\quad t \mapsto e^{2\pi i t n} \\
\epsilon \colon \mathbb{R} \to \mathbb{S}^1, &\quad t \mapsto e^{2\pi i t} \\
\widetilde{\omega}_n \colon I \to \mathbb{R}, &\quad t \mapsto t \cdot n \\
\tau_m \colon \mathbb{R} \to \mathbb{R}, &\quad t \mapsto t + m
\end{aligned}$$

First, note that ϵ is a covering map, and the preimage $\epsilon^{-1}(z)$ consists of infinitely many points. Next, note that $\tilde{\omega}_n$ is a lift of ω, as we have $\omega_n = \epsilon \circ \tilde{\omega}_n$. Here, τ_m is an example of a **deck transformation** of ϵ.

Definition 2.3.2. Let $p \colon \tilde{X} \to X$ be a covering. A **deck transformation** is a homeomorphism $\tau \colon \tilde{X} \to \tilde{X}$ such that $p \circ \tau = p$, i.e., it gives an isomorphism of the covering to itself. We denote by $\mathrm{Deck}(p)$ the set of all deck transformations of a covering.

> **Exercise 2.3.3.** $\mathrm{Deck}(p)$ is a group where the action is composition of a map.

Finally, note that we have

$$\widetilde{\omega_m} \cdot (\tau_m \circ \widetilde{\omega_n}) \simeq \widetilde{\omega}_{n+m}.$$

via the linear homotopy

$$f_t(s) = (1-t)\widetilde{\omega}_{n+m}(s) + t(\widetilde{\omega}_m \cdot (\tau_m \circ \widetilde{\omega}_n))(s).$$

Proposition 2.3.4

The map

$$\begin{aligned}
\Phi \colon \mathbb{Z} &\to \pi_1(\mathbb{S}^1, 1) \\
n &\mapsto [\omega_n]
\end{aligned}$$

is a group homomorphism.

Proof.

$$\begin{aligned}
\Phi(n+m) &= [\omega_{n+m}] \\
&= [\epsilon \circ \widetilde{\omega}_{n+m}] \\
&= [\epsilon \circ (\widetilde{\omega}_m \cdot (\tau_m \circ \widetilde{\omega}_n))] \\
&= [(\epsilon \circ \widetilde{\omega}_m] \cdot [\epsilon \circ (\tau_m \circ \widetilde{\omega_n})] \\
&= [\epsilon \circ \widetilde{\omega}_m] \cdot [(\epsilon \circ \widetilde{\omega}_n] \\
&= [\omega_m] \cdot [\omega_n] \\
&= \Phi(n) \cdot \Phi(m).
\end{aligned}$$

$\square$

Theorem 2.3.5

The map $\Phi\colon \mathbb{Z} \to \pi_1(\mathbb{S}^1, 1)$ sending $n \mapsto [\omega_n]$ is a group isomorphism.

Proof. We know that this is a group homomorphism already, so we need to show that this is a bijection. We start with surjectivity. For all $[\alpha] \in \pi_1(\mathbb{S}^1, 1)$, there exists an n such that $\alpha \simeq \omega_n$. By the path lifting property, there exists a unique $\tilde{\alpha} \in I \to \mathbb{R}$ such that $\tilde{\alpha}(0) = \omega$ and $p \circ \tilde{\alpha} = \alpha$. $\tilde{\alpha}(1) \in \epsilon^{-1}(1) \in \mathbb{Z}$, so let $n = \tilde{\alpha}(1)$. But then, $\tilde{\alpha} \simeq \tilde{\omega}_n$, which are both paths in $\mathbb{R}$ between 0 to n. Because homotopies descend, then $\alpha \simeq \omega_n$.

Next is injectivity. We want to show that if $\Phi(n) = [c_1]$, then $n = 0$. Note that $\Phi(n) = [\omega_n]$, so we have that $\omega_n \simeq c_1$ with (some) homotopy f_t ($f_0 = \omega_n$, $f_1 = c_1$, $f_t(0) = f_t(1) = 1$, for all t). Define $g\colon I \times \{0\} \to \mathbb{R}$, given by $g(s, 0) = \tilde{\omega}_n(s) = sn$. Since coverings satisfy the homotopy lifting property, there exists a unique $\tilde{F}\colon I \times I \to \mathbb{R}$ such that $\tilde{f}_0 = g, p \circ \tilde{f}_1 = c_1, p \circ \tilde{F} = F$. We know that $\tilde{f}_1(s) \in \mathbb{Z} = p^{-1}(1)$, it follows that $\tilde{f}_1(s) = m$ is constant, for some integer m. We also have that $p \circ \tilde{f}_t(0) = f_t(0) = 1$ and $p \circ \tilde{f}_t(1) = f_t(1) = 1$. Then, $\tilde{f}_t(0), \tilde{f}_t(1) \in \mathbb{Z}$. Moreover, continuity tells us these two quantities are the same; hence, $\tilde{f}_1(0) = \tilde{f}_0(0) = 0$. Then, $\tilde{f}_1(s) = m$, for all s, and so

$$\tilde{f}_1(0) = m = 0.$$

Then,

$$m = \tilde{f}_1(1) = \tilde{f}_0(1) = 1 \cdot n = n,$$

so we get that $0 = m = n$, concluding the proof. □

§2.3.i The Winding Numbers

Here, we follow Lee's book as an alternative way to compute that fundamental group of the circle. The general idea is the same, but here we provide some results that are unique to the circle.

Proposition 2.3.6

Each point $z \in \mathbb{S}^1$ has a neighborhood U with the following property: $\epsilon^{-1}(U)$ is a countable union of disjoint open intervals $\tilde{U}_n$ with the property that ϵ restricts to a homeomorphism from $\tilde{U}_n$ to U.

Proof. This is a computation from the definition of ϵ. We can cover $\mathbb{S}^1$ by the four open subsets:

$$X_+ = \{(x + iy) \colon x > 0\}$$
$$X_- = \{(x + iy) \colon x < 0\}$$
$$Y_+ = \{(x + iy) \colon y > 0\}$$
$$Y_- = \{(x + iy) \colon y < 0\}$$

The preimage of each of these sets is a countable union of disjoint open intervals of length $\frac{1}{2}$, on each of which ϵ has a continuous *local* inverse. For instance, $\epsilon^{-1}(X_+)$ is the union of the intervals $(n - \frac{1}{4}, n + \frac{1}{4})$ for $n \in \mathbb{Z}$, and for each of these intervals, a continuous local inverse $\epsilon^{-1}\colon X_+ \to (n - \frac{1}{4}, n + \frac{1}{4})$ is given by

$$\epsilon^{-1}(x + iy) = n + \frac{1}{2\pi} \sin^{-1}(y) \text{ on } X_+.$$

Other local inverses are given as follows:

$$\epsilon^{-1}(x+iy) = n + \frac{1}{2} - \frac{1}{2\pi}\sin^{-1}(y) \text{ on } X_-$$
$$\epsilon^{-1}(x+iy) = n + \frac{1}{2\pi}\cos^{-1}(y) \text{ on } Y_+$$
$$\epsilon^{-1}(x+iy) = n - \frac{1}{2\pi}\cos^{-1}(y) \text{ on } Y_-$$

Every point of $\mathbb{S}^1$ is in at least one of the four sets listed above, so this completes the proof. □

Definition 2.3.7. If $q\colon X \to Y$ is any surjective continuous map, a **section** of q is a continuous map $\sigma\colon Y \to X$ such that $q \circ \sigma = id_Y$ (σ is a right-inverse).

Definition 2.3.8. If $U \subseteq Y$ is a subset, a **local section** of q over U is a continuous map $\sigma\colon U \to X$ such that $q \circ \sigma = id_U$.

Example 2.3.9

The local inverses constructed in the proof of proposition 2.3.6 were examples of local sections of ϵ over the sets $X_\pm, Y_\pm$.

The next corollary shows that every evenly covered open subset of the circle admits lots of local sections.

Corollary 2.3.10 (Local Section Property of the Circle)

Let $U \subseteq \mathbb{S}^1$ be any evenly covered open subset. For any $z \in U$ and any r in the fiber of ϵ over z, there is a location section σ of ϵ over U such that $\sigma(z) = r$.

Proof. Given $z \in U$ and $r \in \epsilon^{-1}(z)$, let $\tilde{U} \subseteq \mathbb{R}$ be the component of $\epsilon^{-1}(U)$ containing r. By definition of an evenly covered open subset, $\epsilon\colon \tilde{U} \to U$ is a homeomorphism. Therefore, $\sigma = (\epsilon\mid_{\tilde{U}})^{-1}$ is the desired local section. □

Here are now two corollaries that follows from the unique lifting property and the homotopy lifting property introduced in pervious sections.

Corollary 2.3.11 (Path Lifting Property of the Circle)

Suppose $f\colon I \to \mathbb{S}^1$ is any path, and $r_0 \in \mathbb{R}$ is any point in the fiber of ϵ over $f(0)$. Then, there exists a unique lift $\tilde{f}\colon I \to \mathbb{R}$ of f such that $\tilde{f}(0) = r_0$, and any other lift differs form $\tilde{f}$ by addition of an integer constant.

Proof. A path f can be viewed as a homotopy between two maps from a point space $\{*\}$ into $\mathbb{S}^1$, namely $* \mapsto f(0)$ and $* \mapsto f(1)$. Therefore, the existence and uniqueness of $\tilde{f}$ follows from the homotopy lifting property. To prove the final statement of the corollary, suppose $\tilde{f}'$ is any other lift of f. Then, from the fact that $\epsilon(\tilde{f}(S)) = f(s) = \epsilon(\tilde{f}'(s))$ implies that $\tilde{f}(s) - \tilde{f}'(s)$ is an integer for each s. Since $\tilde{f}0\tilde{f}'$ is a continuous function from the connected space I into the discrete space $\mathbb{Z}$, it must be constant. □

Corollary 2.3.12 (Path Homotopy Criterion for the Circle)

Suppose f_0, f_1 are paths in $\mathbb{S}^1$ with the same initial and terminal points, and $\tilde{f}_0, \tilde{f}_1 \colon I \to \mathbb{R}$ are lifts of f_0, f_1, respectively with the same initial point. Then, $f_0 \sim f_1$ if and only if $\tilde{f}_0$ and $\tilde{f}_1$ have the same terminal point.

Proof. If $\tilde{f}_0$ and $\tilde{f}_1$ have the same terminal point, then they are path-homotopic, since $\mathbb{R}$ is simply connected. It follows that $f_0 = \epsilon \circ \tilde{f}_0$ and $f_1 = \epsilon \circ \tilde{f}_1$ are also path-homotopic.

Conversely, suppose that $f_0 \sim f_1$, and let $H \colon I \times I \to \mathbb{S}^1$ be a path homotopy between them. Then, the homotopy lifting property implies that H lifts to a homotopy $\tilde{H} \colon I \times H \to \mathbb{R}$ such that $\tilde{H}_0 = \tilde{f}_0$. Since H is stationary on $\{0,1\}$, so is $\tilde{H}$, meaning that it is a path homotopy. The path $\tilde{H}_1 \colon I \to \mathbb{R}$ is a lift of f_1 starting at $\tilde{f}_0(1)$, so by uniqueness of lifts it must be equal to $\tilde{f}_1$. Therfore, $\tilde{f}_1$ is path-homotpic to $\tilde{f}_0$, implying in particular that they have the same terminal point. □

The Winding Number

Suppose $f \colon I \to \mathbb{S}^1$ is a loop based at a point $z_0 \in \mathbb{S}^1$. If $\tilde{f} \colon I \to \mathbb{R}$ is any lift of f, then $\tilde{f}(1)$ and $\tilde{f}(0)$ are both points in the fiber $\epsilon^{-1}(z_0)$ so they differ by an integer. Since any other lift differs from $\tilde{f}_z$ by an additive constant, the difference $\tilde{f}(1) - \tilde{f}(0)$ is an integer that depends only on f, and not on the choice of lift. This integer is denoted by $N(f)$, and is called the **winding number** of f.

Theorem 2.3.13

Two loops in $\mathbb{S}^1$ based at the same point are path-homotopic if and only if they have the same winding number.

Proof. Suppose f_0 and f_1 are loops in $\mathbb{S}^1$ based at the same point. By the path lifting property (corollary 2.3.11), they have lifts $\tilde{f}_0, \tilde{f}_1 \colon I \to \mathbb{R}$ starting at the same point, and by the path homotopy criterion (corollary 2.3.12) these lifts end at the same point if and only if $f_0 \sim f_1$. By definition of the winding number, this means that f_0 and f_1 have the same winding number if and only if they are path-homotopic. □

Theorem 2.3.14 (Fundamental Group of the Circle)

The group $\pi_1(\mathbb{S}^1, 1) \cong \mathbb{Z}$.

Proof. Define the maps $J \colon \mathbb{Z} \to \pi_1(\mathbb{S}^1, 1)$ and $K \colon \pi_1(\mathbb{S}^1, 1) \to \mathbb{Z}$ by

$$J(n) = [\omega]^n, \quad K([f]) = N(f).$$

Since the winding number of a loop depends only on its homotopy class, K is well defined. Since $[\omega]^{n+m} = [\omega]^n[\omega]^m$, J is a homomorphism (considering $\mathbb{Z}$ a an additive group). TO prove the theorem, it suffices to show that J is an isomorphism, which we do by showing that K is a two-sided inverse for J.

For this purpose, it is convenient to use a concrete representative of each path class $[\omega]^n$, defined as follows. For any integer n, let $\alpha_n \colon I \to \mathbb{S}^1$ be the loop

$$\alpha_n(s) = e^{2\pi i n s}.$$

It's easy to see that $\alpha_1 = \omega$, $\alpha_{-1} = \overline{\omega}$, and α_n is a reparametrization of $\alpha_{n-1} \cdot \omega$, so by induction, $[\alpha_n] = \omega^n$, for each n. By direct computation, the path $\tilde{\alpha}_n \colon I \to \mathbb{R}$ given by $\tilde{\alpha}_n(s) = ns$ is a lift of α_n, so the winding number of α_n is $\tilde{\alpha}_n(1) - \tilde{\alpha}_n(0) = n$.

To prove that $K \circ J = id_{\mathbb{Z}}$, let $n \in \mathbb{Z}$ be arbitrary, and note that $K(J(n)) = K([\omega]^n) = K([\alpha_n]) - N(\alpha_n) = n$. To prove that $J \circ K - id_{\pi_1(\mathbb{S}^1,1)}$, suppose $[f]$ is any element of $\pi_1(\mathbb{S}^1, 1)$, and let n be the winding number of f. Then, f and α_n are path-homotopic since they are loops based at 1 with the same winding number, so $J(K([f])) = J(n) = [\omega]^n = [\alpha_n] = [f]$. □

§2.3.ii Applications

Retractions and Brouwer's Theorem for $n = 2$

Here is a result on induced maps of retractions and inclusions:

Proposition 2.3.15

Let (X, A) be a pair of spaces such that there exists a retract $r \colon X \to A$. Let $\iota \colon A \hookrightarrow X$ be the inclusion map. Let $r_* \colon \pi_1(X, x_0) \to \pi_1(A, x_0)$ and $\iota_* \colon \pi_1(A, x_0) \to \pi_1(X, x_0)$ be their respective induced homomorphisms between the fundamental groups. Then:

1. r_* is surjective.
2. ι_* is injective.
3. if $\iota \circ r \sim id_X$ (equivalently, r is a weak deformation retract), then r_* and ι_* are isomorphisms.

Proof. (1 and 2): Since $id_{\pi_1(A,x_0)} = (r \circ \iota)_* = r_* \circ \iota_*$ then ι_* has to be injective and r_* has to be surjective.

(3): To show that r_* is an isomorphism if r is a *weak* deformation retract it is enough to show that r_* is injective. Let $[\gamma] \in \pi_1(X, x_0)$ and assume $r_*([\gamma]) = [r \circ \gamma] = [c_p]$, i.e. $r \circ \gamma \sim c_p$. We have that $\iota \circ r \circ \gamma$ is a loop in X which is homotopic to the constant path via the homotopy $\iota \circ f$, where F is the homotopy $r \circ \gamma \simeq e$. Therefore, γ is also homotopic to the constant path since

$$\iota \circ r \circ \gamma \sim \gamma.$$

This can be seen by the homotopy $G(s,t) = H(\gamma(s), t)$, where H is the homotopy from $\iota \circ r$ to id_X. □

Example 2.3.16

A first application of the statements about retraction is that if there cannot be a surjective homomorphism $\pi_1(X, x_0) \to \pi_1(A, x_0)$ for a pair of spaces (X, A) then X cannot retract to A. Moreover, if the two fundamental groups are not isomorphic then X does not deformation retract to A.

Proposition 2.3.17

There is no retraction $\mathbb{D}^2 \to \mathbb{S}^1$.

Proof. $\pi_1(\mathbb{D}^2, 0) = 0$ and $\pi_1(\mathbb{S}^1, 1) \cong \mathbb{Z}$, so there cannot be any surjection $\pi_1(\mathbb{D}^2, 0) \to \pi_1(\mathbb{S}^1, 1)$. Therefore, we cannot have a retraction. $\square$

This allows us to prove Brouwer's fixed point theorem for $n = 2$.

Theorem 2.3.18 (Brouwer's Fixed Point Theorem)

Every map $f\colon \mathbb{D}^2 \to \mathbb{D}^2$ has a fixed point.

Proof. Assume that $f\colon \mathbb{D}^2 \to \mathbb{D}^2$ has no fixed points, i.e. $f(x) \neq x$ for all $x \in \mathbb{D}^2$. We can parametrize the unique line passing through x and $f(x)$ by

$$L_x(t) = tx + (1-t)f(x),$$

which will hit the boundary of the disk in exactly two points: one for negative t and one for positive t. Now define

$$r(x) = L_x(\mathbb{R}_{\geq 0}) \cap \partial\mathbb{D}^2.$$

This defines a map $r\colon \mathbb{D}^2 \to \mathbb{S}^1$ such that $r(x) = x$ for all $x \in \partial D^2 \cong \mathbb{S}^1$. This is continuous, so it is a retraction, which is a contradiction. $\square$

Product Spaces

Exercise 2.3.19. $f\colon Z \to X \times Y$ are continuous if and only if the projections $p_X \circ f\colon Z \to X$ and $p_Y \circ f\colon Z \to Y$ are continuous.

Corollary 2.3.20

$f\colon I \to X \times Y$ is a path if and only if $p_X \circ f\colon I \to X$ and $p_Y \circ f\colon I \to Y$ are paths.

Proof. This follows from the previous exercise. $\square$

Proposition 2.3.21

Let $(X, x_0), (Y, y_0)$ be pointed spaces. Then,

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X_1, x_0) \times \pi_1(Y, y_0).$$

Proof. Note that by the previous corollary, we have the bijection. Thus, it suffices to check the group homomorphism property. This can be seen by focusing coordinate-wise. $\square$

Example 2.3.22

$\pi_1(\mathbb{T}^2, (1,1)) = \pi_1(\mathbb{S}^1 \times \mathbb{S}^1, (1,1)) \cong \mathbb{Z} \times \mathbb{Z}$.

Remark 2.3.23 — Note that $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$.

Here is an interesting application of the proposition above.

Theorem 2.3.24 (Fundamental Theorem of Algebra)
A polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{C}[x]$ has a complex root.

Proof. Assume that there exists a polynomial such that $p(\lambda) \neq 0, \forall \lambda \in \mathbb{C}$. For all $r \in R$, we can define

$$g_r(s) = \frac{\frac{p(re^{2\pi i s})}{p(r)}}{\left|\frac{p(re^{2\pi i s})}{p(r)}\right|}.$$

This is a loop in $\mathbb{S}^1$. Then, we have $g_r(0) = g_r(1) = 1$. For certain r, we will show that $g_r \sim c_p$ via the homotopy $f_t = g_{tr}$. Moreover, for certain rs, this is homotopic to ω_n via the homotopy

$$\tilde{f}_t(s) = \frac{\frac{p_t(re^{2\pi i s})}{p_t(r)}}{\left|\frac{p_t(re^{2\pi i s})}{p_t(r)}\right|}, \text{ where } p_t(z) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0).$$

$\tilde{f}_1 = g_r, \tilde{f}_0 = \omega_n$. $\tilde{f}_t$ is well-defined for $r > \max\{|a_{n-1}| + \cdots + |a_0|, 1\}$. Then,

$$\begin{aligned} |z|^n &> (|a_{n-1}| + \cdots + |a_0|)|z|^{n-1} \\ &> |a_{n-1}||z|^{n-1} + \cdots + |a_0| \\ &\geq |a_{n-1}z^{n-1} + \cdots + a_0| \\ &\geq t|a_{n-1}z^{n-1} + \cdots + a_0|, \end{aligned}$$

for $t \in [0, 1]$. Thus, the denominator is never zero. Therefore, $\omega_n \simeq c_p$ which is only true when $n = 0$. ☐

§2.4 Universal Coverings and the Correspondence Theorem

Definition 2.4.1. Let $q_1 \colon E_1 \to X, q_2 \colon E_2 \to X$ be two coverings of the same topological space X. A **covering homomorphism** from q_1 to q_2 is a continuous map $\phi \colon E_1 \to E_2$ such that $q_2 \circ \phi = q_1$:

$$\begin{tikzcd} E_1 \arrow[rr, "\phi"] \arrow[dr, "q_1"'] & & E_2 \arrow[dl, "q_2"] \\ & X & \end{tikzcd}$$

Remark 2.4.2 — Note that covering homomorphisms may be viewed as lifts.

Proposition 2.4.3
Let $p \colon \tilde{X} \to X$ be a covering. Let $x_0 \in X, \tilde{x}_0 \in p^{-1}(x_0)$.

1. The induced map $p_* \colon \pi_1(\tilde{X}, \tilde{x}_0) \to \pi_1(X, x_0)$ is injective.
2. If $[\alpha] \in \pi_1(X, x_0)$ and $\tilde{\alpha}$ is a lift such that $\tilde{\alpha}(0) = \tilde{x}_0$, then $\tilde{\alpha}$ is a loop if and only if $[\alpha] \in p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

Proof. (1): We want to show if $p_*([\tilde{\alpha}]) = [c_{x_0}]$, then $\tilde{\alpha} \sim c_{\tilde{x}_0}$. By the path lifting property, we know $\tilde{\alpha}$ is the unique lift of $\alpha = p \circ \tilde{\alpha}$ such that $\tilde{\alpha}(0) = \tilde{x}_0$. Specifically, we know $\alpha \sim c_{x_0}$ via the homotopy F such that $f_0 = \alpha$ and $f_1 = c_{x_0}$. By the homotopy lifting property, I can lift it to $\tilde{F}$ between $\tilde{\alpha}$ and $c_{\tilde{x}_0}$ (from the fact that constant paths lift to constant paths).

(2): ($\implies$): If $\tilde{\alpha}$ is a loop, then $p \circ \tilde{\alpha}$ is a loop $(p \circ \tilde{\alpha}) \in p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

($\impliedby$): Let $[\alpha] \in p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. Then, there exists a loop $\tilde{\gamma} \in \tilde{X}$ such that $[p \circ \tilde{\gamma}] = [\alpha]$. If $\tilde{\alpha}$ is the lift of α such that $\tilde{\alpha}(0) = \tilde{x}_0$, then

$$p \circ \tilde{\alpha} = \alpha \sim p \circ \tilde{\gamma} = \gamma.$$

Then, letting F be a homotopy between α and γ, we get that $\tilde{F}$ is the lifted homotopy $(p \circ \tilde{F} = F\ )$ between $\tilde{\alpha}$ and $\tilde{\gamma}$, so we get $\tilde{\alpha}(1) = \tilde{x}_0$. □

Remark 2.4.4 — The above proposition proves that $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ is a subgroup of $\pi_1(X, x_0))$ that is isomorphic to $\pi_1(\tilde{X}, \tilde{x}_0))$.

Recall the following definition, which says that if we have a covering with the base space being connected, the fiber is of constant cardinality.

Definition 2.4.5. Let $p\colon \tilde{X} \to X$ be a covering such that X is connected. Then, $\forall x \in X$,

$$\deg(p) = |p^{-1}(x)|$$

is called a **degree** of the covering.

Proposition 2.4.6

Let $p\colon \tilde{X} \to X$ be a covering, where $X, \tilde{X}$ are path-connected with $x_0 \in X, \tilde{x}_0 \in p^{-1}(x_0)$. Then,

$$\deg(p) = [\pi_1(X, x_0) : p_*(\pi_1(\tilde{X}, \tilde{x}_0))].$$

Proof. We call $G = \pi_1(X, x_0)$ and $H = p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ for clarity $(H \leq G)$. We want to define

$$\Phi\colon G/H \to p^{-1}(x_0).$$

Let $[\alpha] \in G$ and a lift $\tilde{\alpha}$ such that $\tilde{\alpha}(0) = \tilde{x}_0, \tilde{\alpha}(1) = \tilde{x}_1 \in p^{-1}(x_0)$. If we also had $[\beta] \in G$ such that $[\alpha] = [\beta]$, we have that $\tilde{\beta}(1) = \tilde{x}_1$. Particularly, we note that $\tilde{x}_1$ doesn't depend on the representative. Let $[h] \in H$, we know it lifts to a loop $\tilde{h}$ in $\tilde{X}$, and $h \cdot \alpha$ lifts to a path $\tilde{h} \cdot \tilde{\alpha}$ from $\tilde{x}_0$ to $\tilde{x}_1$. We can therefore define the following map:

$$\begin{aligned} \Phi\colon G/H &\to p^{-1}(x_0) \\ H[\alpha] &\mapsto \tilde{\alpha}(1) \end{aligned}$$

This is well-defined, as the end points do not depend on the representative of the map. Now we just need to show injectivity and surjectivity.

- Injectivity: Assume that $\Phi(H[\beta]) = \Phi(H[\alpha])$ i.e., $\tilde{\alpha}(1) = \tilde{\beta}(1)$. Thus, $\tilde{\alpha} \cdot \overline{\tilde{\beta}}$ is a well-defined loop in $\tilde{X}$ at $\tilde{x}_0$. Therefore,

  $$p_*([\tilde{\alpha} \cdot \overline{\tilde{\beta}}]) = [p \circ (\tilde{\alpha} \cdot \overline{\tilde{\beta}})] = [(p \circ \tilde{\alpha}) \cdot (p \circ \overline{\tilde{\beta}})] = [\alpha] \cdot [\overline{\beta}] \in H.$$

  Thus, $H[\alpha] = H[\alpha] \cdot [\overline{\beta}] \cdot [\beta] = H[\beta]$.

- Since $\tilde{X}$ is path-connected, there exists a path from $\tilde{x}_0$ to any other point in $p^{-1}(x_0)$. Each such path projects down to a loop α in X. Then, Φ maps the corresponding $H[\alpha]$ in G/H to $\tilde{\alpha}(1)$, showing surjectivity.

$\square$

Definition 2.4.7. We say that a topological space X has a property P **locally** if $\forall x \in X$ and for all open neighborhoods U of x, there exist an open neighborhood $V \subseteq U$ of x such that V has the property.

Theorem 2.4.8 (Lifting Criterion)

Let $p\colon (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering and $f\colon (Y, y_0) \to (X, x_0)$ be a continuous map with Y path connected and locally path connected. Then, there exists $\tilde{f}\colon (Y, y_0) \to (\tilde{X}, x_0)$ if and only if $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

Proof. ($\implies$): $f = p \circ \tilde{f}$, so $f_* = p_* \circ \tilde{f}_*$ by functoriality.

($\impliedby$): Assume $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. We first define a function $\tilde{f}\colon (Y, y_0) \to (\tilde{X}, \tilde{x}_0)$ (and that this is well-defined). For all $y \in Y$, choose a path between y_0 and y, call it γ. $f \circ \gamma$ is a path from x_0 to $f(y)$. By the path lifting property, we can lift it to the path $\widetilde{f \circ \gamma}$ in $\tilde{X}$ starting at $\tilde{x}_0 \in p^{-1}(x_0)$. Define

$$\tilde{f}\colon (Y, y_0) \to (\tilde{X}, \tilde{x}_0), \quad y \mapsto \widetilde{f \circ \gamma}(1)$$

We now show that this is well-defined. Let us choose another path γ' between y_0 and y. Define

$$h_0 = (\underbrace{f \circ \gamma'}_{x_0 \to f(y)}) \cdot (\underbrace{\overline{f \circ \gamma}}_{f(y) \to x_0}).$$

This is a loop based at x_0. We then have that

$$[h_0] \in f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\tilde{X}, \tilde{x}_0)).$$

Therefore, we can lift it to a loop $\tilde{h}_0$ based at $\tilde{x}_0$ where the first half is $\widetilde{f \circ \gamma'}$ and the second part is $\widetilde{\overline{f \circ \gamma}}$. Therefore, $\widetilde{f \circ \gamma'}(1) = \widetilde{f \circ \gamma}(1)$, so this map is well defined.

Next, $f = p \circ \tilde{f}$, so it is indeed a lift.

Finally, we show that $\tilde{f}$ is continuous. Take $y \in Y$ and let $U \subset X$ be an evenly covered neighborhood of $f(y)$. Let $\tilde{U} \subset \tilde{X}$ be the sheet over U such that $\tilde{f}(y) \in \tilde{U}$. Let $V \subseteq Y$ be a path connected neighborhood of y such that $f(V) \subseteq U$. Fix a path γ from y_0 to y. Let $y' \in V$ and let ν be a path from y to y'. $(f \circ \gamma) \cdot (f \circ \nu)$ is a path in X from x_0 to $f(y')$. $\widetilde{f \circ \nu}$ is a path in $\tilde{U}$, so

$$\widetilde{f \circ \nu} = (p\,|_{\tilde{U}})^{-1}(f \circ \nu).$$

Therefore,

$$\tilde{f}\,|_V = (p\,|_{\tilde{U}})^{-1} \circ f,$$

and so $\tilde{f}$ is continuous on V.

$$\tilde{f}(y') = (\widetilde{(f \circ \gamma) \cdot (f \circ \nu)})(1) = \widetilde{f \circ \nu}(1)$$

$\square$

Definition 2.4.9. A topological space X is **semilocally simply connected** if $\forall x \in X$ there exists an open neighborhood U of x such that

$$\iota_* \colon \pi_1(U, x) \to \pi_1(X, x)$$

is trivial ($\iota \colon U \to X$ is the inclusion).

Example 2.4.10 (Hawaiian Ring)
The **Hawaiian ring** is defined as

$$E = \bigcup_{n \geq 1} C_n,$$

where $C_n = \{(x, y) \in \mathbb{R}^2 \mid (x - \frac{1}{n})^2 + y^2 = \frac{1}{n^2}$. This is not semilocally simply connected.

Definition 2.4.11. A covering space $p \colon \tilde{X} \to X$ is a **universal covering** if $\tilde{X}$ is simply connected.

Proposition 2.4.12
If $p \colon (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ is a universal cover, then X is semilocally simply connected.

Proof. Let $U \subseteq X$ be an evenly covered neighborhood of $x_0 \in X$. Let $\tilde{U}$ be a sheet over U. Let γ be a loop in U based at x_0 which lifts to a loop $\tilde{\gamma} \in \tilde{U}$ based at $\tilde{x}_0$. Then $\tilde{\gamma}$ is homotopic to the constant loop at $\tilde{x}_0$ (since $\tilde{X}$ is simply connected). Composing the homotopy with p, we get that γ is homotopic to the constant loop at x_0 in X. Thus, $\pi_1(U, x_0) \to \pi_1(X, x_0)$ is trivial. □

Theorem 2.4.13 (Existence of the Universal Covering Space)
Let X be path connected, locally path connected, and semilocally simply connected. Then, there exists a universal cover $p \colon \tilde{X} \to X$.

Proof. We first present the sketch. Suppose for a moment that X has a universal covering $q \colon \tilde{X} \to X$. The key fact is that once we choose base points $\tilde{x}_0 \in \tilde{X}$ and $x_0 = q(\tilde{x}_0) \in X$, the fiber $q^{-1}(x)$ over an $x \in X$ is in one-to-one correspondence with path classes from x_0 to x. To see why, we define a map E from the set of such path classes to $q^{-1}(x)$ by sending $[f]$ to the terminal point of the lift of f starting at $\tilde{x}_0$. Since lifts of homotopic paths have the same terminal point by the monodromy theorem, E is well defined. E is surjective, because given any $\tilde{x}$ in the fiber over x, there is a path $\tilde{x}$ from $\tilde{x}_0$ to $\tilde{x}$, and then $q \circ \tilde{f}$ a path from x_0 to x whose lift ends at $\tilde{x}$. Injectivity of E follows from the fact that $\tilde{X}$ is simply connected: if f_1, f_2 are two paths from x_0 to x whose lifts $\tilde{f}_1, \tilde{f}_2$ end at the same point, then $\tilde{f}_1$ and $\tilde{f}_2$ are path-homotopic, and therefore so are $f_1 = q \circ \tilde{f}_1$ and $f_2 = q \circ \tilde{f}_2$.

Now let X be a path connected, locally path connected, and semilocally simply connected space. Choose any base point $x_0 \in X$. Guided by the observation in the preceding paragraph, we define $\tilde{X}$ to be the set of path classes of paths in X starting at x_0, and define $q\colon \tilde{X} \to X$ by $q([f]) = f(1)$. We prove that $\tilde{X}$ has the required properties in a series of steps.

Step 1-*Topologize* $\tilde{X}$: We define a topology on $\tilde{X}$ by constructing a basis. For each $[f] \in \tilde{X}$ and each simply connected open subset $U \subseteq X$ containing $f(1)$, define the set $[f \cdot U] \subseteq \tilde{X}$ by

$$[f \cdot U] = \{[f \cdot a] \mid a \text{ is a path in } U \text{ starting at } f(1)\}.$$

Let $\mathcal{B}$ denote the collection of all sets $[f \cdot U]$. We show that $\mathcal{B}$ is a basis for a topology. First, since X is locally simply connected, for each $[f] \in \tilde{X}$, there exists a simply connected open subset U containing $f(1)$, and clearly $[f] \in [f \cdot U]$. Therefore ,the union of all the set in $\mathcal{B}$ is $\tilde{X}$.

We then check the intersection condition. Suppose $[h] \in \tilde{X}$ is in the intersection of two basis sets $[f \cdot U], [g \cdot V] \in \mathcal{B}$. This means that $h \sim f \cdot a \sim g \cdot b$, where a is a path in U and b is a path in V. Let W be a simply connected neighborhood of $h(1)$ contained in $U \cap V$ (note that such a neighborhood exists because X has a basis of simply connected open subsets). If $[h \cdot c]$ is any element of $[h \cdot W]$, then $[h \cdot c] = [f \cdot a \cdot c] \in [f \cdot U]$ because $a \cdot c$ is a path in U. Similarly, $[h \cdot c] = [g \cdot b \cdot c] \in [g \cdot V]$. Therefore, $[h \cdot W]$ is a basis set contained in $[f \cdot U] \cap [g \cdot V]$, which proves that $\mathcal{B}$ is a basis. From now on, we endow $\tilde{X}$ with the topology generated by $\mathcal{B}$.

Step 2-$\tilde{X}$ *is path connected*: Let $[f] \in \tilde{X}$ be arbitrary. We will show that there is a path from $\tilde{X}$ from $\tilde{x}_0$ to $[f]$, where $\tilde{x}_0 = [c_{x_0}]$. For each $0 \leq t \leq 1$, define $f_t\colon I \to X$ by

$$f_t(s) = f(ts),$$

so f_t is a path in X from x_0 to $f(t)$. Then, define $\tilde{f}\colon I \to \tilde{X}$ by

$$\tilde{f}(t) = [f_t].$$

Clearly, $\tilde{f}(0) = [f_0] = \tilde{x}_0$, and $\tilde{f}(1) = [f_1] = [f]$. Therefore, we need to only show that $\tilde{f}$ is continuous. For this, it suffices to show that the preimage under $\tilde{f}$ of ever basis subset $[h \cdot U] \subseteq \tilde{X}$ is open. Let $t_0 \in I$ be a point such that $\tilde{f}(t_0) \in [h \cdot U]$. This means that $f_{t_0} \sim h \cdot c$ for some path c lying in U, and in particular that $f(t_0) = f_{t_0}(1) \in U$. For each $0 \leq t \leq 1$, define a path $f_{t_0 t}$ by

$$f_{t_0 t}(s) = f(t_0 + s(t - t_0)).$$

This path just follows f from $f(t_0)$ to $f(t)$, so $f_{t_0} \cdot f_{t_0 t}$ is easily seen to be path homotopic to f_t. Then, by continuity of f, there is some $\delta > 0$ such that $f(t_0 - \delta, t_0 + \delta) \subseteq U$. If $t \in (t_0 - \delta, t_0 + \delta)$, then

$$f_t \sim f_{t_0} \cdot f_{t_0 t} \sim h \cdot c \cdot f_{t_0 t},$$

from which it follows that

$$\tilde{f}(t) = [f_t] = [h \cdot c \cdot f_{t_0 t}] \in [h \cdot U].$$

This shows that $\tilde{f}^{-1}[h \cdot U]$ contains the set $(t_0 - \delta, t_0 + \delta)$, so $\tilde{f}$ is continuous.

Step 3-q *is a covering map*: Let $U \subseteq X$ be any simply connected open subset. We will show that U is evenly covered. Choose any point $x_1 \in U$. We begin by showing that $q^{-1}(U)$ is the disjoint union of the sets $[f \cdot U]$ as $[f]$ varies over all the distinct path classes from x_0 to x_1. It follows immediately from the definition of q that $q([f \cdot U]) \subseteq U$, so $\cup_{[f]}[f \cdot U] \subseteq q^{-1}(U)$. Conversely, if $[g] \in q^{-1}(U)$, then $g(1) = q([g]) \in U$, so there is a path b in U from $g(1)$ to x_1, and $[g] = [g \cdot b \cdot b^{-1}] \in [(g \cdot b) \cdot U]$. This proves that

$$q^{-1}(U) = \cup_{[f]}[f \cdot U].$$

Particularly, this shows that q is continuous: X has a basis of simply connected open subsets, and the preimage under q of each such set is a union of basis sets and therefore open. And q is clearly surjective, because each $x \in X$ is equal to $q([g])$ for any path g from x_0 to x.

Next, we show that q is a homeomorphism from each set $[f \cdot U]$ to U. It is surjective because for each $x \in U$ there is a path a from $f(1)$ to x in U, so $x = q([f \cdot a]) \in q([f \cdot U])$. To see that it is injective, let $[g], [g'] \in [f \cdot U]$, and suppose $q([g]) = q([g'])$, or in other words, $g(1) = g'(1)$. Then, by definition of $[f \cdot U]$, $g \sim f \cdot a$ and $g' \sim f \cdot a'$ for some paths a, a' in U from $f(1)$ to $g(1)$. Since U is simply connected, $a \sim a'$ and therefore $[g] = [g']$. Finally, q is an open map because it takes basis open subsets to open subsets, and therefore $q \colon [f \cdot U] \to U$ is a homeomorphism.

Each set $[f \cdot U]$ is open by definition, and each is path connected because it is homeomorphic to the path connected set U. It follows that $\tilde{X}$ is locally path connected. To complete the proof that q is a covering map, we need to show that for any two paths f and f' from x_0 to x_1, the sets $[f \cdot U]$ and $[f' \cdot U]$ are either equal or disjoint. If they are not disjoint, there exists $[g] \in [f \cdot U] \cap [f' \cdot U]$, so $g \sim f \cdot a \sim f' \cdot a'$ for paths a, a' in U from x_1 to $g(1)$. Since U is simply connected, $a \sim a'$, which implies $f \sim f'$, and therefore $[f \cdot U] = [f' \cdot U]$.

***Step 4**-$\tilde{X}$ is simply connected*: Suppose $F \colon I \to \tilde{X}$ is a loop based at $\tilde{x}_0$. Let $f = q \circ F$, so F is a lift of f. If we write $\tilde{f}(t) = [f_t]$ as in step 2, then $q \circ \tilde{f}(1) = q([f_t]) = f_t(1) = f(t)$, so $\tilde{f}$ is also a lift of f starting at $\tilde{x}_0$. By the unique lifting property, $F = \tilde{f}$. Since F is a loop,

$$[c_{x_0}] = \tilde{x}_0 = F(1) = \tilde{f}(1) = [f_1] = [f],$$

so f is null-homotopic. By the monodromy theorem, this means that F is null-homotopic. □

Proposition 2.4.14

Let X be path connected, locally path connected, and semilocally simply connected. Then, for all $H \leq \pi_1(X, x_0)$, there exist a covering space $p \colon X_H \to X$ such that $p_*(\pi_1(X_H, \tilde{x}_0)) = H$.

Proof. This follows by the injectivity theorem 2.2.9 (or look at proposition 5.5 in notes). □

Definition 2.4.15. Given coverings $p_1 \colon (\tilde{X}, \tilde{x}_1) \to (X, x_0)$ and $p_2 \colon (\tilde{X}, \tilde{x}_2) \to (X, x_0)$, a **basepoint preserving isomorphism** is an isomorphism of covers such that $h(\tilde{x}_1) = \tilde{x}_2$.

Proposition 2.4.16

Let X be path connected and locally path connected, and let $x_0 \in X$. Then, two path connected covering spaces $p_1\colon (\tilde{X}_1, \tilde{x}_1) \to (X, x_0)$ and $p_2\colon (\tilde{X}, x_2) \to (X, x_0)$ are isomorphic via an isomorphism $h\colon \tilde{X}_1 \to \tilde{X}_2$ mapping $\tilde{x}_1 \in p_1^{-1}(x_0)$ to $\tilde{x}_2 \in p_2^{-1}(x_0)$ if and only if

$$(p_1)_*(\pi_1(\tilde{X}_1, \tilde{x}_1)) = (p_2)_*(\pi_1(\tilde{X}_2, \tilde{x}_2))$$

Proof. ($\implies$): Since $p_1 = p_2 \circ h$, we have

$$(p_1)_*(\pi_1(\tilde{X}_1, \tilde{x}_1) \subseteq (p_2)_*(\pi_1(\tilde{X}_2, \tilde{x}_2)),$$

and since $p_2 = p_q \circ h^{-1}$, we have the reverse inclusion:

$$(p_2)_*(\pi_1(\tilde{X}_2, \tilde{x}_2) \subseteq (p_1)_*(\pi_1(\tilde{X}_1, \tilde{x}_1))$$

($\impliedby$): Assume equality:

$$(p_1)_*(\pi_1(\tilde{X}_1, \tilde{x}_1)) = (p_2)_*(\pi_1(\tilde{X}_2, \tilde{x}_2))$$

We can lift p_1 and p_2 to a continuous map

$$\begin{aligned} \tilde{p}_1\colon (\tilde{X}_1, \tilde{x}_1) \to (\tilde{X}_2, \tilde{x}_2) \\ \tilde{p}_2\colon (\tilde{X}_2, \tilde{x}_2) \to (\tilde{X}_1, \tilde{x}_1), \end{aligned}$$

so that $p_1 \circ \tilde{p}_2 = p_2$ and $p_2 \circ \tilde{p}_1 = p_1$. Then, $\tilde{p}_1 \circ \tilde{p}_2$ fixes the point $\tilde{x}_2 \in \tilde{X}_2$. By the unique lifting property and the fact that $\tilde{p}_1 \circ \tilde{p}_2$ fix $\tilde{x}_2$, we have that $\tilde{p}_1 \circ \tilde{p}_2 = id_{\tilde{X}_2}$. Similarly, we get $\tilde{p}_2 \circ \tilde{p}_1 = id_{\tilde{X}_1}$. □

Theorem 2.4.17 ('Galois Correspondence')

Let X be path connected, locally path connected, and semi locally simply connected. Then:

1. There is a bijection between subgroups of $\pi_1(X, x_0)$ and path connected covering spaces $p\colon (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ up to base point preserving isomorphisms.
2. Ignoring the basepoints, the correspondence gives a bijection between conjugacy classes of subgroups of $\pi_1(X, x_0)$ and path connected covering spaces $p\colon (\tilde{X}, \tilde{x}_0) \to (X, x_0)$, up to isomorphisms.

Proof. (1): To a covering space $p\colon (\tilde{X}, \tilde{x}_0) \to (X, x_0)$, we associate the subgroup

$$p_*(\pi_1(\tilde{X}, \tilde{x}_0)) \subseteq \pi_1(X, x_0).$$

Using the results above, we know that this is well-defined on the isomorphism classes and it is bijective.

(2): Let $p\colon (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering space and $\tilde{x}_1, \tilde{x}_2 \in p^{-1}(x_0)$. Let

$$H_i = p_*(\pi_1(\tilde{X}, \tilde{x}_i)) \subseteq \pi_1(X, x_0).$$

Consider a path $\tilde{\gamma}$ from $\tilde{x}_1$ to $\tilde{x}_2$. Let $\gamma = p \circ \tilde{\gamma}$ be a loop at x_0. Let $[f] \in \pi_1(X, x_0)$, then $[f] \in H_1$ if and only if the lift $\tilde{f}$ is a loop at $\tilde{x}_1$. Then, $\tilde{\gamma}^{-1} \cdot \tilde{f} \cdot \tilde{\gamma}$ is a loop at $\tilde{x}_2$. Thus,

$$p_*(\tilde{\gamma}^{-1} \cdot \tilde{f} \cdot \tilde{\gamma}) = \gamma^{-1} \cdot f \cdot \gamma,$$

hence, $[\gamma]^{-1} \cdot [f] \cdot [\gamma] \in H_2$. Therefore, $[\gamma]^{-1} H_1 [\gamma] \subseteq H_2$ and similarly $[\gamma]^{-1} H_2 [\gamma] \subseteq H_1$. Conversely, let $H_1 \subseteq \pi_1(X, x_0)$ as above and $[\delta] \in \pi_1(X, x_0)$ be an arbitrary element. Let $\tilde{\gamma}$ be a lift of δ such that $\tilde{\delta}(0) = \tilde{x}_0$ and define $\tilde{x}_3 = \delta(1)$. Then, the same construction yields

$$p_*(\pi_1(\tilde{X}, \tilde{x}_3)) = [\delta]^{-1} H_1 [\delta].$$

□

3 Seifert-Van Kampen Theorem

§3.1 The Wedge Product

Definition 3.1.1. Let $\{(X_\alpha, x_\alpha)\}_\alpha$ be a collection of pointed topological spaces. The **wedge product** of this collection is defined as

$$\bigvee_\alpha (X_\alpha, x_\alpha) = \bigsqcup_\alpha X_\alpha/(x_\alpha \sim x_\beta),$$

that is, the disjoint union of the X_α with the points x_α all identified.

Remark 3.1.2 — When we write $/(x_\alpha \sim x_\beta)$, we mean that we're *gluing* the space X_α at the same point.

Notation 3.1.3. Note that we may omit the basepoints from the notation when it is not important.

We work a bit with $\mathbb{S}^1$ and the figure-eight to familiarize ourselves with this.

Example 3.1.4

The *figure-eight* is $\mathbb{S}^1 \vee \mathbb{S}^1$.

Example 3.1.5

The set *bouquet* is given by $\mathbb{S}^1 \vee \mathbb{S}^1 \vee \mathbb{S}^1$.

Remark 3.1.6 — $\mathbb{S}^1 \vee \mathbb{S}^1 \vee \mathbb{S}^1$ is not the same as $(\mathbb{S}^1 \vee \mathbb{S}^1) \vee \mathbb{S}^1$! The base points do play a role, in the latter we are identifying only two of them.

If we denote the two loops in the figure-eight $\mathbb{S}^1 \vee \mathbb{S}^1$ by a and b, and their inverses at $\bar{a}$ and $\bar{b}$, then intuitively, every loop can be written as a *word*. For instance,

$$aaab\bar{b}b\bar{a}a\bar{a}bb$$

means go around a three times, then around b, around b backwards and again around b, and so on. We can further see that a loop described like this can be *reduced* to a homotopic loop: loops of the form $b\bar{b}$ are homotopic to the constant loop, so we can replace them with c_p, and we can then remove the constant loops from the expression. It follows that the loop above is homotopic to a reduced loop of the form

$$a^3b\bar{a}b^2 \text{ or } a^3ba^{-1}b^2.$$

From this observation one can conjecture the form of the fundamental group of the figure-eight: take two circles $A = \mathbb{S}^1$ and $B = \mathbb{S}^1$, joint them at a point x_0, take generators a and b of the fundamental groups of A and B (which are isomorphic to $\mathbb{Z}$), and describe the fundamental group of the figure-eight as the set of reduced words on a and b, where the inverse of a word is the word obtained by changing the order of letters and replacing every letter with its inverse (for example, the inverse of aba^2b^{-3} is $b^3a^{-2}b^{-1}a^{-1}$), and taking as multiplication the concatenation of words, followed by a reduction. We formalize this process next, using the concept of a free product.

§3.2 Free Product of Groups

Definition 3.2.1. Let $\{G_\alpha\}_\alpha$ be a collection of groups. A **word** on these groups is a finite sequence $g_1 \dots g_m$ of elements of the $g_i \in G_{\alpha_i}$, and m is the **length** of the word.

Definition 3.2.2. The **empty word** is denoted by ϵ.

Definition 3.2.3. The **product** of two words is the concatenation,

$$(g_1 \dots g_m) * (h_1 \cdot g_n) = g_1 \dots g_m h_1 \dots g_n$$

Definition 3.2.4. A word $g = g_1 \dots g_m$ is called **reduced** if $g_1 \neq e_{\alpha_i}$ (the unit element of the group G_{α_i}) and for any two consecutive letters g_i, g_{i+1}, $\alpha_i \neq \alpha_{i+1}$ (that is, consecutive letters are not from the same group).

Given any word g on the groups $\{G_\alpha\}_\alpha$, we can reduce it to a reduced word g' as follows.

1. If $g_i = e_{\alpha_i}$, then remove it from g.
2. If $\alpha_i = \alpha_{i+1}$, then replace $g_i g_{i+1}$ with the group element $g_i \cdot g_{i+1}$ from G_{α_i}.

Remark 3.2.5 — As every such operation reduces the length of the word by one, the process has to terminate. Moreover, a word is reduced if and only it can't be reduced further by the above two operations.

Remark 3.2.6 — A word g can be reduced to a word g' in different ways, depending on the order in which the operations are applied. It is not yet obvious that every word reduces to a unique reduced word.

On the set of reduced words we can define a multiplication as follows. Given reduced words $g = g_1 \dots g_m$ and $h = h_1 \dots h_n$ construct a new reduced word $g \cdot h$ by taking the concatenation $g \cdot h$ and then reducing the word recursively as follows.

$$g \cdot h = \begin{cases} g * h & \text{if } g \text{ and } h \text{ are not in the same group} \\ g_1 \dots g_{m-1}(g_m \cdot h_1)h_2 \dots h_n & \text{if } g_m, h_1 \in G_\alpha \text{ and } g_m \cdot h_1 \neq e_\alpha \\ g_1 \dots g_{m-1} \cdot h_2 \dots h_n & \text{if } g_m \cdot h_1 = e_\alpha \end{cases},$$

where $*$ is the concatenation of words. This process eventually leads to a reduced word, denoted by $g \cdot h$. Define the set

$$*_\alpha G_\alpha = \{\text{reduced words on } \{G_\alpha\}_\alpha\}.$$

Theorem 3.2.7

The pair $(*_\alpha G_\alpha, \cdot)$ is a group, called the **free product** of $\{G_\alpha\}_\alpha$. The unit element is the empty word $\epsilon = []$, and the inverse of an element $g_1 \dots g_m$ is $g_m^{-1} \dots g_1^{-1}$.

Proof. Checking that the inverse has the given form is straight-forward. Checking associativity of the operation requires some work.

The verification that $g \cdot h$ is again a reduced word follows from the definition: if the concatenation $g * h$ is not reduced, then it is replaced by a shorter word. As words have finite length, this process has to terminate in a reduced word. That the empty word is the unit element follows from the definition of the product '$\cdot$'. That the inverse element has the given form is obvious, but can be shown formally by induction: if $m = 1$, then $g \cdot g^{-1} = \epsilon$ (by the definition of the product) and assuming the statement holds for $m-1$, then

$$g_1 \cdot \cdots \cdot g_m \cdot g_m^{-1} \cdot \cdots \cdot g_1 = \epsilon.$$

To get the group structure, we just need to show associativity.

To prove this, set $W = *_\alpha G_\alpha$ for the set of reduced words, and consider the group of bijections $\mathrm{Sym}(W)$. We will *embed* W into $\mathrm{Sym}(W)$ via an injective map L that is compatible with multiplication, i.e., $L(g \cdot h) = L(g) \circ L(h)$, and from this the associativity in $\mathrm{Sym}(W)$ will naturally lead to the associativity of the product $\cdot$ in W. To start with, for every element $g \in G_\alpha$ we have a map L_g, the left multiplication, such that $L_g(h) = g \cdot h$ for a word $h \in W$. If $g_1, g_2 \in G_\alpha$ and $h = h_1 \dots h_m$, then one verifies that

$$L_{g_1} \circ L_{g_2} = L_{g_1 g_2}.$$

Hence, $L_{g^{-1}} = L_g^{-1}$, so that $L_g \in \mathrm{Sym}(W)$. For any word $g = g_1 \dots g_m$, the map

$$\begin{aligned} L \colon W &\to \mathrm{Sym}(W) \\ g &\mapsto L_{g_1} \circ \cdots \circ L_{g_m} = L_{g_1 \dots g_m} \end{aligned}$$

is injective, since for any $g \in W$, $L_g(\epsilon) = g$, hence if $g \neq h$ in W, $L_g \neq L_h$ in $\mathrm{Sym}(W)$. Now, we know that composition $L_g \circ L_h$ obeys the same rules as the product '$\cdot$': if $g = g_1 \dots g_m$ and $h = h_1 \dots h_n$ are reduced words, then

$$L_g \circ L_h = \begin{cases} L_{g*h} & \text{if } g, h \text{ are not in the same group} \\ L_{g_1} \circ \dots L_{g_{m-1}} \circ L_{g_m h_1} \circ L_2 \circ \cdots \circ L_n & \text{if } g, h \in G_\alpha \text{ and } g_m h_1 \neq e \\ L_{g_1} \circ \dots L_{g_{m-1}} \circ L_2 \circ \cdots \circ L_n & \text{if } g, h \in G_\alpha \text{ and } g_m h_1 = e \end{cases}$$

From this it follows that $L_{g \cdot h} = L_g \circ L_h$. The associativity then follows from:

$$\begin{aligned} L_{(g\cdot h)\cdot k} &= L_{g \cdot h} \circ L_k \\ &= (L_g \circ L_h) \circ L_k) \\ &= L_g \circ (L_h \circ L_k) \\ &= L_g \circ L_{h \cdot k} \\ &= L_{g \cdot (h \cdot k)} \end{aligned}$$

By the injectivity of L, we have associativity. □

Example 3.2.8

$\pi_1(\mathbb{S}^1 \vee \mathbb{S}^1) \cong \pi_1(\mathbb{S}^1) * \pi_1(\mathbb{S}^1) \cong \mathbb{Z} \times \mathbb{Z}$.

Remark 3.2.9 — The situation similar to the above example where

$$\pi_1((X,x_0) \vee (Y,y_0)) \cong \pi_1(X,x_0) * \pi_1(Y,y_0)$$

is desirable, but this is not true in general. For instance, the fundamental group of the Hawaiian earring is a counterexample to this. We define the Hawaiian earring as

$$X = \{C_n \mid n \in \mathbb{N}\},$$

where C_n is a circle centered at $(\frac{1}{n}, 0)$ with radius $\frac{1}{n}$. Its fundamental group is uncountable, but $*_n \pi_1(C_n)$ is countable.

Lemma 3.2.10

Let $\{\phi_\alpha\}_\alpha$ be a collection of group homomorphisms $\phi_\alpha \colon G_\alpha \to G$. Then there exists a unique map

$$*_\alpha \phi_\alpha \colon *_\alpha G_\alpha \to G$$

such that $(*_\alpha \phi_\alpha) \circ \iota_\beta = \phi_\beta$, for all β. Here, $\iota_\alpha \colon G_\alpha \to *_\alpha G_\alpha$ is given by $e_\alpha \mapsto \epsilon$, $e_\alpha \neq g_\alpha \mapsto g_\alpha$.

Proof. Define $*_\alpha \phi_\alpha \colon *_\alpha G_\alpha \to G$ by

$$(*_\alpha \phi_\alpha)(g_1 \dots g_m) = \phi_{\alpha_1}(g_1) \dots \phi_{\alpha_m}(g_m),$$

where $g_i \in G_{\alpha_i}$. This clearly satisfies the property $*_\alpha \phi_\alpha \circ \iota_\alpha = \phi_\alpha$. Moreover, since every ϕ_α is a group homomorphism, for $g_i, g_{i+1} \in G_\alpha$ we get $\phi(g_i)\phi(g_{i+1}) = \phi(g_i g_{i+1})$ and $\phi(e_\alpha) = e$, so that $*_\alpha \phi_\alpha$ is compatible with the operations bringing a word into reduced form. Therefore, $*_\alpha \phi_\alpha(g \cdot h) = (*_\alpha \phi_\alpha)(g)(*_\alpha \phi_\alpha)(h)$ and we have a group homomorphism. The requirement that the restriction to the G_α satisfies $*_\alpha \phi_\alpha \circ \iota_\alpha = \phi_\alpha$ leaves one with no other choice than to define the homomorphism as the above. □

§3.3 The Theorem

We now apply the free product to topology. The goal is to reduce the computation of the fundamental group of an open cover to the fundamental groups of the individual sets in the cover. Let $X = \cup_\alpha X_\alpha$ be an open cover and denote by $\iota_\alpha \colon X_\alpha \to X$ and the inclusion maps. Assume that $x_0 \in \cap_\alpha X_\alpha$. The inclusion maps induce maps

$$(\iota_\alpha)_* \colon \pi_1(X_\alpha, x_0) \to (X, x_0)$$

of the fundamental groups with base x_0. By lemma 3.2.10, these maps induce a unique map

$$\Phi = *_\alpha (\iota_\alpha)_* \colon *_\alpha \pi_1(X_\alpha, x_0) \to \pi_1(X, x_0),$$

and these maps are compatible with the inclusion $\iota_\alpha \colon \pi_1(X_\alpha, x_0) \to *_\alpha \pi_1(X_\alpha, x_0)$, meaning that

$$*_\alpha (\iota_\alpha)_* \circ \iota_\alpha = (\iota_\alpha)_*.$$

The theorem will say that if the pairwise intersections $X_\alpha \cap X_\beta$ are path-connected, then the induced map Φ is surjective. In general, however, it will not be injective: the reason is that loops in $X_\alpha \cap X_\beta$ are accounted for twice in $*_\alpha \pi_1(X_\alpha, x_0)$, once as an element of

$\pi_1(X_\alpha, x_0)$ and once as an element of $\pi_1(X_\beta, x_0)$. To remedy this, we have to factor such loops out, and for this we need to study the inclusions

$$\iota_{\alpha\beta}\colon X_\alpha \cap X_\beta \to X_\alpha, \quad \iota_{\beta\alpha}\colon X_\alpha \cap X_\beta \to X_\beta$$

with the induced maps $(\iota_{\alpha\beta})_*, (\iota_{\beta\alpha})_*$, respectively, of fundamental groups. The whole setup is summarized in the following 'Seifert-van Kampen' commutative diagram:

$$\begin{tikzcd}
 & \pi_1(X_\alpha, x_0) \arrow[dr, "\iota_\alpha"] \arrow[drr, bend left, "(\iota_\alpha)_*"] & & \\
\pi_1(X_\alpha \cap X_\beta, x_0) \arrow[ur, "(\iota_{\alpha\beta})_*"] \arrow[dr, "(\iota_{\beta\alpha})_*"'] & & *_\alpha \pi_1(X_\alpha, x_0) \arrow[r, "\Phi"] & \pi_1(X, x_0) \\
 & \pi_1(X_\beta, x_0) \arrow[ur, "\iota_\beta"] \arrow[urr, bend right, "(\iota_\beta)_*"'] & &
\end{tikzcd}$$

Theorem 3.3.1

Let $X = \cup_\alpha X_\alpha$ be an open cover and assume $x_0 \in \cap_\alpha X_\alpha$. Then:

1. If for all α, β, $X_\alpha \cap X_\beta$ is path connected, then
$$\Phi\colon\ *_\alpha\, \pi_1(X_\alpha, x_0) \to \pi_1(X, x_0)$$
is surjective.

2. If for all α, β, γ, the intersection $X_\alpha \cap X_\beta \cap X_\gamma$ is path connected, then
$$\pi_1(X, x_0) \cong \left(*_\alpha \pi_1(X_\alpha, x_0)\right)/N,$$
where $N = \ker(\Phi)$. Here, N is explicitly defined as the normal closure of
$$U = \{(\iota_{\alpha\beta})_*(w)(\iota_{\beta\alpha})_*(w^{-1}) \mid w \in \pi_1(X_\alpha \cap X_\beta, x_0)\} \subset *_\alpha \pi_1(X_\alpha, x_0).$$

Proof Idea. The surjectivity is shown by *factoring out* a loop γ in X based at x_0 as a concatenation $\gamma_1 \dots \gamma_m$, where $\gamma_i \in X_{\alpha_i}$. Next, consider Φ:

$$\begin{aligned}
\Phi\colon\ & *_\alpha\, \pi_1(X_\alpha, x_0) \to \pi_1(X, x_0) \\
& [\gamma_1] \cdot \dots \cdot [\gamma_m] \mapsto (\iota_{\alpha_1})_*([\gamma_1]) \dots (\iota_{\alpha_m})_*([\gamma_m]) = [\gamma_1 \dots \gamma_m] = [\gamma]
\end{aligned}$$

This is not a unique factorization. By factoring over the kernel N, we get an isomorphism. □

Remark 3.3.2 — When we say that N is the normal closure of U, we mean that N the smallest normal subgroup of $*_\alpha \pi_1(X_\alpha, x_0)$ containing U.

We now interpret what this theorem is saying. For $[\gamma] \in \pi_1(X, x_0)$, we have $\gamma \simeq \gamma_1 \dots \gamma_m$ for $\gamma_i \in X_{\alpha_i}$ for a collection of loops $\{\gamma_i\}_i$. Therefore, the induced map from the free product looks something like

$$\begin{aligned}
\Phi\colon\ & *_\alpha\, \pi_1(X_\alpha, x_0) \to \pi_1(X, x_0) \\
& [\gamma_1] \dots [\gamma_m] \mapsto (\iota_{\alpha_1})_*([\gamma_1]) \dots (\iota_{\alpha_m})_*([\gamma_m]) = [\gamma_1 \dots \gamma_m] = [\gamma].
\end{aligned}$$

To show that this map is surjective, one needs to factor out every loop in X based at x_0 as a concatenation of loops $\gamma_1 \dots \gamma_m$, where each loop is contained in an open set from the cover. The map is not injective because such a factorization is not unique; for instance, we may think of loops in an intersection. By quotienting by the kernel, we then get an isomorphism.

Example 3.3.3

Consider $\mathbb{S}^1 \vee \mathbb{S}^1 \vee \mathbb{S}^1$. Let $A, B \subset \mathbb{S}^1 \vee \mathbb{S}^1 \vee \mathbb{S}^1$ be open covers that contains only two of the three circles with $A \neq B$. We have $\pi_1(\mathbb{S}^1 \vee \mathbb{S}^1) \cong \mathbb{Z} * \mathbb{Z}$, $\pi_1(\mathbb{S}^1 \vee \mathbb{S}^1 \vee \mathbb{S}^1) \cong \mathbb{Z} * \mathbb{Z} * \mathbb{Z}$, and so

$$\pi_1(A, x_0) * \pi_1(B, x_0) \cong \mathbb{Z} * \mathbb{Z} * \mathbb{Z} * \mathbb{Z} \ncong \pi_1(\mathbb{S}^1 \vee \mathbb{S}^1 \vee \mathbb{S}^1, x_0).$$

This is because we are over-counted a loop, so we have an inclusion/ exclusion type set up here. Let $w = [\gamma] \in \pi_1(X_A \cap X_B, x_0)$. This is represented in $*_A \pi_1(X_A, x_0) = \pi_1(X_A, x_0) * \pi_1(X_B, x_0)$ both as

$$(\iota_{AB})_*(w) \in \pi_1(X_A, x_0)$$
$$(\iota_{BA})_*(w) \in \pi_1(X_B, x_0).$$

Hence we need a way to quotient out these loops in order not to count them twice in the free group. Ultimately, we have $\pi_1(A \cap B, x_0) \cong \mathbb{Z} = \langle [a] \rangle$.

Example 3.3.4

Let $\mathbb{S}^n = U_1 \cup U_2$, where $U_1 = \mathbb{S}^n \setminus \{(0, \dots, 0, 1)\}$ and $U_2 = \mathbb{S}^n \setminus \{(0, \dots, 0, -1)\}$. Note that if $n \geq 2$, we have that $U_1 \cap U_2$ is homeomorphic to $\mathbb{R}^n \setminus \{0\}$, which is path connected. Both U_1 and U_2 are homeomorphic to $\mathbb{R}^n$, so they are contractible, so

$$\pi_1(U_1) \cong \pi_1(U_2) \cong 0.$$

Thus, the Seifert-van Kampen theorem tells us that $\pi_1(\mathbb{S}^n) = 0$ as well.

Note that this argument doesn't extend to $\mathbb{S}^1$, since $U_1 \cap U_2 = \mathbb{R} \setminus \{0\}$ is not path-connected.

Remark 3.3.5 — This tells us that $\mathbb{R}^n \setminus \{x_1\}$ is simply connected. In fact, we can show that $\mathbb{R}^n \setminus \{x_1, \dots, x_k\}$ is simply connected. The way we show this is as follows. We use that $\mathbb{R}^n$ is Hausdorff. We put disjoint balls around each point, and then join them with paths in some order. Then, $\mathbb{R}^n \setminus \{x_1, \dots, x_k\}$ deformation retracts to the boundaries of these balls with the paths that connect them. This is then homotopy equivalent to a wedge sum of k-copies of $\mathbb{S}^{n-1}$, and so we get simply connectedness.

Exercise 3.3.6. Let $X \subset \mathbb{R}^3$ be the union of n lines through the origin. Compute $\pi_1(\mathbb{R}^3 \setminus X)$.

Solution. We have that $\mathbb{R}^3$ minus n lines deformation retracts to the unit sphere $\mathbb{S}^2$ with $2n$ points removed. Then, one of these points can be used to create a stereographic projection so that we're dealing now with $\mathbb{R}^2$ minus $2n - 1$ points removed. Then, this is just a wedge sum of $\mathbb{S}^1$ $2n - 1$ times, and so the fundamental group is $F_{2n-1} = *_{2n-1}\mathbb{Z}$. □

Example 3.3.7

Let $X = \mathbb{R}^2$, $A_1 = \mathbb{R}^2 \setminus \{(1,0)\}$ and $A_2 = \mathbb{R}^2 \setminus \{(-1,0)\}$. We have that $x_0 = (0,0) \in A_1 \cap A_2$. Then, we have

$$\pi_1(A_1 \cap A_2, x_0) \cong \mathbb{Z} * \mathbb{Z}$$
$$\pi_1(A_1, x_0) \cong \pi_1(A_2, x_0) \cong \mathbb{Z}$$
$$\pi_1(\mathbb{R}^2, x_0) = 0.$$

Let $\omega = [\gamma] \in \pi_1(A_1 \cap A_2, x_0)$ be a loop that winds around $(1,0)$. The loop γ gives rise to different elements in each of the groups considered:

- $A_1 \cap A_2 \cong \mathbb{S}^1 \vee \mathbb{S}^1$ and the fundamental group $\pi_1(A_1 \cap A_2, x_0)$ is the free group on two generators a and b, with $[\gamma] = a$ one of them
- $A_1 \simeq \mathbb{S}^1$, and $(\iota_{A_1 A_2})_*([\gamma])$ is a generator of $\pi_1(A_1, x_0) \cong \mathbb{Z}$
- $A_2 \simeq \mathbb{S}^1$, but $(\iota_{A_2 A_1})_*([\gamma]) = e$, the constant loop in $\pi_1(A_2, x_0) \cong \mathbb{Z}$
- $A_1 \cup A_2 = \mathbb{R}^2$ and the image of γ under both $(\iota_{A_1})_* \circ (\iota_{A_1 A_2})_*$ and $(\iota_{A_2})_* \circ (\iota_{A_2 A_1})_*$ is the unit element in the trivial group $\pi_1(\mathbb{R}^2, x_0)$. In $\pi_1(A_1, x_0) * \pi_1(A_2, x_0)$, the concatenation of elements of $\pi_1(A_1, x_0)$ with elements of $\pi_1(A_2, x_0)$ does not reduce, unless at least one of these is the unit element. In our case:

$$(\iota_{A_1 A_2})_*([\gamma]) \cdot (\iota_{A_2 A_1})_*([\gamma])^{-1} = (\iota_{A_1 A_2})_*([\gamma]) e_{A_2} = (\iota_{A_1 A_2})_*([\gamma]).$$

§3.4 Application to CW-Complexes

As an application of the Seifert-Van Kampen theorem, we look at CW complexes. Here are some advanced results that we will refer to without proof:

- Any smooth manifold can be equipped with a CW complex structure.
- Any topological manifold with dimension greater than 4 can be equipped with a CW complex structure.

Definition 3.4.1. A **CW complex** is a topological space which is built inductively as follows:

1. the **zero skeleton** X^0, a discrete set
2. Given X^{n-1} and a collection of disk $\{D^n_a\}_a$ ($D^n_\alpha \cong \mathbb{D}^n$, $\mathbb{S}^{n-1}_\alpha = \partial D^n_\alpha$) with attaching maps:
$$\phi_\alpha \colon \mathbb{S}^{n-1}_\alpha \to X^{n-1}$$
Define $X^n = (X^{n-1} \sqcup (\sqcup_\alpha D^n_\alpha))/\sim$, quotienting by the equivalence relation $x \sim \phi_\alpha(x)$, for all $x \in \mathbb{S}^{n-1}_\alpha$.
3. Define $X = \cup_n X^n$ equipped with the weak topology, i.e., $U \subset X$ is open if and only if $U \cap X^n \subset X^n$ is open, for all n.

Definition 3.4.2. We have

- X^n is called an n-**skeleton**.

- D^n_α is called an n-**cell**.
- $e^n_\alpha = D^n_\alpha \setminus S^n_\alpha$ is called an **open** n-**cell**.

Remark 3.4.3 — We can have multiple CW-complex structure for the same topological space. There is a notion of minimal CW-complex structure, where minimal refers to a CW-complex structure with the minimal number of cells.

Definition 3.4.4. A CW complex is **finite-dimensional** if $X = X^n$, for some n. In this case, the largest n for which there are cells in the complex is the **dimension** of the complex.

Remark 3.4.5 — CW complexes need not to be finite.

Remark 3.4.6 — The dimension of a CW-complex is well-defined; it is not locally contractible, as it is not locally path connected.

Example 3.4.7

We could take $X^0 = \mathbb{Z}$ and the attaching maps defined by $\phi_n(0) = n$ and $\phi_n(1) = n+1$. The resulting CW complex is homeomorphic to $\mathbb{R}$.

Example 3.4.8

The torus has one 0-cell, two 1-cells, and one 2-cell, as we can see by *folding* a square.

Example 3.4.9

Consider $\mathbb{RP}^n = \mathbb{S}^n/(x \sim -x)$. We have that $\mathbb{RP}^0 = \{\text{point}\}$, and $\mathbb{RP}^n = \mathbb{RP}^{n-1} \sqcup \mathbb{D}^n/(x \sim \phi(x))$. That is, we are gluing the boundary of $\mathbb{D}^n$ to the $\mathbb{S}^{n-1}$ of $\mathbb{RP}^{n-1}$. Inductively, we get that

$$\mathbb{RP}^n = e^0 \cup e^1 \cup \cdots \cup e^n.$$

That is, it is a CW-complex with an open cell in each dimension.

Definition 3.4.10. A **subcomplex** of a CW complex X is the closure of a collection of open cells in X.

Example 3.4.11

The topologist's sine curve is not a CW complex.

Definition 3.4.12. A topological space is called **normal** if, for any two closed subsets, they have disjoint open neighborhoods.

Definition 3.4.13. A topological space is called **Hausdorff**, if there exists disjoint open neighborhoods for any two distinct points.

Here are some results that we won't prove.

Properties 3.4.14

We have:

1. CW complexes are normal (and hence Hausdorff).

2. CW complexes are locally contractible.

Proposition 3.4.15

Let X be a path connected topological space and for some n, let $\phi_\alpha^n \colon \mathbb{S}_\alpha^{n-1} \to X$ be some attaching maps. Define

$$Y = X \sqcup (\sqcup_\alpha D_\alpha^n)/(n \sim \phi_\alpha^n(x)).$$

Let $x_0 \in X$, then:

- For $n = 2$,
$$\pi_1(X, x_0)/N \cong \pi_1(Y, x_0),$$
where N is the normal subgroup generated by $[\gamma_\alpha]$, where $[\gamma_\alpha] \in \ker(\iota_*)$ and $\iota \colon X \hookrightarrow Y$. and ι_* is the induced map on the fundamental groups.

- For $n > 2$,
$$\pi_1(X, x_0) \cong \pi_1(Y, x_0).$$

Remark 3.4.16 — This says that attaching 2-cells could change the topology (for example, it could cover holes). However, attaching higher dimensional cells does not change anything.

Theorem 3.4.17

Let X be a path connect CW complex, with $x_0 \in X^2$. Then, the inclusion map $\iota \colon X^2 \to X$ induces an isomorphism

$$\pi_1(X^2, x_0) \cong \pi_1(X, x_0).$$

Example 3.4.18

We compute $\pi_1(\mathbb{RP}^2, x_0) \cong \mathbb{Z}/2$.

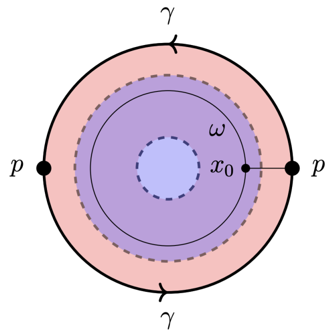
Solution. We note that we may consider the CW-complex structure on $\mathbb{RP}^2$ by having two e^0, two e^1, and one e^2 so that we can get our desired 'Northern hemisphere', which is exactly $\mathbb{RP}^2$. Then, consider this CW complex structure, but with a closed disk taken out of the interior (from e^2); call this second space A. We also consider a space B that is *like* a closed disk sitting inside $\mathbb{S}^1$, where the disk isn't touching $\mathbb{S}^1$. In particular, we let this closed disk in B and closed disk in A have a nonempty intersection. Now take a basepoint $x_0 \in A \cap B$. $A \cap B$ deformation retracts to a circle, so its fundamental group is $\mathbb{Z}$ and is generated by a loop represented by ω. B is contractible, so has trivial fundamental group, and A is homeomorphic to the Möbius strip (which can be seen if we *break apart* the A into half by adding in one cells to the center opening). Thus, to compute $\pi_1(\mathbb{RP}^2, x_0)$, we first compute $\pi_1(A, x_0) * \pi_1(B, x_0) \cong \mathbb{Z}$. We also need to compute the normal subgroup N:

$$N = \langle\langle\{(\iota_{A,B})_*[\omega](\iota_{B,A})_*[\omega]^{-1}\}\rangle\rangle = \langle[\omega]\rangle.$$

Since ω corresponds to the outer-circle of the Möbius strip and γ corresponds to its inner circle, we have that $\langle[\omega]\rangle = \langle[\gamma]^2\rangle$. Consequently,

$$\pi_1(\mathbb{RP}^2, x_0) \cong \langle[\gamma]\rangle / \langle[\gamma]^2\rangle \cong \mathbb{Z}/2\mathbb{Z}.$$

This is all summarized in the diagram below:



□

§3.5 Higher Homotopy Groups

Definition 3.5.1. The **homotopy groups** $\pi_n(X, x_0)$ $(n \geq 0)$ are groups of base points preserving homotopies of continuous maps $\phi\colon I^n \to X$ such that

$$\phi(\partial I^n) = x_0.$$

Example 3.5.2

For $n = 1$, this is the fundamental group.

Example 3.5.3

For $n = 0$, $\pi_0(X, x_0)$ is not a group, and it's just the set of path components of X.

Remark 3.5.4 — For $n \geq 2$, $\pi_n(X, x_0)$ are abelian.

Example 3.5.5

For $\mathbb{S}^2$, we know:

$$\pi_1(\mathbb{S}^2) = 0, \pi_2(\mathbb{S}^2) = \mathbb{Z}, \pi_3(\mathbb{S}^2) = \mathbb{Z}, \pi_4(\mathbb{S}^2) = \mathbb{Z}/2\mathbb{Z}, \pi_5(\mathbb{S}^2) = \mathbb{Z}/2\mathbb{Z}, \pi_6(\mathbb{S}^2) = \mathbb{Z}/12\mathbb{Z}.$$

Note that $\mathbb{S}^2$ has no cell in dimensions greater than 2, but we see that $\pi_n(\mathbb{S}^2)$ is nonzero for infinitely many n.

§3.6 Group Representations

Definition 3.6.1. A group G has a **group representation** $\langle S \mid R\rangle$, where S is a set of generators and R is a set of relations, if $G = \langle S\rangle/\langle R\rangle$.

Definition 3.6.2. A group $G = \langle S \mid R\rangle$ is **finitely generated** if $|S| < \infty$ and is **finitely presented** if $|S|, |R| < \infty$.

§3.6.i CW-Complexes and Group Representations

Here, we let X be a path-connected CW-complex, and be $X = X^2$. (note that the other cells make no difference for the π_1) Let $x_0 \in X$, and assume that $x_0 \in X^1$. We aim to compute $\pi_1(X, x_0)$.

Step 1: Consider X^1, and take a spanning tree (a tree that doesn't create a cycle) of X^1, call it T. We define

$$a := \{z \in X^1 \setminus T\}.$$

In other words, if we add in a, then we get the entirety of X^1. We claim (but check) that $\pi_1(X^1, x_0) \cong *_{\text{edges in } a}\mathbb{Z}$.

Step 2: Let $e^2_\alpha \subset X^2$ be a 2-cell, with an attaching map $\phi_\alpha\colon \mathbb{S}^1_\alpha \to X^1$. We can define a loop $\gamma_\alpha\colon I \to X^2$ given by

$$\gamma_\alpha(t) = \phi_\alpha(e^{2\pi i t}).$$

If $x_1 = \gamma_0(0) = x_0$, then $g_\alpha\colon I \to X^1$ with $g_\alpha(0) = x_0$ and $g_\alpha(1) = x_1$ (by path connectedness). We have a loop

$$\omega_\alpha = [g_\alpha \cdot \gamma_\alpha \cdot \overline{g}_\alpha] \in \pi_1(X^1, x_0).$$

Therefore, ω_α will correspond to a reduced word u_α in a.

Step 3: We now apply Seifert-Van Kampen. Let

$$A := \left(\cup_\alpha e^2_\alpha\right) \cup \{\text{paths from } x_0 \text{ to each cell}\}$$

and

$$B := X^2 \setminus \cup_\alpha\{y_\alpha\},$$

where y_α is a point in each cell. What B sort of *removes* the two-cells. Then, we get that

$$\pi_1(B, x_0) \cong \pi_1(X^1, x_0) \cong *_{\text{edges in } a}\mathbb{Z}.$$

Noting that as A is contractible, we have that $\pi_1(A, x_0) \cong 0$. Then, we get that

$$\pi_1(X, x_0) \cong \pi_1(A, x_0) * \pi_1(B, x_0)/\langle\langle U\rangle\rangle,$$

where we've modded out by loops that boundaries of 2-cells. ($\langle\langle U\rangle\rangle$ denotes the normal closure of U).

To summarize, we have:

- Every circle in X^1 corresponds to a loop based at x_0, which corresponds to a generator of $\pi_1(X, x_0)$.

- Every loop in X^2 can be represented as a combination of such cycle-paths along edges, these correspond to a reduced word in the generators of $\pi_1(X, x_0)$.
- A loop is null-homotopic if it is homotopic to the boundary of a 2-cell in X^2, such loops correspond to a relator on the set of words in $\pi_1(X^2, x_0)$.

Exercise 3.6.3. Motivate the following statements:

- Möbius strip: $\pi_1(M, x_0) \cong \langle ab, ac\rangle / \langle abac\rangle \cong \langle ab, ac \mid abac\rangle$.
- $\mathbb{RP}^2$: $\pi_1(\mathbb{RP}^2) \cong \langle a \mid a^2\rangle \cong \mathbb{Z}/2\mathbb{Z}$.
- Klein bottle: $\pi_1(K) \cong \langle a, b \mid aba^{-1}b\rangle$.

Theorem 3.6.4

For every group representation G, there exists a path connected two-dimensional CW complex X_G such that

$$\pi_1(X_G) \cong G.$$

Proof. Let $G = \langle S \mid R\rangle$. Construct the one-skeleton X_G as a wedge of circles with one circle for each generator. Every relation in R describes a loop in X^1G. For any relation, take a 2-cell with $\mathbb{S}^1_\alpha$. Define an attaching map $\phi_\alpha\colon \mathbb{S}^1_\alpha \to X^1$ that maps $\mathbb{S}^1_\alpha$ onto a loop specified by a relation. $\square$

Remark 3.6.5 — Given any group, there always exist some S, R so that $G = \langle S \mid R\rangle$. However, there is no general algorithm to compute the presentation for an arbitrary group.

4 Homology

§4.1 Simplicial Homology

Definition 4.1.1. We define a **standard k-simplex** as

$$\Delta_k = [e_0, \ldots, e_k] = \{(x_0, \ldots, x_k) \in \mathbb{R}^{k+1} \mid \sum_{i=0}^{k} x_i = 1, x_i \geq 0\},$$

where the e_i denote the standard basis vectors in $\mathbb{R}^{k+1}$.

We define in the non-standard way as

$$[v_0, \ldots, v_k] = \{\sum_{i=0}^{k} x_i v_i \mid \sum x_i = 1, x_i \geq 0\},$$

which is the convex hull of $v_0, \ldots, v_k$. We may denote this as $\operatorname{conv}\{v_0, \ldots, v_k\}$. We note that there exists a map $\sigma\colon \Delta_k \to [v_0, \ldots, v_k]$ by sending

$$(x_0, \ldots, x_k) \mapsto \sum_{i=0}^{k} x_i v_i.$$

Definition 4.1.2. We say that $v_0, \ldots, v_k$ are in **general position** if they do not lie in any $(k-1)$-dimensional affine subspace. Equivalently if, $v_k - v_{k-1}, \ldots, v_1 - v_0$ are linearly independent.

Example 4.1.3

v_0 and v_1 are in general position if they aren't the same point.

Example 4.1.4

v_0, v_1, v_2 are in general position if they aren't collinear.

Example 4.1.5

v_0, v_1, v_2, v_3 are in general position if they aren't coplanar.

Proposition 4.1.6

$v_0, \ldots, v_k$ are in general position if and only if $\sigma\colon \Delta^k \to [v_0, \ldots, v_k]$ is a homeomorphism.

Proof. ($\implies$): First, we note that σ is surjective and continuous. It suffices to show injectivity, since Δ^k is compact and $[v_0, \ldots, v_k]$ is Hausdorff. Let $(x_0, \ldots, x_k) \neq (x'_0, \ldots, x'_k)$ such that $\sum x_i v_i = \sum x'_i v_i$. Letting $y_i = x_i - x'_i$, we have that

$$\sum_{i=0}^{k} y_i v_i = 0.$$

Moreover, $y_0 = -y_1 - \cdots - y_k$, by the convex hull assumption ($\sum x_i = \sum x_i' = 1$). Therefore,

$$0 = \sum_{i=0}^{k} y_i v_i = y_1(v_1 - v_0) + \cdots + y_k(v_k - v_0).$$

This shows that $(v_1 - v_0), \ldots, (v_k - v_0)$ are linearly dependent, implying that $(v_1 - v_0), \ldots, (v_k - v_{k-1})$ is linearly dependent. This provides a contradiction, so we get injectivity.

($\Longleftarrow$): suppose $v_0, \ldots, v_k$ are not in general position. Then, $v_1 - v_0, \ldots, v_k - v_{k-1}$ are linearly dependent. This means that there exists $\alpha_0, \ldots, \alpha_{k-1}$ not all zero such that

$$\begin{aligned} 0 &= \alpha_0(v_1 - v_0) + \cdots + \alpha_{k-1}(v_k - v_{k-1}) \\ &= -\alpha_0 v_0 + \sum_{i=0}^{k-2}(\alpha_i - \alpha_{i+1})v_{i+1} + \alpha_{k-1}v_k. \end{aligned}$$

Then, $\sigma(-\alpha_0, \alpha_1 - \alpha_0, \ldots, \alpha_{k-1}) = 0$, so by injectivity, $\alpha_0 = \cdots = \alpha_{k-1} = 0$. $\square$

Definition 4.1.7. Let $[v_0, \ldots, v_n]$ be an n-simplex. If we remove one of $(n+1)$ vertices, and consider the convex hull of the remaining n vertices, we have a **face** of $[v_0, \ldots, v_n]$.

Notation 4.1.8. We write this as $[v_0, \ldots, \hat{v}_i, \ldots, v_n]$, where the $\hat{v}_i$ denotes the missing vertex.

Definition 4.1.9. The union of all $(k-1)$ dimensional faces is the **boundary** of the simplex. We write this as $\partial([v_0, \ldots, v_k]) = \cup_{i=0}^{k}[v_0, \ldots, \hat{v}_i, \ldots, v_k]$.

Definition 4.1.10. We define the **interior** of the simplex as $\mathrm{Int}(\Delta^k) = [v_0, \ldots, v_k] \setminus \partial([v_0, \ldots, v_k])$.

Definition 4.1.11. A **Δ-complex structure** on a space X is a collection of maps $\sigma_\alpha \colon \Delta^k \to X$ such that:

- $\sigma_\alpha |_{\mathrm{Int}(\Delta^k)} \colon \mathrm{Int}(\Delta^k) \to X$ must be injective and each point in X must be the image under exactly one of the σ_α.
- If σ_α is in the collection, then the restriction of σ_α to a face of Δ^k is also in the collection.
- $U \subset X$ is open if and only if $\sigma_\alpha^{-1}(U) \subseteq \Delta^k$ is open, for all α.

Example 4.1.12

Consider the torus, $\mathbb{S}^1 \times \mathbb{S}^1$. We put a Δ-complex structure on I^2 (the square used to construct the torus), and compare it with the quotient map $q \colon I^2 \to \mathbb{S}^1 \times \mathbb{S}^1$. We label the vertices v_0, v_1, v_2, v_3, and connect the vertices v_0 and v_2 by a 1-simplex.

- For 2-simplices, we have $q \circ [v_0, v_1, v_2]$ and $q \circ [v_0, v_3, v_1]$.
- For 1-simplices, we have $q \circ [v_0, v_1], q \circ [v_1, v_2], q \circ [v_0, v_2]$. (note that the quotient rules out $q \circ [v_0, v_3]$ and $q \circ [v_3, v_2]$).
- For 0-simplices, we have $q \circ [v_0]$. (note that the quotient rules out $q \circ [v_1], q \circ [v_2], q \circ [v_3]$)

Definition 4.1.13. Let X be a topological space with a Δ-complex structure. We define $\Delta_n(X)$ to be the free abelian group with basis the n-simplex from $\Delta_n(X)$;

$$\Delta_n(X) = \{\sum_\alpha m_\alpha \sigma_\alpha^n \mid m_\alpha \in \mathbb{Z} \text{ and finitely many are nonzero}\},$$

where σ_α^n denote the n-simplices from a Δ-simplex structure.

Definition 4.1.14. We define the **boundary operator** on $\Delta_n(X)$ as $\partial_n \colon \Delta_n(X) \to \Delta_{n-1}(X)$, with

$$\partial_n(\sigma_\alpha^n) = \sum_{j=0}^{n} (-1)^j (\sigma_\alpha^n |_{[e_0,\ldots,\hat{e}_j,\ldots,e_n]}).$$

Lemma 4.1.15

We have that $\partial_{n-1} \circ \partial_n = 0$.

Proof. This is just a computation. □

Definition 4.1.16. A **chain complex** of abelian groups is a diagram (C_*, ∂_*):

$$\ldots \longrightarrow C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \ldots \xrightarrow{\partial_1} C_0 \longrightarrow 0\ ,$$

such that $\partial_{n-1} \circ \partial_n = 0$, i.e., $\operatorname{Im}(\partial_n) \subseteq \ker(\partial_{n-1})$. Elements of C_n are called n**-chains** and ∂_n are called **boundary homomorphisms**.

The previous lemma tells us that we can form a **chain complex**:

$$\ldots \longrightarrow \Delta_n(X) \xrightarrow{\partial_n} \Delta_{n-1}(X) \xrightarrow{\partial_{n-1}} \ldots \xrightarrow{\partial_1} \Delta_0(X) \longrightarrow 0$$

Definition 4.1.17. The n**-th simplicial homology group** of X is the quotient $H_n^\Delta(X) = \ker(\partial_n)/\operatorname{Im}(\partial_{n+1})$.

Example 4.1.18

We have

$$H_n^\Delta(\mathbb{S}^1 \times \mathbb{S}^1) = \begin{cases} \mathbb{Z} & \text{if } n = 0, 2 \\ \mathbb{Z}^2 & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}.$$

Exercise 4.1.19. Compute the simplicial homology groups of the cylinder, the circle, the Möbius strip, and the Klein bottle.

§4.2 Singular Homology

There are two issues with the simplicial homology. Firstly, $\Delta_n(X)$ depends on a choice of Δ-complex structure. We want the simplicial homology groups to be independent of this choice! Secondly, the simplicial homology is not obviously functorial. That is, if $f \colon X \to Y$, whether the induced map $f_* \colon H_n^\Delta(X) \to H_n^\Delta(Y)$ satisfies $(id_X)_* = id_{H_n^\Delta(X)}$ and $(f \circ g)_* = f_* \circ g_*$.

Definition 4.2.1. A **singular** n**-simplex** in a space X is a continuous map $\sigma \colon \Delta^n \to X$.

Remark 4.2.2 — Here we are only requiring the map to be continuous, so we are not assuming that the image is nicely embedded. For example, it does not have to look like a simplex at all.

Definition 4.2.3. Given a topological space X, we define $C_n(X)$ to be the free abelian group with basis the set of singular n-simplices on X.

$$C_n(X) = \{\sum_\alpha m_\alpha \sigma_\alpha^n \mid m_\alpha \in \mathbb{Z}, \text{ finitely many nonzero}\},$$

where σ_α^n are singular n-simplices.

Remark 4.2.4 — Note that $C_n(X)$ is much bigger than $\Delta_n(X)$.

Definition 4.2.5. We define the **boundary operator** on $C_n(X)$ as $\partial_n \colon C_n(X) \to C_{n-1}(X)$ defined by

$$\partial_n(\sigma_\alpha^n) = \sum_{j=0}^{n} (-1)^j \sigma_\alpha^n \mid_{[e_0,\dots,\hat{e}_j,\dots,e_n]} .$$

Similar to before, we have that we can form a chain complex.

Lemma 4.2.6

$\partial_{n-1} \circ \partial_n = 0$.

The chain complex is as follows:

$$\dots \longrightarrow C_n(X) \xrightarrow{\partial_n} C_{n-1}(X) \xrightarrow{\partial_{n-1}} \dots \xrightarrow{\partial_1} C_0(X) \longrightarrow 0$$

Definition 4.2.7. The n**-th singular homology group** of X is defined as

$$H_n(X) = \ker(\partial_n)/(\partial_{n+1}).$$

Definition 4.2.8. $c \in Z_n(X) = \ker(\partial_n)$ is called a **cycle**.

Definition 4.2.9. $c \in B_n(X) = \operatorname{Im}(\partial_{n+1})$ is called a **boundary**.

Definition 4.2.10. $c \in C_n(X)$ is called a **chain**.

Remark 4.2.11 — Given this terminology, the homology focuses on cycles that are not in the boundary. Using this notation, we have $H_n(X) = Z_n/B_n$.

Definition 4.2.12. Given a continuous map of topological spaces $f \colon X \to Y$, we can define $f_\# \colon C_n(X) \to C_n(Y)$ by

$$f_\# \left(\sum_\alpha m_\alpha \sigma_\alpha^n \right) = \sum_\alpha m_\alpha (f \circ \sigma_\alpha^n).$$

Further, we may define its induced map $f_* \colon H_n(X) \to H_n(Y)$ as

$$c + B_n(X) \mapsto f_\#(c) + B_n(Y).$$

Notation 4.2.13. We write $[c] \in H_n(X)$, where we have a representative $c \in Z_n$, for the class of maps $c + B_n \in H_n(X)$.

Definition 4.2.14. We say that $c_1, c_2 \in [c] \in H_n(X)$ are **homologous** if $c_1 - c_2 \in B_n(X)$.

Lemma 4.2.15

We have that $\partial f_\# = f_\# \partial$.

Proof. Exercise. □

This shows that f induces a map of chain complexes, since commutativity shows that the squares in the following diagram commute.

$$\begin{tikzcd}
\dots \arrow[r] & C_{n+1}(X) \arrow[r, "\partial^X_{n+1}"] \arrow[d, "f_\#"] & C_n(X) \arrow[r, "\partial^X_n"] \arrow[d, "f_\#"] & C_{n-1}(X) \arrow[r] \arrow[d, "f_\#"] & \dots \\
\dots \arrow[r] & C_{n+1}(Y) \arrow[r, "\partial^Y_{n+1}"] & C_n(Y) \arrow[r, "\partial^Y_n"] & C_{n-1}(Y) \arrow[r] & \dots
\end{tikzcd}$$

As a consequence, we have that $f_\#(\ker(\partial^X_n)) \subset \ker(\partial^Y_n)$, since $\partial^Y(f_\# \alpha) = f_\#(\partial^X \alpha) = 0$ by linearity and commutativity. By the same reasoning, $f_\#(\mathrm{Im}(\partial^X_n)) \subseteq \mathrm{Im}(\partial^Y_n)$. Therefore, $f_\# \colon H_n(X) \to H_n(Y)$ that maps $c + B_n(X) \mapsto f_\#(c) + B_n(Y)$.

Lemma 4.2.16

$f_\#$ is well defined.

Proof. Let $c_1, c_2 \in \ker(\partial_n)$ such that $c_1 + B_n(X) = c_2 + B_n(X)$. Then,

$$\begin{aligned}
B_n(Y) &= f_*((c_1 - c_2) + B_n(X)) \\
&= f_\#(c_1 - c_2) + B_n(Y) \\
&= f_\#(c_1) - f_\#(c_2) + B_n(Y).
\end{aligned}$$

Then, we have that

$$f_\#(c_1) + B_n(Y) = f_\#(c_2) + B_n(Y).$$

Therefore,

$$f_*(c_1 + B_n(X)) = f_*(c_2 + B_n(X)).$$

□

Proposition 4.2.17

f_* is functorial.

Proof. Let $f \colon Y \to X, g \colon Z \to Y$. We want to show that $(f \circ g)_* = f_* \circ g_*$. Take any $c + B_n(Z) \in G_n(Z)$. Then,

$$\begin{aligned}
(f \circ g)_*(c + B_n(Z)) &= (f \circ g)_\#(c) + B_n(X) \\
(f_* \circ g_*)(c + B_n(Z)) &= f_\#(g_\#(c)) + B_n(X).
\end{aligned}$$

Take any $\sum_\alpha m_\alpha \sigma_\alpha \in C_n(Z)$. Then,

$$\begin{aligned}(f\circ g)_\#(\sum_\alpha m_\alpha\sigma_\alpha) &= \sum_\alpha m_\alpha((f\circ g)\circ\sigma_\alpha)\\ &= \sum_\alpha m_\alpha(f\circ g\circ\sigma_\alpha)\\ &= \sum_\alpha m_\alpha f_\#(g_\#(\sigma_\alpha))\\ &= (f_\#\circ g_\#)(\sum_\alpha m_\alpha\sigma_\alpha.\end{aligned}$$

Now take any $c + B_n(X) \in H_n(X)$. We want to show that $(id_X)_* = id_{H_n(X)}$. We have that

$$(id_X)_*(c + B_n(X)) = id_\#(c) + B_n(X) = c + B_n(X).$$

Then, $(id_X)_* = id_{H_n(X)}$. □

Corollary 4.2.18

If X and Y are homeomorphic, then $H_n(X) \cong H_n(Y)$, for all n.

Proof. Let $f\colon X \to Y$ be a homeomorphism, and let $g = f^{-1}$, which we know exists. Then, we have that

$$g_*\circ f_* = (g\circ f)_* = (id_X)_* = id_{H_n(X)}.$$

We have that same for the reverse composition, so we're done. □

Proposition 4.2.19

Let $X = \{\text{point}\}$. Then, $H_n(X) = 0$ for all $n > 0$ and $H_0(X) = \mathbb{Z}$.

Proof. For all n, there exists a unique $\sigma^n\colon \Delta^n \to \{\text{point}\}$, which is the constant map. Therefore, for all n, $C_n(X) = C_n(\{\text{point}\}) \cong \mathbb{Z}$. We now calculate the boundary. We have

$$\begin{aligned}\partial(\underbrace{m\cdot\sigma^n}_{\substack{\in C_n(X)\\ m\in\mathbb{Z}}}) &= \sum_{j=0}^n m(-1)^j\sigma^n\,|_{[e_0,\dots,\hat{e}_j,\dots,e_n]}\\ &= \sum_{j=0}^n m(-1)^j\sigma^{n-1}.\end{aligned}$$

Thus,

$$\partial_n = \begin{cases}0 & n \text{ odd}\\ m\sigma^{n-1} & n \text{ even}\end{cases}.$$

Thus, we have the sequence:

$$\dots \xrightarrow{id} \mathbb{Z} \xrightarrow{0} 0 \xrightarrow{id} \dots \longrightarrow \mathbb{Z} \longrightarrow 0$$

Therefore, we get the result. □

Proposition 4.2.20

X is path connected if and only if $H_0(X) \cong \mathbb{Z}$.

Proof. ($\implies$): Assume that X is path connected. Then, a map from the 0-simplex to X is just a choice of point. For all $x \in X$, $[x]$ is equal to the image of the map

$$\begin{aligned} \sigma \colon \Delta^0 &\to X \\ e_0 &\mapsto x. \end{aligned}$$

Since X is path connected, we have that for every two points $x, x' \in X$, there exists a path γ such that $\gamma(0) = x$, $\gamma(1) = x'$. This tells us that there exists $\sigma' \colon \Delta^1 \to X$, $[e_0, e_1] \mapsto [x, x']$ such that $\partial\sigma' = [x'] - [x]$. Now take any 0-chain in X, $\sum m_\alpha[x_\alpha]$ for $m_\alpha \in \mathbb{Z}$. Then, from what we just said, there exists σ_α^1 such that $\partial\sigma_\alpha^1 = [x_\alpha] - [x_0]$, fr some $x_0 \in X$. Therefore,

$$\sum m_\alpha[x_\alpha] - \sum m_\alpha[x_0] = \sum m_\alpha([x_\alpha] - [x_0]) = \sum m_\alpha \partial\sigma_\alpha^1 = \partial(\sum m_\alpha \sigma_\alpha^1).$$

Now, note that up to a boundary, any 0-chain is a multiple of $[x_0]$. We can then construct the isomorphism

$$\begin{aligned} H_0(X) = \underbrace{Z_0(X)/B_1(X)}_{=C_0(X)/B_1(X)} &\xrightarrow{\sim} \mathbb{Z} \\ m[x_0] &\mapsto m, \end{aligned}$$

and so it follows that $H_0(X) \cong \mathbb{Z}$.

($\impliedby$): If $H_0(X) \cong \mathbb{Z}$, then for all $[x_1], [x_0] \in C_0(X)$ such that there exists $\sigma^1 \in C_1(X)$ such that $\partial\sigma^1 = [x_1] - [x_0]$. σ^1 then provides a path from x_0 to x_1, so X is path connected. $\square$

Remark 4.2.21 — Let $X = \sqcup_i X_i$, where each X_i is a path connected component of X. Since simplices are path connected, their images must fall into an X_i. In particular, we may write

$$C_n(X) = \oplus_i C_n(X_i).$$

More importantly, we have that $\partial(C_n(X_i)) \subseteq C_{n-1}(X_i)$. By this, we may write that

$$H_n(X) \cong \oplus_i H_n(X_i).$$

Corollary 4.2.22

$H_0(X) \cong \mathbb{Z}^n$, where n is the number of path connected components.

§4.3 Reduced Homology Group

Definition 4.3.1. Let us define the **augmented chain complex**:

$$\dots \longrightarrow C_n(X) \xrightarrow{\partial_n} C_{n-1}(X) \xrightarrow{\partial_{n-1}} \dots \xrightarrow{\partial_1} C_0(X) \xrightarrow{\epsilon} \mathbb{Z} \longrightarrow 0\ ,$$

where we define $\epsilon \colon C_0(X) \to \mathbb{Z}$ as the map that sends $\sum n_i\sigma_i \mapsto \sum n_i$, where $n_i \in \mathbb{Z}$.

Definition 4.3.2. For a topological space X, the **reduced homology group** of X are

$$\tilde{H}_n(X) = \begin{cases} H_n(X) & n > 0 \\ \ker(\epsilon)/\mathrm{Im}(\partial_1) & n = 0 \end{cases}$$

Let us first check that $\tilde{H}_n(X)$ is well-defined; in particular, we just need to check that when $n = 0$, this definition makes sense, i.e., $\mathrm{Im}(\partial_1) \subset \ker(\epsilon)$: let $c = \sum m_\alpha \sigma_\alpha \in C_1(X)$. Then,

$$\begin{aligned} \epsilon(\partial c) &= \epsilon\left(\sum m_\alpha \sigma_\alpha \mid_{[e_1]} - \sum m_\alpha \sigma_\alpha^\alpha \mid_{[e_0]}\right) \\ &= \epsilon\left(\sum m_\alpha \sigma_\alpha \mid_{[e_1]}\right) - \epsilon\left(\sum m_\alpha \sigma_\alpha \mid_{[e_0]}\right) \\ &= \sum m_\alpha - \sum m_\alpha = 0. \end{aligned}$$

We now ask what the relationship between $H_n(X)$ and $\tilde{H}_n(X)$ is. We have the following diagram:

$$\begin{tikzcd} \ker(\epsilon) \arrow[r, hook] \arrow[d] & C_0(X) \arrow[r, "\epsilon"] \arrow[d] & \mathbb{Z} \arrow[d] \\ \tilde{H}_0(X) \arrow[r, hook] & H_0(X) \arrow[r, "\bar{\epsilon}"] & \mathbb{Z} \end{tikzcd}$$

Here the downwards arrows are by taking the respective quotient. Here, we have $\ker(\bar{\epsilon}) \cong \tilde{H}_0(X)$. This gives

$$H_0(X) \cong \mathbb{Z} \oplus \tilde{H}_0(X).$$

This is because the short exact sequence *splits*.

Definition 4.3.3. A short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

splits if $B \cong A \oplus C$.

It actually turns out that in the diagram above, the lower half is an exact sequence.

Example 4.3.4

$\tilde{H}_n(\{\text{point}\}) = 0$ for all n.

§4.3.i Some Homological Algebra

Definition 4.3.5. A sequence of abelian groups A_i and homomorphism ρ_i

$$\cdots \longrightarrow A_n \xrightarrow{\rho_n} A_{n-1} \xrightarrow{\rho_{n-1}} \cdots \xrightarrow{\rho_1} A_0 \longrightarrow 0$$

is **exact** if $\ker(\rho_i) = \mathrm{Im}(\rho_{i+1})$, for all i.

Remark 4.3.6 — Exactness implies that $\ker(\rho_n)/\mathrm{Im}(\rho_{i+1}) = 0$, for all i.

Remark 4.3.7 — if $\rho_{n+1} = 0$, then $\mathrm{Im}(\rho_{n+1}) = \ker(\rho_n) = 0$; thus, ρ_n is injective.

Remark 4.3.8 — If $\rho_n = 0$, then $\ker(\rho_n) = \mathrm{Im}(\rho_{n+1}) = A_n$, so ρ_{n+1} is injective.

Definition 4.3.9. A sequence of abelian groups and homomorphisms

$$0 \longrightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \longrightarrow 0$$

with α injective and β surjective, is called a **short exact sequence**.

Remark 4.3.10 — In fact, for all short exact sequences β induces an isomorphism

$$C \cong B/\mathrm{Im}(\alpha) \cong B/A.$$

Example 4.3.11

The following is a short exact sequence:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0$$

§4.3.ii Back to Some Topology

Definition 4.3.12. We say that (X, A) is a **good pair** if A is a nonempty closed subspace of X and a deformation retract of some neighborhood in X.

Example 4.3.13

$(\mathbb{D}^n, \partial\mathbb{D}^n)$ are a good pair.

Example 4.3.14

Let X be a CW complex, and A a subcomplex. Then, (X, A) is a good pair.

Theorem 4.3.15

Let (X, A) be a good pair. Then, there exists a long exact sequence

$$\begin{tikzcd}
\dots \arrow[r] & \tilde{H}_n(A) \arrow[r, "i_*"] & \tilde{H}_n(X) \arrow[r, "q_*"] & \tilde{H}_n(X/A) \arrow[dll, "\partial"] & \\
& \tilde{H}_{n-1}(A) \arrow[r, "i_*"] & \tilde{H}_{n-1}(X) \arrow[r, "q_*"] & \tilde{H}_{n-1}(X/A) \arrow[dll, dashed, "\partial"] & \\
& \tilde{H}_0(A) \arrow[r, "i_*"] & \tilde{H}_0(X) \arrow[r, "q_*"] & \tilde{H}_0(X/A) \arrow[r] & 0
\end{tikzcd}$$

Here, i_* is induced by $i\colon A \hookrightarrow X$, q_* is induced by the quotient $q\colon X \to X/A$. The map ∂ is called the **connecting homomorphism** which we'll define in the proof.

Example 4.3.16

We calculate the homology and reduced homology for the spheres using the good pair $(\mathbb{D}^n, \partial\mathbb{D}^n)$.

Solution. We note that $(\mathbb{D}^n, \partial\mathbb{D}^n)$ is a good pair. Then, $X \simeq Y$ implies that $H_n(X) \cong H_n(Y)$. Then,

$$H_*(\mathbb{D}^n) \cong H_*(\{\text{point}\}) = \begin{cases} \mathbb{Z} & n = 0 \\ 0 & \text{otherwise} \end{cases}.$$

Then, from the theorem we get the following chain:

$$\begin{tikzcd}
\dots \arrow[r] & \tilde{H}_k(\mathbb{S}^{n-1}) \arrow[r] & \tilde{H}_k(\mathbb{D}^n) \arrow[r, "q_*"] & \tilde{H}_k(\mathbb{S}^n) \arrow[dll] & \\
& \tilde{H}_{k-1}(\mathbb{S}^{n-1}) \arrow[r, "i_*"] & \tilde{H}_{k-1}(\mathbb{D}^n) \arrow[r, "q_*"] & \tilde{H}_{k-1}(\mathbb{S}^n) & \\
\dots & & & \dots & \\
\dots \arrow[r] & \tilde{H}_0(\mathbb{S}^{n-1}) \arrow[r, "i_*"] & \tilde{H}_0(\mathbb{D}^n) \arrow[r, "q_*"] & \tilde{H}_0(\mathbb{S}^n) \arrow[r] & 0
\end{tikzcd}$$

Here, we note that all the $\tilde{H}_i(\mathbb{D}^n) \cong 0$ as $\mathbb{D}^n$ is contractible. We have that $\mathrm{Im}(q_*) = \ker(\partial) = 0$, so ∂ is injective, meaning that

$$\mathrm{Im}(\partial) \cong \tilde{H}_k(\mathbb{S}^n).$$

Next, we have that $\mathrm{Im}(\partial) = \ker(i_*) = \tilde{H}_{k-1}(\mathbb{S}^{n-1})$, so

$$\tilde{H}_{k-1}(\mathbb{S}^{n-1}) \cong \tilde{H}_k(\mathbb{S}^n).$$

Now, we have that $\mathbb{S}^0 = \{\text{point}\} \sqcup \{\text{point}\}$, so

$$H_k(\mathbb{S}^0) = \begin{cases} 0 & k > 0 \\ \mathbb{Z} \oplus \mathbb{Z} & k = 0 \end{cases}$$

$$\tilde{H}_k(\mathbb{S}^0) = \begin{cases} 0 & k > 0 \\ \mathbb{Z} & k = 0 \end{cases}.$$

Inductively,

$$\mathbb{Z} \cong \tilde{H}_0(\mathbb{S}^0) \cong \tilde{H}_1(\mathbb{S}^1) \cong \ldots \cong \tilde{H}_n(\mathbb{S}^n)$$

and $\tilde{H}_k(\mathbb{S}^n) \cong 0$ if $k \neq n$. Thus,

$$H_k(\mathbb{S}^n) = \begin{cases} 0 & k \neq n \\ \mathbb{Z} & k = 0, n \end{cases}$$

$$\tilde{H}_k(\mathbb{S}^n) = \begin{cases} 0 & k \neq 0, n \\ \mathbb{Z} & k = n \end{cases}.$$

$\square$

Consider now the short exact sequence of chain complexes

$$\begin{tikzcd} 0 \arrow[r] & A_\bullet \arrow[r, "i"] & B_\bullet \arrow[r, "j"] & C_\bullet \arrow[r] & 0 \end{tikzcd},$$

where each $A_\bullet, B_\bullet, C_\bullet$ represents a bigger chain complex:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
\cdots \arrow[r] & A_{n+1} \arrow[r, "\partial^A_{n+1}"] \arrow[d, "i"] & A_n \arrow[r, "\partial^A_n"] \arrow[d, "i"] & A_{n-1} \arrow[r, "\partial^A_{n-1}"] \arrow[d, "i"] & \cdots \\
\cdots \arrow[r] & B_{n+1} \arrow[r, "\partial^B_{n+1}"] \arrow[d, "j"] & B_n \arrow[r, "\partial^B_n"] \arrow[d, "j"] & B_{n-1} \arrow[r, "\partial^B_{n-1}"] \arrow[d, "j"] & \cdots \\
\cdots \arrow[r] & C_{n+1} \arrow[r, "\partial^C_{n+1}"] \arrow[d] & C_n \arrow[r, "\partial^C_n"] \arrow[d] & C_{n-1} \arrow[r, "\partial^C_{n-1}"] \arrow[d] & \cdots \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

Theorem 4.3.17

Consider the setting above. Then there exists a long exact sequence of homology:

$$\begin{tikzcd}
\cdots \arrow[r] & H_n(A_\bullet) \arrow[r, "i_*"] & H_n(B_\bullet) \arrow[r, "j_*"] & H_n(C_\bullet) \arrow[dll, rounded corners, "\partial"] & \\
 & H_{n-1}(A_\bullet) \arrow[r, "i_*"] & H_{n-1}(B_\bullet) \arrow[r, "j_*"] & H_{n-1}(C_\bullet) \arrow[dlll, rounded corners, "\partial"] & \\
\cdots \arrow[r] & H_0(A_\bullet) \arrow[r, "i_*"] & H_0(B_\bullet) \arrow[r, "j_*"] & H_0(C_\bullet) \arrow[r] & 0
\end{tikzcd}$$

where ∂ is the connecting homomorphism (defined in the proof).

Proof. We diagram chase the above. There are six things to verify (which show that this is a long exact sequence); that us, $\mathrm{Im}(i_*) \subset \ker(j_*)$, $\mathrm{Im}(j_*) \subset \ker(\partial)$, $\mathrm{Im}(\partial) \subset \ker(i_*)$, $\ker(j_*) \subset \mathrm{Im}(i_*)$, $\ker(\partial) \subset \mathrm{Im}(j_*)$, $\ker(i_*) \subset \mathrm{Im}(\partial)$.

Firstly, we need to define the connecting homomorphism, ∂, and show that it is well defined (that is, it doesn't depend on a choice). Define $\partial\colon H_n(C_\bullet) \to H_{n-1}(A_\bullet)$ defined by $[c_n] \mapsto \partial[c_n]$. Choose a representative $c_n \in C_n$ such that $[c_n] \in H_n(C_\bullet)$. By surjectivity, there exists $b_n \in B_n$ such that $j(b_n) = c_n$. then, by commutativity $j\partial b_n = \partial(jb_n) = \partial(c_n) = 0$. Then, by exactness, $\partial b_n \in \ker(j) = \mathrm{Im}(i)$, i.e., $\exists a_{n-1} \in A_{n-1}$ such that $ia_{n-1} = \partial b_n$. Finally, by commutativity,

$$i(\partial a_{n-1}) = \partial i a_{n-1} = \partial\partial b_n = 0$$

By injectivity, $\partial a_{n-1} = 0$, so $[a_{n-1}] \in H_{n-1}(A_\bullet)$. Therefore, we define the **connecting homomorphism** as $\partial([c_n]) = [a_{n-1}]$. We move on to well defined-ness. We have made two choices in this construction. That is,

1. We've chosen b_n such that $jb_n = c_n$.

2. We have chosen a representative of $[c_n] \in H_n(C_\bullet)$.

Thus, to show that our definition of ∂ is well-defined, we must show that our choice of c_n and b_n do not matter. We start with another $b'_n \in B_n$ such that $j(b'_n) = c_n$. By construction, we can find $a'_{n-1} \in A_{n-1}$ such that $i(a'_{n-1}) = \partial b'_n$.

$$j(b_n) = j(b'_n) = c_n.$$

Thus, $j(b_n - b'_n) = c_n - c_n = 0$. Thus, $b_n - b'_n \in \ker(j) = \operatorname{Im}(i)$, Thus, there exists $a_n \in A_n$ such that $i(a_n) = b_n - b'_n$. Then,

$$i(\partial a_n) = \partial(ia_n) = \partial(b_n - b'_n) = ia_{n-1} - ia'_{n-1} = i(a_{n-1} - a'_{n-1}).$$

i is injective, so $\partial a_n = a_{n-1} - a'_{n-1}$. Specifically, $[a_{n-1}] = [a'_{n-1}]$, so we're done. Next, we show that our choice of c_n doesn't matter. Choose $c'_n \in [c_n]$. By construction, we have $b'_n \in B_n$ and $a'_{n-1} \in A_{n-1}$. Then, $i(a'_{n-1}) = \partial b'_n$ and $j(b'_n) = c'_n$. Since $c_n, c'_n \in [c_n]$, $c_n - c'_n = \partial c_{n+1}$. By surjectivity, $\exists b_{n+1} \in B_{n+1}$ such that $j(b_{n+1}) = c_{n+1}$. By commutativity, $j(\partial b_{n+1}) = \partial(jb_{n+1}) = \partial c_{n+1} = c_n - c'_n = j(b_n - b'_n)$. Thus,

$$j(\partial b_{n+1} - b_n + b'_n)) = 0.$$

By exactness, $\partial b_{n+1} - b_n + b'_n \in \ker(j) = \operatorname{Im}(i)$. Then, there exists $a_n \in A_n$ such that $i(a_n) = \partial b_{n+1} - (b_n - b'_n)$. By commutativity,

$$\begin{aligned} i(\partial a_n) = \partial(ia_n) = \partial(\partial b_{n+1} - (b_n - b'_n)) = -\partial b_n + \partial b'_n \\ = -ia_{n-1} + ia'_{n-1} = i(a'_{n-1} - a_{n-1}). \end{aligned}$$

Since i is injective, $\partial a_n = a'_{n-1} - a_{n-1}$, and we're done.

Finally, we show exactness. We need to show the following:

- $\operatorname{Im}(i_*) \subseteq \ker(j_*)$
- $\operatorname{Im}(j_*) \subseteq \ker(\partial)$
- $\operatorname{Im}(\partial) \subseteq \ker(i_*)$
- $\ker(j_*) \subseteq \operatorname{Im}(i_*)$
- $\ker(\partial) \subseteq \operatorname{Im}(j_*)$
- $\ker(i_*) \subseteq \operatorname{Im}(\partial)$

These all follow from diagram chasing. (Refer to Lee's Zigzag lemma for this) □

§4.4 Relative Homology Groups

Let (X, A) be a pair of of topological spaces, with $A \subseteq X$. Then, we define the *relative chains*

$$C_n(X, A) = C_n(X)/C_n(A).$$

Note that our boundary operator $\partial_n : C_n(X) \to C_{n-1}(X)$ sends $C_n(A)$ to $C_{n-1}(A)$. This induces a boundary operator:

$$\begin{aligned} \overline{\partial}_n : C_n(X, A) &\to C_{n-1}(X, A) \\ c + C_n(A) &\mapsto \partial_n(c) + C_{n-1}(A). \end{aligned}$$

Definition 4.4.1. Elements of $C_n(X, A)$ are called **relative chains**.

Exercise 4.4.2. Show that $\overline{\partial}_n$ well-defined and that $\overline{\partial}_{n-1} \circ \overline{\partial}_n = 0$.

By the previous exercise, we can define relative homology:

Definition 4.4.3. Given (X, A), we can define **relative homology groups**

$$H_n(X, A) = \ker(\overline{\partial}_n)/\mathrm{Im}(\overline{\partial}_{n+1}).$$

Elements of $\ker(\overline{\partial}_n)$ are called **relative cycles** and elements of $\mathrm{Im}(\overline{\partial}_{n+1})$ are called **relative boundaries**.

Example 4.4.4

Let $X = I \times I$ and $A = \{0\} \times I \cup \{1\} \times I$. Then, a horizontal line σ^1 is a chain in X, but a relative cycle in (X, A).

Remark 4.4.5 — Consider the sequence

$$\begin{tikzcd} 0 \arrow[r] & C_\bullet(A) \arrow[r, "i_\#"] & C_\bullet(X) \arrow[r, "j_\#"] & C_\bullet(X, A) \arrow[r] & 0, \end{tikzcd}$$

where $i \colon A \hookrightarrow X$. $i_\#$ is injective and $j_\#$ is the quotient, so it is surjective, so we have a short exact sequence of chain complexes. Therefore, we have a long exact sequence on homology:

$$\begin{tikzcd}
\ldots \arrow[r] & H_n(A) \arrow[r, "i_*"] & H_n(X) \arrow[r, "j_*"] & H_n(X, A) \arrow[dll, "\partial"] \\
 & H_{n-1}(A) \arrow[r, "i_*"] & H_{n-1}(X) \arrow[r, "j_*"] & H_{n-1}(X, A) \arrow[dlll, "\partial"] \\
\ldots \arrow[r] & H_0(A) \arrow[r, "i_*"] & H_0(X) \arrow[r, "j_*"] & H_0(X, A) \arrow[r] & 0
\end{tikzcd}$$

Remark 4.4.6 — There is a completely analogous long exact sequence of reduced homology groups for a pair (X, A) with $A \neq \emptyset$. This comes from applying the preceding machinery to the short exact sequence of chain complexes formed by the short exact sequences

$$\begin{tikzcd} 0 \arrow[r] & C_\bullet(A) \arrow[r] & C_\bullet(X) \arrow[r] & C_\bullet(X, A) \arrow[r] & 0, \end{tikzcd}$$

in nonnegative dimensions, augmented by the short exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \mathbb{Z} \arrow[r, "\mathbf{1}"] & \mathbb{Z} \arrow[r] & 0 \arrow[r] & 0 \end{tikzcd}$$

in dimension -1. In particular, this means $\tilde{H}_n(X, A)$ is the same as $H_n(X, A)$ for all n, when $A \neq \emptyset$.

Theorem 4.4.7

If $f, g \colon X \to Y$ are two maps such that $f \simeq g$, then $f_* = g_*$.

Proof. We have the homotopy $C\colon X \times I \to Y$, $f_0 = f$, $f_1 = g$. Further, take the singular simplex $\sigma\colon \Delta^n \to X$. We build $F \circ (\sigma \times id)\colon \Delta^n \times I \to x \times I \to Y$. We define the *prism operator* $P\colon C_n(X) \to C_{n+1}(Y)$ by

$$P(\sigma) = \sum_{i=0}^{n} (-1)^i (F \circ (\sigma \times id))\Big|_{[v_0,\dots,v_i,w_i,\dots,w_n]}.$$

Claim 4.4.8. $\partial P(\sigma) = g_\# \sigma - f_\# \sigma - P(\partial\sigma)$

Proof. We first note that $g_\#\sigma$ points to the 'top face' and that $f_\#\sigma$ points to the 'bottom face'. We have that

$$\begin{aligned}\partial P(\sigma) &= \partial\left(\sum (-1)^i F \circ (\sigma \times id)\Big|_{[v_0\dots v_i w_i \dots w_n]}\right)\\ &= \sum_{j\leq i} (-1)^i (-1)^j F \circ (\sigma \times id)\Big|_{[v_0,\dots,\hat{v_j},\dots,v_i,w_i,\dots,w_n]}\\ &\qquad \sum_{j \geq i} (-1)^i (-1)^{j+1} F \circ (\sigma \times id)\Big|_{[v_0,\dots,v_i,w_i,\dots,\hat{w_j},\dots,w_n]}.\end{aligned}$$

The terms with $i = j$ cancel except $F \circ (\sigma \times id)\,|_{[\hat{v_0},w_0,\dots,w_n]}$ and $F \circ (\sigma \times id)\,|_{[v_0,\dots,v_n,\hat{w_n}]}$ (this follows by computation). Note that the former is equal to $f_\#$ and the latter is equal to $g_\#$. Then,

$$\partial P(\sigma) = g_\#(\sigma) - f_\#(\sigma) + \sum_{i<j} \dots + \sum_{j<i} \dots .$$

Further, $P(\partial\sigma)$ remains to be the two summed terms at the end. □

By the claim, we can now finish off the proof. If $\partial P(c) = g_\# c - f_\# c - P(\partial c)$ for $c \in C_n(X)$, then if c is a cycle ($\partial c = 0$), then

$$\partial P(c) = g_\#(c) - f_\#(c).$$

Therefore, $[f_\#(c)] = f_*([c]) = g_*([c]) - [g_\#(c)]$, so $f_* = g_*$. □

Definition 4.4.9. Given morphisms of chain complexes $p, q\colon A_\bullet \to B_\bullet$, then they are **chain homotopic** if there exists an operator $P\colon A_n \to B_{n+1}$ such that $p - q = \partial P + P\partial$.

Proposition 4.4.10

Chain homotopic maps induce the same homomorphisms on homology.

Corollary 4.4.11

If $h\colon X \to Y$ is a homotopy equivalence, then $h_*\colon H_n(X) \to H_n(Y)$ is an isomorphism.

Proof. Let $j\colon Y \to X$ be the homotopy inverse of h. We have that $j \circ h \simeq id_X$ and $h \circ j \simeq id_Y$. Specifically, this gives that $(j \circ h)_* = (id_X)_*$ and $(h \circ j)_* = (id_Y)_*$. Further, by functoriality, we have

$$j_* \circ h_* = id_{H_n(X)}, \quad h_* \circ j_* = id_{H_n(Y)}.$$

□

Example 4.4.12

All contractible spaces have the same homology as a point. That is, $H_n(X) = 0$ for all $n > 0$ and $H_0(X) = \mathbb{Z}$.

§4.4.i Excision

Theorem 4.4.13 (Excision Theorem)

Given $Z \subset A \subset X$ such that $\overline{Z} \subset \operatorname{Int}(A)$, then the inclusion of pairs $(X \setminus Z, A \setminus Z) \to (X, A)$ induces an isomorphism of relative homology

$$H_n(X \setminus Z, A \setminus Z) \to H_n(X, A),$$

for all n.

Remark 4.4.14 — This is equivalent to the following statement: given a topological space X with $A, B \subseteq X$ such that $X = \operatorname{Int}(A) \cup \operatorname{Int}(B)$, then the inclusion of pairs

$$(B, A \cap B) \to (X, A)$$

gives an isomorphism on homology ($B = X \setminus Z$).

The proof of the theorem relies on the following proposition:

Proposition 4.4.15

Let $\mathcal{U} = \{U_\alpha\}_\alpha$ be a collection of subspaces of X whose interiors form an open cover of X. Now define the set

$$C_n^{\mathcal{U}} = \{n\text{-chains in } X \text{ where each simplex lies in some } U_\alpha\}.$$

Then, $i\colon C_\bullet^{\mathcal{U}}(X) \to C_\bullet(X)$ is a chain homotopy equivalence (i.e., $\exists s\colon C_\bullet(X) \to C_\bullet^{\mathcal{U}}(X)$ such that $i \circ s$ and $s \circ i$ are chain homotopic to the identities).

Proof. Refer to proposition 2.21 in Hatcher. □

Proof of Excision Theorem (4.4.13). Let A, B be such that $\operatorname{Int}(A) \cup \operatorname{Int}(B) = X$ and $\mathcal{U} = \{A, B\}$. Note that we have $A \cup B = X$. Note that we have $C_n^{\mathcal{U}}(X) = C_n(A) + C_n(B) \subseteq C_n(X)$ and that the inclusion $C_\bullet^{\mathcal{U}}(X) \hookrightarrow C_\bullet(X)$ give the map

$$C_\bullet^{\mathcal{U}}(X, A) = C_\bullet^{\mathcal{U}}(X)/C_\bullet^{\mathcal{U}}(A) \hookrightarrow C_\bullet^{\mathcal{U}}(X)/C_\bullet(A).$$

The inclusion is a chain homotopy equivalence

$$\begin{aligned} C_n(B, A\cap B) = C_n(B)/(C_n(A)\cap C_n(B)) &\cong (C_n(A)+C_n(B))/C_n(A) \\ &= C_n^{\mathcal{U}}(X)/C_n(A) = C_n^{\mathcal{U}}(X,A). \end{aligned}$$

Therefore, this induces an isomorphism on homology $H_n(B, A\cap B)\cong H_n(X,A)$. ☐

Lemma 4.4.16

Let X be a topological space with $x_0\in X$. Then,

$$H_n(X,\{x_0\})\cong \tilde{H}_n(X), \forall n.$$

Proof. We consider the L.E.S. of the pair $(X,\{x_0\})$. Then, we have the following:

$$\begin{tikzcd} \cdots \arrow[r] & \tilde{H}_n(\{x_0\}) \arrow[r, "i_*"] & \tilde{H}_n(X) \arrow[r, "j_*"] & H_n(X,\{x_0\}) \arrow[dll, "\partial"] \\ & \tilde{H}_{n-1}(\{x_0\}) \arrow[r, "i_*"] & \tilde{H}_{n-1}(X) \arrow[r, "j_*"] & H_{n-1}(X,\{x_0\}) \arrow[dlll, "\partial"] \\ \cdots \arrow[r] & \tilde{H}_0(\{x_0\}) \arrow[r, "i_*"] & \tilde{H}_0(X) \arrow[r, "j_*"] & H_0(X,\{x_0\}) \arrow[r] & 0 \end{tikzcd}$$

However, we know that $\tilde{H}_n(\{x_0\})=0$ for all n. Therefore, by exactness, we get that $H_n(X,\{x_0\})\cong\tilde{H}_n(X)$. ☐

Remark 4.4.17 — In the above, we are not requiring $(X,\{x_0\})$ to be a good pair.

Proposition 4.4.18

If (X,A) is a good pair, then the quotient map $q\colon (X,A)\to(X/A,A/A)$ induces an isomorphism on homology

$$H_n(X,A)\xrightarrow{\sim} H_n(X/A,A/A)\xrightarrow{\sim}\tilde{H}_n(X/A).$$

Proof. Let V be a neighborhood of A in X that deformation retracts onto A. We have a commutative diagram

$$\begin{tikzcd} H_n(X,A) \arrow[r, "1"] \arrow[d, "4"] & H_n(X,V) \arrow[r, "5"] \arrow[d] & H_n(X\setminus A, V\setminus A) \arrow[d, "3"] \\ H_n(X/A,A/A) \arrow[r, "2"] & H_n(X/A,V/A) \arrow[r, "6"] & H_n((X/A)\setminus(A/A),(V/A)\setminus(A/A)) \end{tikzcd}.$$

First note that the maps 5 and 6 are isomorphisms by the excision theorem. To prove the theorem we want to show that 4 (induced by the quotient map) is an isomorphism and to do so we show that $1,2,3$ are isomorphism and the rest follows by commutativity of the diagram (note that the diagram is commutative since it is induced by a commutative diagram of inclusions and quotient maps).

Consider the long exact sequence of the triple (X, V, A) induced by the following short exact sequence of complexes,

$$0 \longrightarrow C_\bullet(V, A) \xrightarrow{\phi} C_\bullet(X, A) \xrightarrow{\psi} C_\bullet(X, V) \longrightarrow 0.$$

We have that ϕ is injective since $C_\bullet(V) \to C_\bullet(X)$ is injective and we are quotienting by the same equivalence relation (so it is still an inclusion). On the other hand, ψ is surjective because we are quotienting by the same set $C_\bullet(X)$ by a stronger equivalence relation. This induces a long exact sequence on homology:

$$\begin{tikzcd}
\dots \arrow[r] & H_n(V, A) \arrow[r, "i_*"] & H_n(X, A) \arrow[r, "q_*"] & H_n(X, V) \arrow[dll, "\partial"] \\
& H_{n-1}(V, A) \arrow[r, "i_*"] & H_{n-1}(X, A) \arrow[r, "q_*"] & H_{n-1}(X, V) \arrow[dll, "\partial"] \\
\dots \arrow[r] & H_0(V, A) \arrow[r, "i_*"] & H_0(X, A) \arrow[r, "q_*"] & H_0(X, V) \arrow[r] & 0
\end{tikzcd}$$

Note that sine $i\colon A \to V$ is a homotopy equivalence, $H_n(V, A) \cong 0$ for all n. his can be seen by considering the long exact sequence on homology of the pair (V, A). Thus, 1 is an isomorphism. By the same argument, $H_n(V/A, A/A) \cong 0$ by looking at the long exact sequence for the triple $(X/A, V/A, A/A)$:

$$\begin{tikzcd}
\dots \arrow[r] & H_n(V/A, A/A) \arrow[r, "i_*"] & H_n(X/A, A/A) \arrow[r, "q_*"] & H_n(X/A, V/A) \arrow[dll, "\partial"] \\
& H_{n-1}(V/A, A/A) \arrow[r, "i_*"] & H_{n-1}(X/A, A/A) \arrow[r, "q_*"] & H_{n-1}(X/A, V/A) \arrow[dll, "\partial"] \\
\dots \arrow[r] & H_0(V/A, A/A) \arrow[r, "i_*"] & H_0(X/A, A/A) \arrow[r, "q_*"] & H_0(X/A, V/A) \arrow[r] & 0
\end{tikzcd}$$

Thus, we get that 2 is also an isomorphism. The quotient map $q\colon X \to X/A$ restricted to $X \setminus A$ and $V \setminus A$ is a homeomorphism (since no identification happens outside of A), so we have a homeomorphism of pairs

$$(X \setminus A, V \setminus A) \to (X/A \setminus A/A, V/A \setminus A/A),$$

which induces an isomorphism on homology, which is the map 3. This concludes the proof. □

Corollary 4.4.19

Suppose we have a collection of spaces $\{X_\alpha\}$ with $x_\alpha \in X_\alpha$ such that (X_α, x_α) is a good pair. Then, $i_\alpha\colon X_\alpha \to \bigvee_\alpha X_\alpha$ induce an isomorphism on homology

$$i_* = \oplus_\alpha i_\alpha\colon \oplus_\alpha \tilde{H}_n(X_\alpha) \to \tilde{H}_n(\bigvee_\alpha X_\alpha).$$

Proof. We have that $(X, A) = (\sqcup_\alpha X_\alpha, \sqcup_\alpha \{x_\alpha\})$. We also know that

$$H_n(X, A) \cong \tilde{H}_n(X/A) \cong \tilde{H}_n\left(\bigvee_\alpha X_\alpha\right).$$

Now,

$$H_n(X,A) = H_n(\sqcup_\alpha X_\alpha, \sqcup_\alpha\{x_\alpha\}) = \oplus_\alpha H_n(X_\alpha, \{x_\alpha\}) \cong \oplus_\alpha \tilde{H}_n(X_\alpha/\{x_\alpha\}) \cong \oplus_\alpha \tilde{H}_n(X_\alpha).$$

□

Applications of Excision

Theorem 4.4.20 (Brouwer Invariance of Domain)
If nonempty open sets $U \subset \mathbb{R}^m$ and $V \subset R^n$ are homeomorphic, then $m = n$.

Proof. Refer to Hatcher theorem 2.26. □

Example 4.4.21
We will show by induction on n that the identity map in $id_n \colon \Delta^n \to \Delta^n$, viewed as a singular n simplex, is a cycle generating $H_n(\Delta^n, \partial\Delta^n)$. That it is a cycle is clear since we are considering relative homology. When n $= 0$ it certainly represents a generator. For the induction step, let $\Lambda \subset \Delta^n$ be the union of all but one of the $(n-1)$ dimensional faces of Δ^n. Then we claim there are isomorphisms

$$H_n(\Delta^n, \partial\Delta^n) \xrightarrow{\sim} H_{n-1}(\partial\Delta^n, \Lambda) \xleftarrow{\sim} H_{n-1}(\Delta^{n-1}, \partial\Delta^{n-1})$$

The first isomorphism is a boundary map in the long exact sequence of the triple $(\Delta^n, \partial\Delta^n, \Lambda)$, whose third terms $H_i(\Delta^n, \Lambda)$ are zero since Δ^n deformation retracts onto Λ, hence $(\Delta^n, \Lambda) \cong (\Lambda, \Lambda)$. The second isomorphism is induced by the inclusion $i \colon \Delta^{n-1} \to \partial\Delta^n$ as the face not contained in Λ. When $n = 1$, i induces an isomorphism on relative homology since this is true already at the chain level. When $n > 1$, $\partial\Delta^{n-1}$ is nonempty so we are dealing with good pairs and i induces a homeomorphism of quotients $\Delta^{n-1}/\partial\Delta^{n-1} \cong \partial\Delta^n/\Lambda$. The induction step then follows since the cycle in is sent under the first isomorphism to the cycle ∂i_n which equals $\pm i_{n-1}$ in $C_{n-1}(\partial\Delta^n, \Lambda)$.

§4.5 Equivalence of Simplicial and Singular Homology

Theorem 4.5.1 (Five Lemma)
In a commutative diagram of abelian groups as below, if the two rows are exact and $\alpha, \beta, \delta, \epsilon$ are isomorphisms, then γ is also an isomorphism.

$$\begin{tikzcd}
\dots \arrow[r] & A \arrow[r] \arrow[d, "\alpha"] & B \arrow[r] \arrow[d, "\beta"] & C \arrow[r] \arrow[d, "\gamma"] & D \arrow[r] \arrow[d, "\delta"] & E \arrow[r] \arrow[d, "\epsilon"] & \dots \\
\dots \arrow[r] & A' \arrow[r] & B' \arrow[r] & C' \arrow[r] & D' \arrow[r] & E' \arrow[r] & \dots
\end{tikzcd}$$

Proof. Exercise. □

Remark 4.5.2 — Note that the above can be approached if we dice up the problem into two:

1. γ is surjective if β and δ are surjective and ϵ is injective.
2. γ is injective if β and δ are injective and α is surjective.

This means that we don't require α to be injective, nor do we require ϵ to be surjective.

Theorem 4.5.3

Let (X, A) be a pair of Δ-complex and subcomplex. Then, the inclusion of chain complexes $\Delta_\bullet(X, A) \subset C_\bullet(X, A)$ induces an isomorphism of homology groups

$$H_n^\Delta(X, A) \cong H_n(X, A).$$

Proof. First suppose $A = \emptyset$ and that X is finite dimensional. For X^k, the k skeleton of X, consisting of all simplices of dimension k or less, we have a commutative diagram of exact sequences:

$$\begin{tikzcd}
H_{n+1}^\Delta(X^k, X^{k+1}) \arrow[r] \arrow[d] & H_n^\Delta(X^{k-1}) \arrow[r] \arrow[d] & H_n^\Delta(X^k) \arrow[r] \arrow[d] & H_n^\Delta(X^k, X^{k-1}) \arrow[r] \arrow[d] & H_{n-1}^\Delta(X^{k-1}) \\
H_{n+1}(X^k, X^{k+1}) \arrow[r] & H_n(X^{k-1}) \arrow[r] & H_n(X^k) \arrow[r] & H_n(X^k, X^{k-1}) \arrow[r] & H_{n-1}(X^{k-1})
\end{tikzcd}$$

Let us first show that the first and fourth vertical maps are isomorphisms for all n. The simplicial chain group $\Delta_n(X^k, X^{k-1})$ is zero for $n \neq k$, and is free abelian with basis the k simplices of X when $n = k$. Hence $H_n^\Delta(X^k, X^{k-1})$ has exactly the same description. The corresponding singular homology groups $H_n(X^k, X^{k-1})$ can be computed by considering the map $\Phi\colon \sqcup_\alpha (\Delta_\alpha^k, \partial\Delta_\alpha^k) \to (X^k, X^{k-1})$ formed by the characteristic maps $\Delta^k \to X$ for all the k simplices of X. Since Φ induces a homeomorphism of quotient spaces $\sqcup_\alpha \Delta_\alpha^k / \partial\Delta_\alpha^k \cong X^k/X^{k-1}$, it induces isomorphisms on all singular homology groups. Thus $H_n(X^k, X^{k-1})$ is zero for $n \neq k$, while for $n = k$ this group is free abelian with basis represented by the relative cycles given by the characteristic maps of all the k simplices of X, in view of the fact that $H_k(\Delta^k, \partial\Delta^k)$ is generated by the identity map $\Delta^k \to \Delta^k$, as we showed in example 4.4.21. Therefore the map $H_n^\Delta(X^k) \to H_n(X^k)$ is an isomorphism.

By induction on k we may assume the second and fifth vertical maps in the preceding diagram are isomorphisms as well. The following frequently quoted basic algebraic lemma will then imply that the middle vertical map is an isomorphism, finishing the proof when X is finite-dimensional and $A = \emptyset$.

We next consider the case that X is infinite dimensional, where we will use the following fact: A compact set in X can meet only finitely many open simplices of X, that is, simplices with their proper faces deleted. This is a general fact about CW complexes proved in the appendix of Hatcher, but here is a direct proof for Δ complexes. If a compact set C intersected infinitely many open simplices, it would contain an infinite sequence of points xi each lying in a different open simplex. Then the sets $U_i = X \setminus \cup_{j \neq i}\{x_j\}$, which are open since their preimages under the characteristic maps of all the simplices are clearly open, form an open cover of C with no finite subcover.

This can be applied to show the map $H_n^\Delta(X) \to H_n(X)$ is surjective. Represent a given element of $H_n(X)$ by a singular n cycle Z. This is a linear combination of finitely many singular simplices with compact images, meeting only finitely many open simplices of X, hence contained in X^k for some k. We have shown that $H_n^\Delta(X^k) \to H_n(X^k)$ is an isomorphism, in particular surjective, so z is homologous in X^k (hence in X) to a simplicial cycle. This gives surjectivity. Injectivity is similar: If a simplicial n cycle z is the boundary of a singular chain in X, this chain has compact image and hence must lie in some X^k, so z represents an element of the kernel of $H_n^\Delta(X^k) \to H_n(X^k)$. But we know this map is injective, so z is a simplicial boundary in X^k, and therefore in X.

It remains to do the case of arbitrary X with $A \neq \emptyset$, but this follows from the absolute case by applying the five-lemma to the canonical map from the long exact sequence of simplicial homology groups for the pair (X, A) to the corresponding long exact sequence of singular homology groups. □

§4.6 Cellular Homology

§4.6.i Degree

Here is a motivating example.

Example 4.6.1

Note that we had $H_n(\mathbb{D}^n, \partial\mathbb{D}^n) \cong \tilde{H}_n(\mathbb{S}^n)$, as $\mathbb{D}^n/\partial\mathbb{D}^n \cong \mathbb{S}^n$. We now want to find explicit generators of each of the groups.

Solution. Note that we have a homeomorphism $(\Delta^n, \partial\Delta^n) \to (\mathbb{D}^n, \partial\mathbb{D}^n)$. Thus, we look at $H_n(\Delta^n, \partial\Delta^n)$ and aim to find a generator of this, as we have

$$H_n(\Delta^n, \partial\Delta^n) \cong \tilde{H}_n(\mathbb{S}^n) \cong \mathbb{Z}.$$

Now, we focus on the identity map $i_n \colon \Delta^n \to \Delta^n$ and an n-chain in $C_n(\Delta^n)$. The claim is that i_n is a relative cycle, and $[i_n]$ is the generator of $H_n(\Delta^n, \partial\Delta^n) \cong \mathbb{Z}$. We have

$$\partial(\mathrm{Int}(C_n(\partial\Delta^n))) = \partial\mathrm{Int}(C_{n-1}(\partial\Delta^n)) = C_{n-1}(\partial\Delta^n).$$

We now write $[i_n]$ as the homology class of i_n in $H_n(\Delta^n, \partial\Delta^n)$. We prove our claim via induction. For $n = 0$, we have that Δ^0 is just a point, and its boundary is empty. Therefore, $H_0(\Delta^0, \partial\Delta^0) \cong \mathbb{Z}$. Therefore, $i_0 \colon \{\text{point}\} \to \{\text{point}\}$ generates $H_0(\Delta^0)$ since it's the only choice. Now assume that $H_{n-1}(\Delta^{n-1}, \partial\Delta^{n-1})$ is generated by $[i_{n-1}]$. We note that we have

$$H_n(\Delta^n, \partial\Delta^n) \overset{(1)}{\cong} H_{n-1}(\partial\Delta^n, \Lambda) \overset{(2)}{\cong} H_{n-1}(\Delta^{n-1}, \partial\Delta^{n-1}),$$

where Λ denotes the union of all $(n-1)$ faces of Δ^n except one. We now prove the isomorphism (1): We have that the map $H_n(\Delta^n, \partial\Delta^n) \to H_{n-1}(\partial\Delta^n, \Lambda)$ is the connecting homomorphism in the long exact sequence of the triple $(\Delta^n, \partial\Delta^n, \Lambda)$. The short exact sequence associated to this long exact sequence is

$$0 \longrightarrow C_\bullet(\partial\Delta^n, \Lambda) \longrightarrow C_\bullet(\Delta^n, \Lambda) \longrightarrow C_\bullet(\Delta^n, \partial\Delta^n) \longrightarrow 0.$$

We note that the second map is injective because it is induction by the inclusion $\partial\Delta^n \hookrightarrow \Delta^n$. The third is surjective because we are quotienting by 'more stuff'. We now note that $H_k(\Delta^n, \Lambda) \cong 0$ for all k, as can be seen through the long exact sequence of (Δ^n, Λ).

$$\cdots \longrightarrow \tilde{H}_k(\Lambda) \longrightarrow \tilde{H}_k(\Delta^n) \longrightarrow H_k(\Delta^k, \Lambda) \longrightarrow \cdots$$

Then, Δ^n and Λ are contractible, so $\tilde{H}_k(\Lambda)$ and $\tilde{H}_k(\Delta^n)$ are 0 (note we used $\tilde{H}_k$ here to disregard what occurs at $k = 0$). Considering the long exact sequence

$$\cdots \longrightarrow H_n(\partial\Delta^n, \Lambda) \longrightarrow H_n(\Delta^n, \Lambda) \longrightarrow H_n(\Delta^n, \partial\Delta^n) \longrightarrow \cdots$$

Then, $H_k(\Delta^n, \Lambda)$ are all 0, so exactness then tells us that the map $H_n(\Delta^n, \partial\Delta^n) \to H_{n-1}(\partial\Delta^n, \Lambda)$ is an isomorphism.

We next show that (2) is an isomorphism. For this, we consider the morphism induced by the inclusion $i\colon \Delta^{n-1} \to \partial\Delta^n$. Then, we note that i induces a homeomorphism on the spaces $\Delta^{n-1}/\partial\Delta^{n-1} \cong \partial\Delta^n/\Lambda$, and so it follows that we have an induced isomorphism: $\tilde{H}_{n-1}(\Delta^{n-1}/\partial\Delta^{n-1}) \cong \tilde{H}_{n-1}(\partial\Delta^n/\Lambda)$, implying that $H_{n-1}(\Delta^{n-1}/\partial\Delta^{n-1}) \cong H_{n-1}(\partial\Delta^n/\Lambda)$. To summarize what we have, we have the following situation

$$\begin{array}{ccccc} H_n(\Delta^n, \partial\Delta^n) & \xrightarrow{\sim} & H_{n-1}(\partial\Delta^n, \Lambda) & \xrightarrow{\sim} & H_{n-1}(\Delta^{n-1}, \partial\Delta^{n-1}) \\ [i_n] & \longmapsto & \partial\,[i_n] & \longmapsto & \pm\,[i_{n-1}] \end{array}$$

Now, what is a generator of $\tilde{H}_n(\mathbb{S}^n) \cong \mathbb{Z}$? Now, we may consider $\mathbb{S}^n$ to be equal to $\Delta_1^n \cup \Delta_2^n/\sim$, where $\sim$ is the act of 'gluing along the boundary'. We now claim that $\Delta_1^n - \Delta_2^n$ is a generator. To prove this, we look at the long exact sequence $(\Delta_1^n \cup \Delta_2^n/\sim, \Delta_2^n)$. Explicitly, we have Δ_2^n is contractible, and that $\tilde{H}_k(\Delta_2^n) = 0$ for all k. Thus,

$$\tilde{H}_n(\Delta_1^n \cup \Delta_2^n/\sim) \cong H_n(\Delta_1^n \cup \Delta_2^n)/\sim, \Delta_2^n).$$

In fact, we have $H_n(\Delta_1^n \cup \Delta_2^n)/\sim, \Delta_2^n) \cong H_n(\Delta_1^n, \partial\Delta_1^n)$ because we have a homeomorphism $\Delta_1^n/\partial\Delta_1^n \cong (\Delta_1^n \cup \Delta_2^n)/\Delta_2^n$ (same reason as before). Now, we have

$$\begin{array}{ccccc} \tilde{H}_n(\Delta_1^n \cup \Delta_2^n/\sim) & \xrightarrow{\sim} & H_n(\Delta_1^n \cup \Delta_2^n/\sim, \Delta_2^n) & \xrightarrow{\sim} & H_n(\Delta_1^n, \partial\Delta_1^n) \\ [\Delta_1^n - \Delta_2^n] & \longmapsto & [\Delta_1^n] & \longmapsto & [i_n]\,. \end{array}$$

Thus, $[\Delta_1^n - \Delta_2^n]$ is a generator of $\tilde{H}_n(\mathbb{S}^n)$. □

As a corollary, we have the following.

Corollary 4.6.2

Given a CW complex X with $X = A \cup B$ for subcomplexes A, B, the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ gives an isomorphism on homology:

$$H_n(B, A \cap B) \cong H_n(X, A).$$

Proof. Suppose $A \cap B \neq \emptyset$. Then, because CW pairs are good pairs, we can do a similar argument as before to show that $B/A \cap B$ is homeomorphic to $A \cup B/A$, so it induces an isomorphism on homology. On the other hand, if $A \cap B = \emptyset$, then $X = A \sqcup B$, meaning that

$$C_n(X, A) = C_n(X)/C_n(A) = (C_n(A) \oplus C_n(B))/C_n(A) \cong C_n(B),$$

which concludes the proof. □

Definition 4.6.3. Given $f\colon \mathbb{S}^n \to \mathbb{S}^n$ $(n > 0)$, it induces a homomorphism on homology $f_*\colon H_n(\mathbb{S}^n) \to H_n(\mathbb{S}^n)$. Since $H_n(\mathbb{S}^n) \cong \mathbb{Z}$, then $f_*(\alpha) = d\alpha$ for some $d \in \mathbb{Z}$, which is defined to be the **degree** of f, denoted $\deg(f)$.

Here are some properties of the degree.

Properties 4.6.4 (Properties of the Degree)

Let $f\colon \mathbb{S}^n \to \mathbb{S}^n$.

1. $\deg(id_{\mathbb{S}^1}) = 1$.
2. If f is not surjective, then $\deg(f) = 0$.
3. If $f \simeq g$, then $\deg(f) = \deg(g)$, since $f_* = g_*$.
4. $\deg(f \circ g) = \deg(f) \cdot \deg(g)$, since $(f \circ g)_* = f_* \circ g_*$ (functoriality).
5. If f is a homotopy equivalence, $\exists g$ such that $f \circ g \simeq id$, so $\deg(f \circ g) = \deg(f) \cdot \deg(g) = 1$, so $\deg(f) = \pm 1$.
6. If f is a reflection in a hyperplane through the origin. Then, for a generator $\Delta_1^n - \Delta_2^n$, we get that
$$f_*([\Delta_1^n - \Delta_2^n]) = -[\Delta_1^n - \Delta_2^n] \implies \deg(f) = -1.$$
7. Let $a\colon \mathbb{S}^n \to \mathbb{S}^n$ be the antipodal map. Then, $\deg(a) = (-1)^{n+1}$.
8. If $f\colon \mathbb{S}^n \to \mathbb{S}^n$ has no fixed points, then f is homotopic to the antipodal map, implying that $\deg(f) = (-1)^{n+1}$.

Proof. 2. This is because there exists a point $x \notin \mathrm{Im}(f)$. Thus we get the following commutative diagram:

$$\begin{tikzcd}
0 \cong H_n(\mathbb{S}^n, \{x\}) \arrow[r, "i_*"] & H_n(\mathbb{S}^n) \\
H_n(\mathbb{S}^n) \arrow[u, "f_*"] \arrow[ur, "f_*"] &
\end{tikzcd}$$

In the above i_* is the induced map by the inclusion of $i\colon \mathbb{S}^n \setminus \{x\} \hookrightarrow \mathbb{S}^n$. Thus, the diagram tells us that f_* must be multiplication by 0.

7. This is because a is a composition of $n + 1$ reflections.

8. We can see this fact by constructing a homotopy

$$F(x, t) = f_t(x) = \frac{(1-t)f(x) - tx}{|(1-t)f(x) - tx|}.$$

Note that the bottom is only zero when $f(x) = \frac{t}{1-t}x$. The only time when the norms of these are equal is when $\frac{t}{1-t} = 1$, which is when $t = \frac{1}{2}$. $\square$

§4.6.ii Cellular Boundary Formula

The idea here is to look at pairs (X^n, X^{n-1}), (X^{n-1}, X^{n-2}), and so on. Here X^n represents an n-skeleton. Given a CW complex X, we can construct the **cellular chain complex** as follows.

$$\begin{tikzcd}
 & & H_n(X^n) \arrow[dr, "j_n"] & & & & H_{n-2}(X^{n-2}) \\
\cdots \arrow[r] & H_{n+1}(X^{n+1}, X^n) \arrow[ur, "\partial_{n+1}"] \arrow[rr, "d_{n+1}"] & & H_n(X^n, X^{n-1}) \arrow[dr, "\partial_n"] \arrow[rr, "d_n"] & & H_{n-1}(X^{n-1}, X^{n-2}) \arrow[ur] \arrow[r] & \cdots \\
H_{n+1}(X^{n+1}) \arrow[ur] & & & & H_{n-1}(X^{n-1}) \arrow[ur, "j_{n-1}"] & &
\end{tikzcd}$$

In the above, we have more arrows going in and out of $H_k(X^k)$ diagonally. The **cellular cell complex** is the horizontal line in the middle, i.e.,

$$\cdots \longrightarrow H_{n+1}(X^{n+1}, X^n) \xrightarrow{d_{n+1}} H_n(X^n, X^{n-1}) \xrightarrow{d_n} H_{n-1}(X^{n-1}, X^{n-2}) \xrightarrow{d_{n-1}} \cdots$$

Lemma 4.6.5

$d_n \circ d_{n+1} = 0$.

Proof. We have that $d_n \circ d_{n+1} = j_{n-1} \circ \partial_n \circ j_n \circ \partial_{n+1} = 0$. This is because the inner two maps are consecutive maps. $\square$

Thus, the complex above is exact. Now, shifting gears, note that we have

$$H_n(X^n, X^{n-1}) \cong \tilde{H}_n(X^n/X^{n-1}) \cong \tilde{H}_n\left(\vee_\alpha \mathbb{S}^n_\alpha\right) \cong \oplus_\alpha \mathbb{Z}.$$

where α are the n-cells of X. Therefore,

$$H_n(X^n, X^{n-1}) \cong \oplus_\alpha \mathbb{Z} = \mathbb{Z}^N,$$

where N is the number of n-cells from previous results.

Definition 4.6.6. We define the **cellular homology groups** of X as $H_n^{\mathrm{CW}}(X) = \ker(d_n)/\operatorname{Im}(d_{n+1})$.

Lemma 4.6.7

Let X be a CW complex. Then,

1. $H_k(X^n, X^{n-1}) = 0$ if $k \neq n$ and $H_n(X^n, X^{n-1}) \cong \mathbb{Z}^m$, where m is the number of n-cells of X.
2. $H_k(X^n) = 0$ for $k > n$.
3. For $k < n$, $X^n \hookrightarrow X$ induces an isomorphism $i_*\colon H_k(X^n) \to H_k(X)$.

Proof. (1): For $n > 0$, we have

$$H_k(X^n, X^{n-1}) = \oplus_\alpha \tilde{H}_k(\mathbb{S}^n_\alpha) = \begin{cases} 0 & k < n \\ \mathbb{Z}^m & k = n \end{cases}.$$

(2): We have a long exact sequence of the pair (X^n, X^{n-1}). Let $k > n$. Then, $H_{k+1}(X^n, X^{n-1})$ and $H_k(X^n, X^{n-1})$ are 0. Therefore, $H_k(X^{n-1}) \cong H_k(X^n)$ and $H_k(X^{n-1}) \cong H_k(X^{n-2})$, and continuing, we get

$$H_k(X^n) \cong \ldots \cong H_k(X^0) \cong .$$

(3) We have a long exact sequence of (X^{n+1}, X^n) for $k < n$. Now, noting that $H_{k+1}(X^{n+1}, X^n) \cong 0$ and $H_k(X^{n+1}, n) \cong 0$, we have that $H_k(X^n) \cong H_k(X^{n+1})$ for $k < n$. Continuing this, we get that

$$H_k(X^n) \cong H_k(X^{n+1}) \cong \ldots$$

Now note that if X is finite-dimensional, there must be some $N \in \mathbb{Z}$ such that $X = X^N$, so this comes to end end:

$$H_k(X^n) \cong H_k(X^{n+1}) \cong \ldots \cong H_k(X^N) = H_k(X).$$

Therefore, the inclusion $i_*\colon H_k(X^n) \to H_k(X)$ induces an isomorphism. Now, what if X is note finite dimensional? If X is infinite dimensional, a singular chain X will have a compact image, so will intersect only finitely many cells. Then, a k-cycle in X will correspond to a k-cycle in X^m for some m. (trust this fact) By the finite dimensional case, the k-cycles in X^m will be homologous to a k-cycle in X^n for $n > k$. That is, $H_k(X^n) \to H_k(X)$ is surjective. If a k-cycle in X^n bounds a chain in X, this chain will lie in some X^m for $m \in \mathbb{Z}$. By the finite dimensional argument, this is homologous to a chain in X^n, so the map $H_k(X^n) \to H_k(X)$ is injective. ∎

Theorem 4.6.8 (Equivalence of Cellular and Singular Homolody)
For a CW complex X, $H_n^{\mathrm{CW}}(X) \cong H_n(X)$.

Proof. By the third statement of the previous lemma, we know that $X^n \hookrightarrow X$ induces an isomorphism $H_k(X^n) \cong H - k(X)$ for $k < n$. In particular, $H_n(X^{n+1}) \cong H_n(X)$. We look at the long exact sequence of (X^{n+1}, X^n).

$$\ldots H_{n+1}(X^{n+1}, X^n) \xrightarrow{\partial_{n+1}} H_n(X^n) \xrightarrow{i_n} H_n(X^{n+1}) \xrightarrow{j_n} H_n(X^{n+1}, X^n) \longrightarrow \ldots$$

Since $H_n(X^{n+1}, X^n) \cong 0$, i_n is surjective: $\mathrm{Im}(i_n) = H_n(X^{n+1})$. Now, by the first isomorphism theorem,

$$H_n(X^{n+1}) = \mathrm{Im}(i_n) \cong H_n(X^n)/\ker(i_n) \cong H_n(X^n)/\mathrm{Im}(\partial_{n+1}).$$

Now we focus on the long exact sequence of the pair (X^n, X^{n-1}).

$$\ldots \longrightarrow H_n(X^{n-1}) \xrightarrow{i_n} H_n(X^n) \xrightarrow{j_n} H_n(X^n, X^{n-1}) \xrightarrow{\partial_n} \ldots$$

$H_n(X^{n-1}) \cong 0$, and so $\ker(j_n) = 0$ so it is injective. This injective then gives

$$H_n(X^n)/\mathrm{Im}(\partial_{n+1}) \cong j_n(H_n(X^n))/j_n(\mathrm{Im}(\partial_{n+1})).$$

Thus, $j_n(H_n(X^n)) = \ker(\partial_n)$ by exactness. We can also write $j_n(\mathrm{Im}(\partial_{n+1})) = \mathrm{Im}(j_n \circ \partial_{n+1}) = \mathrm{Im}(d_n)$ (in the definition of the cellular complex). By a similar argument, for (X^{n-1}, X^{n-2}), we have $\ker(\partial_n) = \ker(j_{n-1} \circ \partial_n) = \ker(d_n)$. Therefore,

$$\begin{aligned} H_n(X) &\cong H_n(X^n)/\mathrm{Im}(\partial_{n+1}) \cong j_n(H_n(X^n))/j_n(\mathrm{Im}(\partial_{n+1})) \\ &\cong \ker(j_[n-1] \circ \partial_n)/\mathrm{Im}(j_n \circ \partial_{n+1}) = \ker(d_n)/\mathrm{Im}(d_{n+1}) \cong H_n^{\mathrm{CW}}(X). \end{aligned}$$

□

Example 4.6.9

We can give $\mathbb{S}^n$ (for $n \geq 2$) a CW complex by one 0-cell and one n-cell. This tells us that

$$H_k^{\mathrm{CW}}(\mathbb{S}^n) \cong \begin{cases} \mathbb{Z} & k = n, 0 \\ 0 & \text{otherwise} \end{cases}$$

Example 4.6.10

We can give $\mathbb{CP}^n$ a CW complex by $e^0 \cup e^2 \cup \cdots \cup e^{2n}$. This tells us that

$$H_k^{\mathrm{CW}}(\mathbb{CP}^n) = \begin{cases} \mathbb{Z} & k \text{ even and } k \leq 2n \\ 0 & \text{otherwise} \end{cases}$$

We now consider how we may explicit express d_n. Recall that we have $H_n(X^n, X^{n-1}) \cong \mathbb{Z}^N$, where N denotes the number of n-cells in X. We have $d_n \colon H_n(X^n, X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$. $X^n/X^{n-1} = \vee_\alpha \mathbb{S}^n_\alpha$, where α is indexing the n-cells, and $X^{n-1}/X^{n-2} = \vee_\beta \mathbb{S}^{n-1}_\beta$, where β is indexing the $n-1$-cells. Then,

$$d_n(e^n_\alpha) = \sum_\beta d_{\alpha\beta} e^{n-1}_\beta.$$

Proposition 4.6.11 (Cellular Boundary Formula)

We have $d_n(e^n_\alpha) = \sum\limits_\beta d_{\alpha,\beta} e^{n-1}_\beta$, where

$$d_{\alpha,\beta} = \deg(\mathbb{S}^{n-1}_\alpha \to X^{n-1} \to \mathbb{S}^{n-1}_\beta \simeq X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta)).$$

In the above, the first map is the attaching map ϕ^n_α of e^n_α and the second map is a quotient map collapsing $X^{n-1} \setminus e^{n-1}_\beta$.

Proof.

$$\begin{tikzcd} H_n(\mathbb{D}^n_\alpha, \partial\mathbb{D}^n_\alpha) \arrow[r, "\partial", "\cong"'] \arrow[d, "(\Phi^n_\alpha)_*"] & \tilde{H}_{n-1}(\mathbb{D}^n_\alpha) \arrow[r, "(\Delta_{\alpha\beta})_*"] \arrow[d, "(\phi^n_\alpha)_*"] & \tilde{H}_{n-1}(\mathbb{S}_\beta)^{n-1} \cong \mathbb{Z} \\ H_n(X^n, X^{n-1}) \arrow[r, "\partial_n"] & \tilde{H}_{n-1}(X^{n-1}) \arrow[r, "j_{n-1}"] & \tilde{H}_{n-1}(X^{n-1}, X^{n-1}) \cong \mathbb{Z} \arrow[u, "P_r"'] \end{tikzcd}$$

Here, we have:

- ∂ is the connecting homomorphism in the long exact sequence $(\mathbb{D}^n_\alpha, \partial\mathbb{D}^n_\alpha)$ and $\mathbb{D}^n_\alpha$ is contractible, so is an isomorphism.
- ∂_n is the connecting homomorphism in the long exact sequence (X^n, X^{n-1}).
- j_{n-1} comes from the long exact sequence of the pair (X^{n-1}, X^{n-2}).
- P_r is projecting onto on of the summands:

$$H_{n-1}(X^{n-1}, X^{n-2}) \cong \tilde{H}_{n-1}(X^{n-1}/X^{n-2}) \cong \bigoplus_\beta \tilde{H}_{n-1}(\mathbb{S}^{n-1}_\beta)$$

- $(\phi^n_\alpha)_*$ is the induced map by the attaching map ϕ^n_α.
- $(\Delta_{\alpha\beta})_*$ is induced by $\Delta_{\alpha\beta} \colon \partial\mathbb{D}^n_\alpha \to \mathbb{S}^{n-1}_\beta$.
- $(\Phi^n_\alpha)_*$is an inclusion

$$H_n(X^n, X^{n-1}) \cong \tilde{H}_n(X^n/X^{n-1}) \cong \bigoplus_\beta \tilde{H}_n(\mathbb{D}^n_\beta/\partial\mathbb{D}^n_\beta) \cong \bigoplus_\beta H_n(\mathbb{D}^n_\beta, \partial\mathbb{D}^n_\beta).$$

Given this, if $[\mathbb{D}^n_\alpha]$ is a generator of $H_n(\mathbb{D}^n_\alpha, \partial\mathbb{D}^n_\alpha)$, then it gets included in $H_n(X^n, X^{n-1})$. Then, we apply $\partial_n([\mathbb{D}^n_\alpha])$ and then project it to one of the summands. □

Example 4.6.12

Let us compute the homology of the torus again.

Solution. We have:

- X^0: one 0-cell, e^0_1
- X^1: two 1-cells, a, b
- X^2: one 2-cell, e^2_1

We have the sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^2 \xrightarrow{d_1} \mathbb{Z} \xrightarrow{0} 0$$

and we want to compute d_1 and d_2. We have that $d_1 \colon H_1(X^1, X^0) \to H_0(X^0, \emptyset) = H_0(X^0)$. The connecting morphism in the long exact sequence (X^1, X^0) satisfies $\partial a = \partial b = 0$. We now want to calculate d_2. We have

$$d_2(e^2_1) = d_a a + d_b b,$$

where $d_a = \deg(\mathbb{S}^1 \to X = \mathbb{S}^1 \vee \mathbb{S}^1 \to X^1/(X^1 \setminus a) \cong a)$ and $d_b = \deg(\mathbb{S}^1 \to X = \mathbb{S}^1 \vee \mathbb{S}^1 \to X^1/(X^1 \setminus b) \cong b)$. In this, each of the $\mathbb{S}^1$ are actually equal to $\mathbb{S}^1_\alpha = \partial e^2_1$. Then, it follows that $d_a = d_b = 0$, and so $d_2 = 0$, Therefore,

$$H_n(\mathbb{T}^2) = \begin{cases} \mathbb{Z} & n = 0, 2 \\ \mathbb{Z}^2 & n = 1 \\ 0 & \text{otherwise} \end{cases}.$$

□

Example 4.6.13

Let us compute the homology of $\mathbb{RP}^2$.

Proof. Our cellular structure is

- X^0: e_1^0, e_2^0
- X^1: a, b
- X^2: e_1^2

Our chain complex is then

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^2 \xrightarrow{d_1} \mathbb{Z} \xrightarrow{2} 0$$

and we want to understand d_1 and d_2. $d_1(a) = e_2^0 - e_1^0$ and $d_1(b) = e_1^0 - e_2^0$. Thus,

$$d_1 = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$d_2(e_1^2) = 2a + 2b$, since going around $\mathbb{S}^1$ once will go through a and b twice. Thus,

$$H_n(\mathbb{RP}^2) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}/2\mathbb{Z} & n = 1 \\ 0 & \text{otherwise} \end{cases}.$$

□

Example 4.6.14

Let us (partially) compute the homology of $\mathbb{RP}^n$.

Solution. Our CW structure is $e^0 \cup e^1 \cup e^2 \cup \cdots \cup e^n$, which comes from $\mathbb{RP}^n = \mathbb{D}^n \cup \mathbb{RP}^{n-1}/\sim$. Our chain complex is then

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_n} \mathbb{Z} \xrightarrow{d_{n-1}} \dots \xrightarrow{d_1} \mathbb{Z} \xrightarrow{d_0} 0$$

Though our original definition of the real projective space is $(\mathbb{R}^{n+1} \setminus 0)/x \sim \lambda x$, we may see that this is equivalent to $\mathbb{S}^n/x \sim -x$. Then, we may denote this to be equal to $\mathbb{S}^n_{\geq 0}$ (the 'upper half') quotiented by $x \sim -x$ on $\partial\mathbb{S}^n \geq 0$. This may be seen to be equal to $\mathbb{D}^n/x \sim -x$ (where the quotient is on $\partial\mathbb{D}^n$). Then, consider

$$\Delta_n \colon \partial\mathbb{D}^n \to X^{n-1}/(X^{n-1} \setminus e^{n-1}) \cong \mathbb{RP}^{n-1}/(\mathbb{RP}^{n-1} \setminus e^{n-1}) \cong \mathbb{RP}^{n-2} \cup e^{n-1}/\mathbb{RP}^{n-2}.$$

The cellular boundary formula provides $d_n(e_\alpha^n) = \sum d_{\alpha\beta} e_\beta^{n-1}$. $\mathbb{RP}^n = \mathbb{D}^n/x \sim -x$ on $\partial\mathbb{D}^n$, where the equivalence relation on $\partial\mathbb{D}^n$ is invariant under an antipodal map $a \colon \mathbb{S}^n \to \mathbb{S}^n$ given by $x \mapsto -x$. Further, Δ_n is invariant under the antipodal map: $\Delta_n = \Delta_n \circ a$. Then,

$$\deg(\Delta_n) = \deg(\Delta_n \circ a) = \deg(\Delta_n)\deg(a) = \deg(\Delta_n) \cdot (-1)^n.$$

It follows that

$$\deg(\Delta_n) = \begin{cases} \deg(\Delta_n) & n \text{ even} \\ -\deg(\Delta_n) & n \text{ odd} \end{cases},$$

and so if n is even, this doesn't say anything but when n is odd, this means that $\deg(\Delta_n) = 0$. Thus, we figure out more information on our chain:

$$\begin{tikzcd} 0 \arrow[r] & \mathbb{Z} \arrow[r, "0"] & \mathbb{Z} \arrow[r, "?"] & \dots \arrow[r, "?"] & \mathbb{Z} \arrow[r, "0"] & \mathbb{Z} \arrow[r] & 0 \end{tikzcd}$$

The maps with '?' will be filled in via local degree calculation, which is the main subject now. □

§4.6.iii Local Degree

Definition 4.6.15. Let $f\colon \mathbb{S}^n \to \mathbb{S}^n$ for $n > 0$. Take $x \in \mathbb{S}^n$ and $y = f(x)$. We say that x is **isolated** in $f^{-1}(y)$ if there exists a neighborhood of x such that $U \cap f^{-1}(y) = \{x\}$.

Now let x be isolated, and $y = f(x)$. We choose neighborhoods U of x and V of y such that $f(U) \subset V$ and $f^{-1}(y) \cap U = \{x\}$. We notice that we have that map $f\colon (U, U \setminus \{x\}) \to (V, V \setminus \{y\})$ induces a map on homology.

$$\begin{tikzcd} H_n(U, U \setminus \{x\}) \arrow[r, "\cong"] \arrow[d, "f_*"] & H_n(\mathbb{S}^n, \mathbb{S}^n \setminus \{x\}) \arrow[r, "\cong"] \arrow[d] & \tilde{H}_n(\mathbb{S}^n) \arrow[r, phantom, "\cong"] \arrow[d] & \mathbb{Z} \arrow[d, "\times N"] \\ H_n(V, V \setminus \{u\}) \arrow[r, "\cong"] & H_n(\mathbb{S}^n, \mathbb{S}^n \setminus \{y\}) \arrow[r, "\cong"] & \tilde{H}_n(\mathbb{S}^n) \arrow[r, phantom, "\cong"] & \mathbb{Z} \end{tikzcd}$$

The maps $H_n(U, U \setminus \{x\}) \to H_n(\mathbb{S}^n, \mathbb{S}^n \setminus \{x\})$ and $H_n(V, V \setminus \{y\}) \to H_n(\mathbb{S}^n, \mathbb{S}^n \setminus \{y\})$ are given by excision. Further, the maps going right from $H_n(\mathbb{S}^n, \mathbb{S}^n \setminus \{x\})$ and $H_n(\mathbb{S}^n, \mathbb{S}^n \setminus \{y\})$ are given by the long exact sequences. The multiplication by an integer N we are referring to is defined to be the **local degree** of f at x, denoted $\deg(f)_x$. We now ask whether this integer is well defined or not. Let $x \in U, U'$ and $y \in V, V'$. Then, we have

$$\begin{tikzcd} \mathbb{Z} \arrow[r, phantom, "\cong"] \arrow[d, "\times N_2"'] & H_n(U, U \setminus x) \arrow[r, "\cong"] \arrow[d] & H_n(\mathbb{S}^n, \mathbb{S}^n \setminus x) & H_n(U', U' \setminus x) \arrow[l, "\cong"'] \arrow[r, phantom, "\cong"] \arrow[d] & \mathbb{Z} \arrow[d, "\times N_1"] \\ \mathbb{Z} \arrow[r, phantom, "\cong"] & H_n(U, V \setminus y) \arrow[r, "\cong"] & H_n(\mathbb{S}^n, \mathbb{S}^n \setminus x) & H_n(V', V' \setminus y) \arrow[l, "\cong"'] \arrow[r, phantom, "\cong"] & \mathbb{Z} \end{tikzcd}$$

We also have isomorphisms from $\mathbb{Z}$ to $H_n(\mathbb{S}^n, \mathbb{S}^n \setminus x)$ (on the top) and $\mathbb{Z}$ to $H_n(\mathbb{S}^n, \mathbb{S}^n \setminus y)$ (on the bottom), so this implies that $N_1 = N_2$ and the local degree is well defined. All the isomorphisms are canonical.

We now ask what a generator of $H_n(U, U \setminus x)$ look like? Define $\Delta \subseteq U$ a geometric n-simplex and consider $x \in \mathrm{Int}(\Delta) \subset \Delta \subset U$. We note that $\sigma\colon \Delta^n \to \Delta$ is a homeomorphism.

Claim 4.6.16. σ is a generator of $H_n(U, U \setminus x)$.

Proof. σ is a cycle in $C_n(U, U \setminus x)$, and so $\partial\sigma \in C_{n-1}(U \setminus x)$. Therefore, $\partial\sigma$ is trivial in the relative chain complex. Writing $|\Delta|$ as the image of σ, by excision, $Z = U \setminus |\Delta|$, then

$$H_n(|\Delta|, |\Delta| \setminus x) \cong H_n(U, U \setminus x).$$

Then, $\sigma\colon \Delta^n \to |\Delta|$ is a homeomorphism, so

$$H_n(\Delta^n, \Delta^n \setminus \text{point}) \cong H_n(|\Delta|, |\Delta| \setminus x),$$

but $H_n(\Delta^n, \Delta^n \setminus \text{point}) \cong H_n(\Delta^n, \partial\Delta^n)$ and $[i_n]$ is a generator of $H_n(\Delta^n, \partial\Delta^n)$. $[i_n]$ maps to $[\sigma]$, and so $[\sigma]$ is a generator of $H_n(|\Delta|, |\Delta| \setminus x)$. □

Definition 4.6.17. For any space X we define $H_n(X, X \setminus \{x\})$ the **local homology groups** of X at x.

Proposition 4.6.18

Let $f\colon \mathbb{S}^n \to \mathbb{S}^n$ and let $f^{-1}(y) = \{x_1, \dots, x_m\}$ for some positive integer m. Then,

$$\deg(f) = \sum_{i=1}^{m} \deg(f)_{x_i}$$

Proof. Let U_i be a neighborhood of x_i, where $U_i \cap U_j = \emptyset$ for $i \neq j$. Pick $y \in V$ such that $f\mid_{(U_i \setminus x_i)} \subset V \setminus y$, for all i. Then,

$$(\bigcup_{i=1}^{m} U_i, \bigcup_{i=1}^{m} U_i \setminus \{x_i\}) \hookrightarrow (\mathbb{S}^n, \mathbb{S}^n \setminus \{x_1, \dots, x_m\})$$

induces an isomorphism on relative homology via excision ($Z = \mathbb{S}^n \setminus \cup_{i=1}^{m} U_i$). Thus,

$$\begin{aligned} H_n(\mathbb{S}^n, \mathbb{S}^n \setminus \{x_1, \dots, x_m\}) &\cong H - n(\bigcup_{i=1}^{m} U_i, \bigcup_{i=1}^{m} U_i \setminus \{x_i\}) \\ &\cong \bigoplus_{i=1}^{m} H_n(U_i, U_i \setminus \{x_i\}) \\ &\cong \bigoplus_{i=1}^{m} H_n(\mathbb{S}^n, \mathbb{S}^n \setminus \{x_i\}) \\ &\cong \bigoplus_{i=1}^{m} \tilde{H}_n(\mathbb{S}^n) \\ &\cong \mathbb{Z}. \end{aligned}$$

The third to last congruence is via excision, and the second to last is by the long exact sequence, and since $\mathbb{S}^n \setminus \{x_i\}$ is contractible. For each i, we have

$$\begin{tikzcd}
 & H_n(U_i, U_i \setminus x_i) \arrow[r, "f_*", "1 \mapsto \deg(f)_{x_i}"'] \arrow[dl, "\cong"'] \arrow[d, "k_i"] & H_n(V, V \setminus y) \arrow[d, "\cong"', "\alpha"] \\
H_n(\mathbb{S}^n, \mathbb{S}^n \setminus x_i) & H_n(\mathbb{S}^n, \mathbb{S}^n \setminus \{x_i, \dots, x_m\}) \arrow[l, "P_i"'] \arrow[r, "f_*"] & H_n(\mathbb{S}^n, \mathbb{S}^n \setminus y) \\
 & \tilde{H}_n(\mathbb{S}^n) \arrow[r, "f_*", "1 \mapsto \deg(f)"'] \arrow[ul, "\cong"] \arrow[u, "j"'] & \tilde{H}_n(\mathbb{S}^n) \arrow[u, "\cong", "\beta"']
\end{tikzcd}$$

Here, we have:

- k_i is the inclusion of a summand: $1 \mapsto (0, \dots, 0, 1, 0, \dots, 0)$, where the 1 is in the i-th position.
- P_i is induced by the inclusion.
- $j\colon 1 \mapsto (1, \dots, 1)$.

Note that the isomorphisms on the top are via excision and the isomorphisms below are via the long exact sequence. Note that the outer square doesn't commute, but the individual squares and triangle commute. We now have

$$\begin{aligned} \beta(\deg(f)) = \beta(f_*(1)) = f_*(j(1)) = f_*(1, \dots, 1) = f_*(\sum_{i=1}^{m} k_i(1)) &= \sum_{i=1}^{m} f_*(k_i(1)) \\ &= \sum_{i=1}^{m} \alpha(f_*(1)) = \sum_{i=1}^{m} \alpha(\deg(f)_{x_i}) = \alpha(\sum_{i=1}^{m} \deg(f)_{x_i}). \end{aligned}$$

With this, we can conclude that $\deg(f) = \sum \deg(f)_{x_i}$. □

Example 4.6.19

Consider $f\colon \mathbb{S}^1 \to \mathbb{S}^1$ given by $z \mapsto z^k$. When $k = 0$, $\deg(f) = 0$. If $k < 0$, we can recover with $k > 0$ by composing with the reflection map. Now let $k > 0$. If $f^{-1}(y) = \{x_1, \dots, x_k\}$. Then, if we consider a small neighborhood U_i about x_i, each of these are mapped homeomorphically by f to an open neighborhood V of y. This local homeomorphism is given by a *stretching* by a factor of k and a *rotation* in the positive/ counter clockwise direction. The process of stretching is homotopic to the identity near x_i and so the local degree of the stretching is $+1$. A rotation is a homeomorphism and so its global and local degree at any point agree. Since the rotation is in the positive direction, we have that it is homotopic to the identity and has degree $+1$. Therefore, $\deg(f)_{x_i} = 1$. We can sum these to conclude that:

$$\sum_i \deg(f)_{x_i} = \deg(f) = k.$$

Let us go back to our example with $\mathbb{RP}^n$ (4.6.14).

Solution. We left off with

$$0 \longrightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{?} \dots \xrightarrow{?} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0$$

and we want to figure out $\deg(\Delta_n)$ for n even. Let $\Delta^n\colon \partial\mathbb{D}^n \to X^{n-1}/(X^{n-1} \setminus e^{n-1})$. We have as follows:

$$\deg(\Delta_n) = \sum_{x \in \Delta_n^{-1}(y)} \deg(\Delta_n)_x = \deg(\Delta)_x + \deg(\Delta_n)_{-x}$$

Then, in the homology, we have

$$\begin{tikzcd} H_n(\partial\mathbb{D}^n, \partial\mathbb{D}^n \setminus \{x\}) \arrow[rr, "a_*"] \arrow[dr, "\deg(\Delta_n)_x"'] & & H_n(\partial\mathbb{D}^n, \partial\mathbb{D}^n \setminus \{-x\}) \arrow[dl, "\deg(\Delta_n)_{-x}"] \\ & H_n(e^{n-1}/\partial e^{n-1}, e^{n-1}/\partial e^{n-1} \setminus \{y\}) & \end{tikzcd},$$

where a_* is the antipodal map. Note that a_* is the multiplication by 1 map, as n is even. Now, $\deg(\Delta_n)_x = \deg(\Delta_n)_{-x} = \pm 1$, and

$$\deg(\Delta_n)_x = \deg(a)\deg(\Delta_n)_{-x} = \deg(\Delta)_{-x}.$$

Therefore,

$$\deg(\Delta_n) = \pm 2.$$

Therefore,

$$H_k(\mathbb{RP}^n) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \text{ or } k = n \text{ is odd} \\ \mathbb{Z}/2 & \text{if } k \text{ is odd and } k < n \\ 0 & \text{otherwise.} \end{cases}$$

□

§4.6.iv Euler Characteristic

Definition 4.6.20. Let F be a finitely generated abelian group. Then, $F \cong \mathbb{Z}^r \oplus (\oplus_i \mathbb{Z}/p_i\mathbb{Z})$. The **rank** of F is the integer r.

We now introduce three definitions of the Euler characteristic (and show they are the same).

Definition 4.6.21. Let X be a Δ-complex that is finite of dimension n. Then, the **Euler characteristic** of X is defined as

$$\chi(X) = \sum_{k=0}^{n} (-1)^k |\{k\text{-simplices in } X\}|.$$

Definition 4.6.22. Let X be a CW complex that is finite of dimension n. Then, the **Euler characteristic** of X is defined as

$$\chi(X) = \sum_{k=0}^{n} (-1)^k |\{k\text{-cells}\}|.$$

Definition 4.6.23. Let X be a topological space with finitely many nonzero homology groups and all finitely generated. Then, the **Euler characteristic** is defined by

$$\chi(X) = \sum_{k=0}^{n} (-1)^k \operatorname{rank}(H_k(X)).$$

Lemma 4.6.24

Given a short exact sequence of finitely generated abelian groups

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

then $\operatorname{rank}(B) = \operatorname{rank}(A) + \operatorname{rank}(C)$.

Proof. Omitted. □

Example 4.6.25

A sequence illustrating the above is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

Proposition 4.6.26

Consider the exact sequence

$$0 \longrightarrow A_n \xrightarrow{d_n} A_{n-1} \xrightarrow{d_{n-1}} \dots \xrightarrow{d_1} A_0 \xrightarrow{d_0} 0\ ,$$

where the A_i are all finitely generated abelian groups. Then,

$$\sum_{k=0}^{n} (-1)^k \operatorname{rank}(A_k) = \sum_{k=0}^{n} (-1)^k \operatorname{rank}(H_k(A_\bullet)).$$

Proof. Consider the following:

$$0 \longrightarrow Z_n \xrightarrow{i_n} A_n \xrightarrow{d_n} B_{n-1} \longrightarrow 0$$

$$0 \longrightarrow B_n \xrightarrow{i} Z_n \xrightarrow{q} H_n \longrightarrow 0$$

Here, $Z_n := \ker(d_n)$ and $B_n = \operatorname{Im}(d_{n+1})$ and $H_n = H_n(A_\bullet)$. Here, i_n and i are inclusions, and q is a quotient map. From these, the sequences must be exact.

Now define $a_n = \operatorname{rank}(A_n)$, $z_n = \operatorname{rank}(Z_n)$, $b_n = \operatorname{rank}(B_n)$, $h_n = \operatorname{rank}(H_n)$. By the previous lemma, it follows that $a_n = z_n + b_{n-1}$ and $z_n = b_n + h_n$. Thus, we have as follows:

$$\sum_{k=0}^{n}(-1)^k h_k = \sum_{k=0}^{n}(-1)^k(z_k - b_k) = \sum_{k=1}^{n}(-1)^k(a_k - b_{k-1} - b_k) + a_0 - b_0.$$

The b_ks then cancel up until b_n. However, b_n is just equal to 0, so this expression ends up equating $\sum\limits_{k=0}^{n}) - 1)^k a_k$. □

Corollary 4.6.27 (Equivalence of Euler Characteristics)

The three definitions above for the Euler characteristic are equivalent.

Proof. Considering the sequence $\Delta_\bullet(X)$, we get that

$$\sum_{k=0}^{n}(-1)^k \operatorname{rank}(\Delta_\bullet(X)) = \sum_{k=0}^{n}(-1)^k \operatorname{rank}(H_k^{\Delta}(X)).$$

Now if we consider $C_\bullet^{\mathrm{CW}}(X)$, we get that

$$\sum_{k=0}^{n}(-1)^k \operatorname{rank}(C_k^{\mathrm{CW}}(X)) = \sum_{k=0}^{n}(-1)^k \operatorname{rank}(H_k^{\mathrm{CW}}(X)).$$

□

§4.7 Mayer-Vietoris

Theorem 4.7.1

Let $X_1, X_2 \subseteq X$ such that $X = X_1 \cup X_2$. Then there exists a long exact sequence

$$\begin{tikzcd}
\dots \arrow[r] & H_n(X_1 \cap X_2) \arrow[r, "i_*"] & H_n(X_1) \oplus H_n(X_2) \arrow[r, "j_*"] & H_n(X) \arrow[dlll, "\partial"] & \\
\dots \arrow[r] & H_0(X_1 \cap X_2) \arrow[r, "i_*"] & H_0(X_1) \oplus H_0(X_2) \arrow[r, "j_*"] & H_0(X) \arrow[r] & 0
\end{tikzcd}$$

Proof. Recall proposition 4.4.15. Look at $\mathcal{U} = \{X_1, X_2\}$, and consider the following sequence:

$$0 \longrightarrow C_\bullet(X_1 \cap X_2) \xrightarrow{i} C_\bullet(X_1) \oplus C_\bullet(X_2) \xrightarrow{j} C_\bullet^{\mathcal{U}}(X) \longrightarrow 0$$

Here, the morphisms i, j are defined as follows. Given the diagram

$$\begin{tikzcd} & X_1 \arrow[dr, "j_1"] & \\ X_1 \cap X_2 \arrow[ur, "i_1"] \arrow[dr, "i_2"'] & & X \\ & X_2 \arrow[ur, "j_2"'] & \end{tikzcd}$$

where i_1, i_2, j_1, j_2 are all inclusions, we define the following:

$$\begin{aligned} i = \begin{bmatrix} (i_1)_\# \\ (i_2)_\# \end{bmatrix} &: C_\bullet(X_1 \cap X_2) \to C_\bullet(X_1) \oplus C_\bullet(X_2) \\ & c \mapsto ((i_1)_\#(c), (i_2)_\#(c)) \\ j = \begin{bmatrix} (j_1)_\# - (j_2)_\# \end{bmatrix} &: C_\bullet(X_1) \oplus C_\bullet(X_1) \to C_\bullet^{\mathcal{U}}(X) \\ & (c_1, c_2) \mapsto (j_1)_\#(c_1) - (j_2)_\#(c_2) \end{aligned}$$

Then, $(j_1)_\#(c_1) - (j_2)_\#(c_2) = 0$ if and only if $c_1 = c_2$. Thus, $\ker(j) \subseteq \mathrm{Im}(i)$. $\mathrm{Im}(i) \subseteq \ker(j)$ since $(j_1)_\#(c) - (j_2)_\#(c) = 0$. Therefore, it is a short exact sequence of chain complexes. This gives a long exact sequence on homology, $H_n^{\mathcal{U}}(X) \cong H_n(X)$. (The connecting homomorphism..?) $\square$

Example 4.7.2

We can use Mayer-Vietoris to compute the homology of the surface of genus two, which we call M_2. Let $X_1, X_2 \subset M_2$, such that they are both homeomorphic to a torus with one hole, and they intersect on a cylinder. Note that this is homotopic to the wedge of two circles (a square with a hole deformation retracts to its boundary). A cylinder is just homotopic to a circle, so we may now apply Mayer-Vietoris we get the following long exact sequence

$$\begin{tikzcd}
\dots \arrow[r] & H_2(X_1 \cap X_2) \arrow[r] & H_2(X_1) \oplus H_2(X_2) \arrow[r] & H_2(X) \arrow[dll, "\partial"] & \\
& H_1(X_1 \cap X_2) \arrow[r] & H_1(X_1) \oplus H_1(X_2) \arrow[r] & H_1(X) \arrow[dll, "\partial"] & \\
& H_0(X_1 \cap X_2) \arrow[r] & H_0(X_1) \oplus H_0(X_2) \arrow[r] & H_0(X) \arrow[r] & 0
\end{tikzcd}$$

We have that $H_2(X_1 \cap X_2) = 0$, $H_2(X_1) \oplus H_2(X_2) = 0$, $H_1(X_1 \cap X_2) = \mathbb{Z}$, $H_1(X_1) \oplus H_1(X_2) = \mathbb{Z}^4$, and $H_0(X_1 \cap X_2) = \mathbb{Z}$ and $H_0(X_1) \oplus H_0(X_2) = \mathbb{Z}^2$. Recall that $\tilde{H}_k(\vee \mathbb{S}^1) \cong \oplus \tilde{H}_k(\mathbb{S}^1)$, and

$$\tilde{H}_k(\mathbb{S}^1) \cong \begin{cases} \mathbb{Z} & k = 1 \\ 0 & \text{otherwise} \end{cases}.$$

Therefore,

$$H_k(\mathbb{S}^1 \vee \mathbb{S}^1) \cong H_k(X_1) \cong H_k(X_2) \cong \begin{cases} \mathbb{Z}^2 & k = 1 \\ \mathbb{Z} & k = 0 \\ 0 & \text{otherwise} \end{cases}.$$

We have that the map

$$H_0(X_1 \cap X_2) \to H_1(X_1) \oplus H_1(X_2)$$

is $[1, 1]^T$ since we are including a path connected space in to two path connected spaces, so we have the identity on each homology group, Moreover, the map

$$H_1(X_1 \cap X_2) \to H_1(X_1) \oplus H_1(X_2)$$

is the zero map, since the circle that generates the first homology group of $X_1 \cap X_2$ is trivial in the punctured torus. Therefore, by exactness we can conclude that $H_2(X) \cong H_1(X_1 \cap X_2) \cong \mathbb{Z}$, and that $H_1(X) \cong H_1(X_1) \oplus H_1(X_2) \cong \mathbb{Z}^4$). And of course $H_0(X) \cong \mathbb{Z}$ since it is path connected.